From Fold to Materials

An Exact, Parameter-Free and Machine-Closed Reconstruction of Materials Science from Smithian Fold Theory

Ernos Labs
Open Source Science Platform and Knowledge Tree

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Third clean-room reconstruction - Materials inventory complete
84 required laws - 21,504 exact candidates
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Paper: CC BY 4.0 - Code: Apache-2.0

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Abstract

This paper reports the complete Materials Science branch of the third clean-room reconstruction of Smithian Fold Theory. The frozen inventory contains 84 obligations in ten ordered subbranches. Their exact grammars execute 21,504 candidates and 21,504 one-for-one decisions, yielding 84 unique survivors, 84 depth-independent closure certificates, 336 passing adverse controls, 84 implementation-distinct reconstructions and 84 post-seal NIST/BIPM correspondence checks. All 84 dependency graphs reach the single premise-free root theorem. The whole prediction surface was sealed before any external Materials source identity was selected. Exact structural consequences include six simple-cubic neighbours; periodic rotation orders one, two, three, four and six with five excluded; seven crystal systems; fourteen Bravais classes; and three acoustic branches. Material-property magnitudes that vary with specimen, method, direction, scale, history or conditions remain conditional records and are not presented as universal constants.

1. Publication, authorship and open-science boundary

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The paper is released under CC BY 4.0 and code under Apache-2.0. Copyright preserves authorship while the licenses preserve open inspection, reuse, criticism and independent replication. The Ernos Labs designation identifies conformance to the published methodological and community standards; copying alone does not confer that designation.

2. Exact scope and meaning of closure

Closure means every obligation in the frozen Materials question surface has one engine-admitted theorem within its exact generated grammar, full dependency trace, adverse-control boundary and registered empirical protocol. It does not mean that every possible material, specimen, processing history or future observation has been enumerated. Biology beyond the material interface, clinical outcomes, Earth-system fate and application engineering remain later or external validation domains.

3. Constitutional arithmetic and prohibited premises

Structural absence is empty One, never numerical zero. Counts are generated positive wholes; exact parts are positive held fractions. Opposition and direction are held fibres, never negative proof quantities. Irrational and imaginary proof values, floating proof arithmetic, a completed infinity, ungenerated continuum, axioms, free or fitted parameters, imported constitutive equations, pretrained models, consensus selection, target-derived rules and application results are prohibited. Host counters can enumerate artifacts but cannot act as scientific values.

4. Dependency spine to the root theorem

Each Materials registration has `root_theorems=(SFT-ROOT-THERE-IS-NO-NOTHING,)`, zero axioms and zero free parameters. Dependencies are accepted only when their immutable model-admission receipts already exist. A separate graph traversal over the materialized registrations confirms 84 of 84 paths reach the root. The path runs through the independently admitted Foundation, Mathematics, Information Science, Physics and Chemistry receipts named in each claim section; no branch name substitutes for an actual dependency identity.

5. Complete pre-source seal

Before NIST or BIPM source identities were chosen, V3 froze the 84 obligations, the exact counting module and the 84 target-blind blueprints. The seal records 21,504 candidates, hashes all three source files and binds every `(claim identity, exact result, prediction label)` tuple. It expressly records that source identities were unselected and target content unopened. Its sealed payload hash is `sha256:da97a6cb6a001964a069b45a5a3698e7ea90f334a08d69c62bd09c46d8112035` and its prediction-set hash is `sha256:ed531eccac30f6915908052ced79e3875d8faf75c1add9eec48c7360d9f6a1e7`.

6. Single fail-closed engine

For each claim, the engine verifies registration, source identity, admitted dependencies, exact candidate identity and cardinality, one decision per candidate, exactly one survivor, closure, minimality, named-shape uniqueness, four adverse controls and implementation-distinct reconstruction. Empirical evidence additionally requires an evaluator-verified derivation seal, capability isolation, target absence before seal, custody release only after a matching prediction seal, all-row retention, an explicit falsification condition and a tampered unfavorable control. A violation halts and preserves a rejection receipt; it cannot enter the model census.

7. Empirical method

Thirty-three byte-frozen records from NIST and BIPM/JCGM supply the independent external boundary. Each claim requires at least two claim-specific discriminators. The predictor cannot read filesystem, network, clock, environment, subprocess, dynamic import or foreign functions. A distinct custodian opens source content only after sealing, verifies every snapshot hash and required feature, constructs the registered observation record and releases it through the portable custody exchange. External authority may falsify a correspondence but cannot alter the already sealed grammar or survivor.

This branch distinguishes exact structural counts from specimen-dependent magnitudes. Six neighbours, allowed rotation orders, seven systems, fourteen Bravais classes and three acoustic branches are exact count/classification claims. Elastic moduli, conductivities, strengths, transition temperatures and analogous quantities depend on material identity, preparation, direction, scale, method and condition; the Fold laws force those dependencies and ledgers, not one fictitious number for every material.

8. Exact structural count certificates

8.1 Six simple-cubic neighbours

Three independently admitted stable spatial generators each carry exactly two held Fold orientations. Their literal product contains six distinct nearest-adjacency positions. No signed coordinate or numerical zero is needed.

8.2 Rotation restriction

The rank-two integral rotation boundary requires Euler-totient degree no greater than two. The factor certificate proves depth independence: if odd prime p divides an order, p-1 divides its degree, leaving only odd prime three; the remaining powers satisfy `a<=2`, `b<=1` and exclude simultaneous `a=2,b=1`. Orders one, two, three, four and six survive; five is the least excluded positive order.

8.3 Seven crystal systems and fourteen Bravais classes

The rank-three metric grammar enumerates three length-equivalence classes times four angle-equivalence classes and retains seven rotation-compatible quotient classes. Crossing those seven classes with five centering forms yields thirty-five candidates; basis and symmetry compatibility retain fourteen. The complete named survivors and exact claim receipts appear in their sections below.

8.4 Three acoustic branches

For one held propagation orientation in rank three, displacement has one propagation-aligned orientation and two independent transverse orientations. The complete displacement census therefore contains three acoustic branches.

9. V1/V2 reconciliation without answer import

The V2 corpus is used only as a question-level completeness obligation. Steps 47, 49, 52, 54, 72, 74, 75, 133, 137, 143, 193 and the Materials portion of 291 are mapped to newly executed V3 claims. None of their prior values, equations or certificates is exposed to the V3 derivation. Step 291's planetary and Tully-Fisher content is explicitly routed to Astronomy and Cosmology rather than hidden inside Materials closure.

10. Reading the exhaustive ledger

Every claim section below records the theorem, exact dependency identities, grammar and boundary, all eight binary axes, the 256-candidate census, sole survivor, base/successor certificate, witnesses, meaning, controls, independent implementation, post-seal source custody, falsifier and immutable evidence identities. The machine-readable JSON remains authoritative and is bundled with this paper.

11. Measurement Identity

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

1. Material system and retained identity

Claim identity: `SFT-MAT-MEAS-MATERIAL-001`

Question and exact theorem. A material is the least complete constituent-and-relation carrier whose composition, bonding organization and declared observation boundary remain distinguishable.

```
complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-constituent-carrier, composition-and-bonding-retained, material-boundary-held and declared-scale-identity.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-constituent-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`composition-and-bonding-retained`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`material-boundary-held`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`declared-scale-identity`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5fd1f0e5902535142cb31d865d5ea64e83b9e08cd7f630cac919c3a01e84d840`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0bbc08685cb1726579e93191fb6f6bde56681445d2d229aa315724a6eb6ade94`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a401a612b8a1d870368aa2d6a487a31c012370a228b62c70915c9c6914706a23`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b2b3e2c6051ce0f6601c2466140f40ceeda0b4cb823d5e91c18daebcf3c5fa50`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:9a39682a714c1e73b5df88972c201028726440e1bf460169e557a45de1513cd5`; certificate `sha256:bbe90ba7d0b746808b9a3dbe8e33c1c7ef995b34f1db2b994dccb8810c85f6ce`; external-validation hash `sha256:f5880932fc39e96a61053ce252e6ffa011365bb09e677042321c321c9247ef32`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:683ba4d17cac39004f018001749fe7c52b9a376f80659e6bca9ecdabdebf44cf`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-REFERENCE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/reference-materials; snapshot `experiments/external_sources/materials/snapshots/nist-reference-materials.html`; `sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9`; scope: material identity, property characterization, uncertainty and traceability.

Comparison and custody records:

- sft-mat-meas-material-001-authority-correspondence: predicted complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-constituent-carrier__composition-and-bonding-retained__material-boundary-held__declared-scale-identity__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material system and retained identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-constituent-carrier, composition-and-bonding-retained, material-boundary-held and declared-scale-identity.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:3e6c63ac46ba6fa3a33796b77a14d8e56af8461367e178d8e4f7133e5691f15a`; derivation seal `sha256:03747f2235206b6276b4df2c863ec64edb6581fe3f24c6cd674312492061e806`; engine receipt `sha256:100e32f861c1188720ddb10a07f5fa28b9a94d7650c859382865519fb64e1be6` at `receipts/engine/model_admitted/SFT-MAT-MEAS-MATERIAL-001-100e32f861c11887.json`; empirical validation `sha256:02058a82c7880fbce8194cc5b4834494351c29d565fe93525deabe152b873680`; measurement receipt `sha256:179da930e46c4b46067ebf11bd97d1a18acda644cb9cc6930c57c5e6e178f935`; isolation `sha256:88f0cd6f457584283c4106d0765538acfd07856df2cdba70cd8d43e5127c8da6`; custody `sha256:9586bdfec7797dd283c176a2c49ac47e128f15b36d4c3a52b4a31b1011477fa2`.

2. Specimen and sampling relation

Claim identity: `SFT-MAT-MEAS-SPECIMEN-001`

Question and exact theorem. A specimen is a source-bound finite part of a material with an explicit sampling map, retained preparation history and declared representativeness boundary.

```
source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundar
y-retained__representativeness-declared__complete-state-transition-boundary-trac
e__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves source-bound-specimen-carrier, specimen-to-material-provenance, sampling-boundary-retained and representativeness-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	source-bound-specimen-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	specimen-to-material-provenance	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	sampling-boundary-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	representativeness-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundary-retained__representativeness-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9c35918d4fd63e38482a40eb699e350a2ee519aa20cf3e4da387a1c8afe55fd9`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:88c6c3fff183744cea93e433facfb571d89ebed74760801f5f079da131b8e6e3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:63c1304a41770cfadab42af9fb67769ed75aaea6637ccef8bc2320976630e09b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:90dbb259614c0b3e1e2f7c8918b80aaf091a937d31290d4872e84c9b43ee85ee`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:f58c2ea33d59fda86da1273df9a7eda41e54ca393804aa6349eb3f8e3140be34`; certificate `sha256:23109cf272ed2894bb8ede6176ff82975653df8ff6a282aa3d2b31fedf0bdcad`; external-validation hash `sha256:ea630c1ae5ba9b7c2b8b409f95a58ea69bcd82f8ea2dba1e62b98b4d681311aa`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:3a695fe79bc147ca2cae3cc48766472922fb8ad57766bc9f0d92fd979e449ac0`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TEXTURE-PHASE-FRACTION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction; snapshot `experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html`; `sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc`; scope: grains, phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-meas-specimen-001-authority-correspondence: predicted source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundary-retained__representativeness-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived source-bound-specimen-carrier__specimen-to-material-provenance__sampling-boundary-retained__representativeness-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered specimen and sampling relation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves source-bound-specimen-carrier, specimen-to-material-provenance, sampling-boundary-retained and representativeness-declared.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:eb6cb5d9d6699b6fa8c5d5a1b80d08a30e5340ee165f372adb13a378fc1c4c9d`; derivation seal `sha256:de5d12aa6b9d38df4dd41e838be989f5efdaa3c54b549b8eda98543e04d1c829`; engine receipt `sha256:64d737e3f4c0fc5a33a13a5af86282e4e74420b103aba81e067d3baa1b55bbac` at `receipts/engine/model_admitted/SFT-MAT-MEAS-SPECIMEN-001-64d737e3f4c0fc5a.json`; empirical validation `sha256:fc00c8fadea60629d2db8350091da01088f9de13941e22ca9288749e6c0cb1bd`; measurement receipt `sha256:8b2eda391cb0daa95d3a916046d65defafa288867eda601ac9119b008d44878f`; isolation `sha256:6582228d88ed2f11aece7eb4830965c3a7cd5b5f166c50c7eaba4e4280125636`; custody `sha256:19398416e13e841a09e8a15c06fb484d7eb5911ea03afcef1ef03a01cd5e28aa`.

3. Material composition record

Claim identity: `SFT-MAT-MEAS-COMPOSITION-001`

Question and exact theorem. Material composition is the exact part-to-whole record of every retained component on one declared amount or support basis.

```
complete-component-support__exact-part-to-whole-composition__component-identitie
s-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-t
race__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-ru
le
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-component-support, exact-part-to-whole-composition, component-identities-retained and basis-and-uncertainty-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-component-support`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`exact-part-to-whole-composition`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`component-identities-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`basis-and-uncertainty-declared`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1495e5970a1ecdcdf9d4372587b7b84cd29df2e68eb10bc9924511e2258d147b`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8a163f92ab47dd17cee0a3c65a94c64a6eafced530cc6453a2784f2177cd17b6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:09fd0a2fc18530f0598ae2dcbf496100a032e91d27f614edab80f0a72aac8539`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ee80dc3cb72bf4024bf186ddbc6b8334e46b57e03c145399957f2b9d82451f92`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:1c370302c7b1df34e6027d61af197f02a43ec583c4b066fe47287bca0dedddfc`; certificate `sha256:ab5e53b673a0f24c340e78feed5d77ad4fdcc972a291abcc90cd54195a54b571`; external-validation hash `sha256:d7de9d562123cd8b519c68f77809dc68f4ce4fbdb53724a2b4f4ed888b238f21`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:ed1f6ea00fbb6f3c4efccac54ff08f15e6cf8eb6a86628d65ea245838980eb27`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MATERIALS-DIVISION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/materials-science-and-engineering-division; snapshot `experiments/external_sources/materials/snapshots/nist-materials-division.html`; `sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3`; scope: materials measurement, composition, interfaces, processing, damage, classes and lifecycle.

Comparison and custody records:

- sft-mat-meas-composition-001-authority-correspondence: predicted complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-component-support__exact-part-to-whole-composition__component-identities-retained__basis-and-uncertainty-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material composition record observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-component-support, exact-part-to-whole-composition, component-identities-retained and basis-and-uncertainty-declared.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

`sha256:cdcbc439c4cfe1e162e40a6b8bc72068658e552c4e4a03c16c6984d5dd8f8340`; derivation seal `sha256:5767520188e33d726e0ef14871b31adc74a3418ab648a59016e22946abc54465`; engine receipt `sha256:b172a8dfe74dc3bb3d5159178251f4b5a4c6de311ac14a084fe99ba9f87bf1a6` at `receipts/engine/model_admitted/SFT-MAT-MEAS-COMPOSITION-001-b172a8dfe74dc3bb.json`; empirical validation `sha256:ff62f3576785762465d2a17af0abcf019afb29cf2e5e89f883176c02f23eb5f1`; measurement receipt `sha256:6879a614933015d04fd3da7337ddce7118ae2db28bb23b7ed439645af4394a62`; isolation `sha256:a133e07a5e3cb3a00fa8d464abe63e91a2c1dc18f20be9c5fd070e6cc73ef503`; custody `sha256:8de12da81639f5178c647e8da843491a125d1a038e8a92f7c88882f323409f6a`.

4. Material phase identity

Claim identity: `SFT-MAT-MEAS-PHASE-001`

Question and exact theorem. A material phase is a maximal connected support whose constituents share one stable recurrence and symmetry observation class at declared conditions.

```
complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-phase-support, shared-recurrence-equivalence, phase-boundary-retained and condition-bounded-phase-class.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-phase-support`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`shared-recurrence-equivalence`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`phase-boundary-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`condition-bounded-phase-class`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1ac7de1af45e524b8c9b0da7b3f8f877fba4a1b78e7a2f01c44861bb0d2901f3`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:77b4f455bc737b32694908d129af28b0479a19481f4c990307d06cbc4a071c00`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:25fc762e494c9db8e616dca4b2a87b825f2d18221bb4b9de422f5e343895e0af`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8fac8c656448533cc88dc6c3a67e5b772dd899d27934e94ed86ceda602fa3351`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:dbb3ede679d2df917b3b1d3925870abcd640c7ac5b6bef94da3b36f5ad31bb09`; certificate `sha256:9b3957f05da97dff4942b040c1abb90b95d8a5e7bc4a8d7656b4ccbfb48f3c04`; external-validation hash `sha256:60dc5f40aab324386373a4beebe4ed741c3e27f21624dfe65830c09f7f607c2a`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:14b6f6f9b39bcb2995bc42d4164785b635c0a872b167d05174462ef7f62a4a06`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-meas-phase-001-authority-correspondence: predicted complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-phase-support__shared-recurrence-equivalence__phase-boundary-retained__condition-bounded-phase-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material phase identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-phase-support, shared-recurrence-equivalence, phase-boundary-retained and condition-bounded-phase-class.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:8a93bd0b5a0225fb750f1e4251ae057e9ba3c649e1e767ba04f9b9ae00b3da99`; derivation seal `sha256:ad68cff33969540719db7d2b44560a371ddaa5bd4de6645407eaff0e0e21db79`; engine receipt `sha256:820c18b5745c2eb4d0f4dd9fb15f1c243d4f34d2c1c7b87d421316ff4426f1ee` at `receipts/engine/model_admitted/SFT-MAT-MEAS-PHASE-001-820c18b5745c2eb4.json`; empirical validation `sha256:3217c3fcbeef32eaab16508631f6a9200e0536e889546fb08ca0b278e7d225a9`; measurement receipt `sha256:40605d06c8c56d5d1cf3d24e347e2f925be621d0d0b1755d25c887234d0af836`; isolation `sha256:7f5971803416f24c7c627e35f7ad99fc03e6b838442f461e642f65437aea1736`; custody `sha256:68aa9c9ad49548681df4ac2ca134e5cae64aa76b8e217219cb3affd796e11a10`.

5. Microstructure as mesoscale organization

Claim identity: `SFT-MAT-MEAS-MICROSTRUCTURE-001`

Question and exact theorem. Microstructure is the complete declared-scale arrangement of phases, grains, interfaces and defects between constituent and bulk observation levels.

```
complete-mesoscale-carrier__constituent-arrangement-relation__interfaces-and-def
ects-retained__resolution-bounded-record__complete-state-transition-boundary-tra
ce__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-mesoscale-carrier, constituent-arrangement-relation, interfaces-and-defects-retained and resolution-bounded-record.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-mesoscale-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	constituent-arrangement-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	interfaces-and-defects-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	resolution-bounded-record	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-mesoscale-carrier__constituent-arrangement-relation__interfaces-and-defects-retained__resolution-bounded-record__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4d9da1817be3190c573b9b4258210ab512e0d2c2e6fbc8e28e954970f97378ae`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f9e6509348500baddf4417a261a58945012068b796a21f30110cb7079c17d2ce`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e7c3258831f4ada06f194a610d6b18187859f9ddb3d850ccf651c1241220632c`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:124c0a150b4c3c2041ea25e07f9a87131a8ee452c614b442f8bf200cf41b0fd9`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:5e3d4ac0287c0720de1b47e900c43f267b74a387e9ca2f75328eb09130bd251f`; certificate `sha256:f076e7ddf7a8650c7b13642e5bae9cb217f53a23479887a911e2b00883c47195`; external-validation hash `sha256:ee24cfe1b083142c03c82b0c868bf8c5112305f35839a1fff8a5f4204e76f52d`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:53a34e89965e7c97cfc0780ecddbf9937112caa7dc849483c429122346ce0f93`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-meas-microstructure-001-authority-correspondence: predicted complete-mesoscale-carrier__constituent-arrangement-relation__interfaces-and-defects-retained__resolution-bounded-record__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-mesoscale-carrier__constituent-arrangement-relation__interfaces-and-defects-retained__resolution-bounded-record__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered microstructure as mesoscale organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-mesoscale-carrier, constituent-arrangement-relation, interfaces-and-defects-retained and resolution-bounded-record.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:8e536fdd3d154712fc1b447a39f6c62acaa780b4680effe34a6c5b81c3de1e3c`; derivation seal `sha256:86d7179cfce073c0669747b856b118570b7f56bdb77f9049719a61d9479239b9`; engine receipt `sha256:2bd2dd39260bb2c978e97390f519d99570ec14a1ed71f2438040642a7a39fb2a` at `receipts/engine/model_admitted/SFT-MAT-MEAS-MICROSTRUCTURE-001-2bd2dd39260bb2c9.json`; empirical validation `sha256:d1d475bce79364060bd227a00440231974a97b9a698adcb5aa593114f145c1fe`; measurement receipt `sha256:687767a7a15260acce4b221cf3dcae4c7de94a93c6de310d64d9ea0763e65430`; isolation `sha256:86f3233608ea9a812c9e9b5f8ceee097f36d34afd46309e024c52f2c0f659472`; custody `sha256:51b87a705394f2363ee5beb0436787fdfaa257ee17792af99a438512bfb55e70`.

6. Material property relation

Claim identity: `SFT-MAT-MEAS-PROPERTY-001`

Question and exact theorem. A material property is a reproducible relation between a declared material state and stimulus/response records, conditional on direction, scale, history and method.

```
material-state-carrier__stimulus-response-relation__condition-and-direction-reta
ined__method-bounded-property__complete-state-transition-boundary-trace__root-bo
und-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves material-state-carrier, stimulus-response-relation, condition-and-direction-retained and method-bounded-property.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`material-state-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`stimulus-response-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`condition-and-direction-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`method-bounded-property`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `material-state-carrier__stimulus-response-relation__condition-and-direction-retained__method-bounded-property__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9d5f10ef32de296ff8fd844e2d412fc5a4fa61e6b436547719a742ceae9254f6`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:eee60e6d8c42048b68a43ec3d3460d5ab672fe4c623828350fe25db95e1e6b0b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5ba5573122548c6b1fbc93f566e3201eeb7d0a7862100fded5a2eb25342d30d5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7a63bce6b4e5bcf0be1bec3e56ba88f15183e47d97e62c5be2ba087f4d9c2c66`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:c102a26be454aff89e7e294cb42102b49138996753da2059894e887ce197b1fd`; certificate `sha256:a8cb28c31491b1f1a5a7f43ee96c82ef56e8f5ebf213bef1a308b35284d49599`; external-validation hash `sha256:8d89cd0128a033bc11d35c7516a69ed90d3575999a75632bee9b7aed56a4b3bd`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:80cd9c287b41551e990818ae07a98fdf4cdd4528a814d66ff7bdd8f8ae2c4a0f`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-REFERENCE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/reference-materials; snapshot `experiments/external_sources/materials/snapshots/nist-reference-materials.html`; `sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9`; scope: material identity, property characterization, uncertainty and traceability.

Comparison and custody records:

- sft-mat-meas-property-001-authority-correspondence: predicted material-state-carrier__stimulus-response-relation__condition-and-direction-retained__method-bounded-property__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-state-carrier__stimulus-response-relation__condition-and-direction-retained__method-bounded-property__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material property relation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves material-state-carrier, stimulus-response-relation, condition-and-direction-retained and method-bounded-property.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:80fd697248d6321e6b90d9794452f6b9fd387ea961db765471df7548a94761d7`; derivation seal `sha256:1d1f8fd9378247d9dfeb92613f2b5fec6cb4c9c3909dc46815a7f416d4db2bdc`; engine receipt `sha256:d13a1540d3b9d11dd631f188e97f4096b58dcc42f19ad0dae0b4d7ad1eb60857` at `receipts/engine/model_admitted/SFT-MAT-MEAS-PROPERTY-001-d13a1540d3b9d11d.json`; empirical validation `sha256:13e08bdc1143aa9d50e2532125113b4afabc13db1c34e8d4079f2f91bb00e27c`; measurement receipt `sha256:7b6727c92a907c19971097f1e19c34a2064840a58b1892285497d238cfffc014`; isolation `sha256:fadc55d82e80bdb6922f6cf2b8e4dc9e0b233a27f6e4324a1db1b348cba2c95d`; custody `sha256:a75d0c32265d651a4565cb6f302dae96060190de2899773c0b4fcb191e5981ee`.

7. Materials uncertainty and metrological traceability

Claim identity: `SFT-MAT-MEAS-TRACEABILITY-001`

Question and exact theorem. A materials measurement is admissible only with a complete result, uncertainty, method, specimen, condition, calibration and unbroken reference chain.

```
complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-measurement-record, result-to-reference-chain, uncertainty-components-retained and calibration-and-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-measurement-record	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	result-to-reference-chain	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	uncertainty-components-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	calibration-and-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

The branch begins by retaining material, specimen, composition, phase, microstructure, property and metrological identities as different carriers with explicit method and scale boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:06fa01f03b954ef5285336e526a90aed8ebffbcb55a443245ef359e1ca0562e4`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:00c29832548d03a45275175072a6636fe6db09d58277db2f6b593eb666bb770b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:664397bfce016fac6e18934501599f9cd4d638f486cf68fccd6a6e85a98e4e09`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:816bbf074bc4def5e8a11de819f574d9f945a4197e01ebce28f46f3b6c1fa869`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:f70bd0aa9ed0e1a4e81927798902ca76757e566963e1a41c71ec14bf9b9f06d8`; certificate `sha256:93eb0026b754f4235aed15c8d33ccada191ea0ddfe7879abed1bee7aa0b05f57`; external-validation hash `sha256:67f1a8e737f8f5da46378dcebd60f9ee9adce75c99f638dc9252cc1ee114c5b4`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:da9c91f2ca355e816db0feb24791e6b9e2f07fd54d1ae885376692c784507db3`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `BIPM-JCGM-VIM-TRACEABILITY-2026-07-24` - Joint Committee for Guides in Metrology / Bureau International des Poids et Mesures; https://jcgm.bipm.org/vim/en/2.41.html; snapshot `experiments/external_sources/materials/snapshots/bipm-vim-traceability.html`; `sha256:3c06dca2ea8ad71516b027b9979e7a8381edd1576606ad10a379591c4cb6dba4`; scope: measurement result, reference chain, calibration and uncertainty.

Comparison and custody records:

- sft-mat-meas-traceability-001-authority-correspondence: predicted complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-measurement-record__result-to-reference-chain__uncertainty-components-retained__calibration-and-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered materials uncertainty and metrological traceability observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered measurement_identity grammar that preserves complete-measurement-record, result-to-reference-chain, uncertainty-components-retained and calibration-and-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

`sha256:0331cf713780ef603d89ab128524abbac7f2778801318dd0b8fd437d65b9ca0e`; derivation seal `sha256:827cfa9bc16633bc9e45d63d74209619bcfd40942f632c5bcdc6d0cf83da8098`; engine receipt `sha256:b0bf205091f5c263309fad96e87e51572c4a3f60f40d61263205778c4509f299` at `receipts/engine/model_admitted/SFT-MAT-MEAS-TRACEABILITY-001-b0bf205091f5c263.json`; empirical validation `sha256:7929a09e55e51aa0aa0d30d435bfa2a2755c8f23d5951673931c13c94d98e889`; measurement receipt `sha256:075d6446b70e416f2a1961f712d1b26ac009326b0eec2664ac4e1edb2398ad06`; isolation `sha256:e79a2ff20201393fcff90b3a3fa417752ae743648e9b0cc3f7eb73989252a169`; custody `sha256:16ebf8cce951c7bf7d037eaa5be0825ab8703ad75184e4291b569d394fb4dff3`.

12. Crystal Quasicrystal

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

8. Generated lattice organization

Claim identity: `SFT-MAT-CRYST-LATTICE-001`

Question and exact theorem. A lattice is a completely generated site-and-adjacency network whose local word recurs under retained translations.

```
complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated_
_finite-support-lattice-class__complete-state-transition-boundary-trace__root-bo
und-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-site-carrier, translation-recurrence-relation, adjacency-word-repeated and finite-support-lattice-class.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-site-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`translation-recurrence-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`adjacency-word-repeated`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`finite-support-lattice-class`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated__finite-support-lattice-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c9646ff6ef73ed550d9dd57d3bd3376e1f2b6d1505670ff5b473a2ecc1f16dc5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4b1c36d0e527ba983cfb859c1ee7030c18d678f7766b0b00850691b28a340ce7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d9773d30b11ed06149d52d2aaade731e6df807698242f853bb43b56cad410efe`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3fffc29ee144f3e4f558b5a20eed2e6c152a3d985a503c009f484019144df44f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:45dc456bf89de22438417c63625a628d65c2cfed0f0a944c50fb3e1b2f26e5cb`; certificate `sha256:4daea4fea105991ed38bba1541939504c775085e2bc923e88d0c73fbdc4374dc`; external-validation hash `sha256:b25db7f0facb055cb4017842334dd5509e5a9064e847f2b0cafb42796cf683ec`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:79e502524e622bf1c982a0b73cbb777eb604a59fdc448a82abdc3696c033cabf`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24` - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot `experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html`; `sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6`; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- sft-mat-cryst-lattice-001-authority-correspondence: predicted complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated__finite-support-lattice-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-site-carrier__translation-recurrence-relation__adjacency-word-repeated__finite-support-lattice-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered generated lattice organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-site-carrier, translation-recurrence-relation, adjacency-word-repeated and finite-support-lattice-class.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:953d8aba886fe20066df18e9a13f9d6984043da3ff9379cd8346719abac8a594`; derivation seal `sha256:8de7be19a12b78faf87780e5d8b279fb8c8e7ba7e9a087d8adb7f85a14bde14a`; engine receipt `sha256:29a30c7d21227319c53ac130261c6ef9c7619d92e841bef00c3bc3f2f8813209` at `receipts/engine/model_admitted/SFT-MAT-CRYST-LATTICE-001-29a30c7d21227319.json`; empirical validation `sha256:53c0b57c7770e4ba2e65a1ded9835b42fcf0fa452df42f4323d3f4cabac1b40b`; measurement receipt `sha256:5f15d36c9fc8bcd15bcaddbcaa0a7f8868ebf2f9bcee0598548c35f646f00bd3`; isolation `sha256:76de27215d27a3fdf1c148dce465d257b19dc89c1c9cae51905839acfd2d905b`; custody `sha256:ffa931b5491e2c28acb540ecd2204c2262a022b75ecf8e633df32ad8b3791436`.

9. Primitive unit-cell identity

Claim identity: `SFT-MAT-CRYST-UNIT-CELL-001`

Question and exact theorem. A primitive unit cell is the least complete labelled cell whose generated translations reproduce every lattice site and relation once.

```
minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis
-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forw
ard-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves minimal-cell-carrier, translation-generates-whole, no-redundant-site-copy and basis-choice-canonicalized.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`minimal-cell-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`translation-generates-whole`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`no-redundant-site-copy`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`basis-choice-canonicalized`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:95bee5e1bc790838af7571ea763454bb76b0b4cfd29afc969b46e5b894b1773a`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:655991b393c1177811a27031f45194e75585bd304199e7024f505539a7895dd8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:52f038e5dccd305e1b474d16809934ad563804788d0130f0902e8985630f440e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a0d6eaaecef72a08b0d991621e1f772a0c5b2da721de6f518654924f3b606102`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:0154a35feeac91950959dc4366d8f1f7fbc32bdd85bba4a14373ce7828cc2de1`; certificate `sha256:07c7eb750e1bfdc7075611d7232bff0c62de4bedbe700286e4d62936adec502a`; external-validation hash `sha256:85798ec472c45e4f89fdfc19152f5e5835f6dbbbd8af31fdda44cfc5c1ec8d11`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:7df3d5e1106dd9be2853c3ceff4a96c592603507e4909374b63428425490fbd0`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24` - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot `experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html`; `sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6`; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- sft-mat-cryst-unit-cell-001-authority-correspondence: predicted minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived minimal-cell-carrier__translation-generates-whole__no-redundant-site-copy__basis-choice-canonicalized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered primitive unit-cell identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves minimal-cell-carrier, translation-generates-whole, no-redundant-site-copy and basis-choice-canonicalized.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:0ee1191c7afc84e64e0772d8b89185e5f90e00eae010409b4705ada433fe51f1`; derivation seal `sha256:1d98de21f5d7e99681d01977448efd664603dde6233b1a0a289b6339db5dae3c`; engine receipt `sha256:062606db8c6509ec2a58c31f865d99c78fe46b8a407e9a0bf823b7abbc04e868` at `receipts/engine/model_admitted/SFT-MAT-CRYST-UNIT-CELL-001-062606db8c6509ec.json`; empirical validation `sha256:0f5045547d470ffff7cff7a76b331bcc87321b131e410d01dfedc202fef8b7fd`; measurement receipt `sha256:370d8291daf74de34741f463bb14759e3ae77c4e70542c12bb03fce0e4694a0c`; isolation `sha256:b54849491c03c4b7826fcc79b350bbff9e6969437a8a7f6ebd296f1e84f0de55`; custody `sha256:6b4d5550b99b1ece2327e422d03af3ab252fd537e9bcd3b86868b07fdce0cfb7`.

10. Crystalline translational order

Claim identity: `SFT-MAT-CRYST-TRANSLATION-001`

Question and exact theorem. Crystalline translational order is invariance of the complete local constituent word and adjacency class under a generated integer translation basis.

```
complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-periodic-support, integer-translation-action, local-word-and-adjacency-invariant and observation-depth-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-periodic-support	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	integer-translation-action	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	local-word-and-adjacency-invariant	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	observation-depth-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2dc5baa4e9c0b21b287632d0b7879ad79b6a20ac0f10e9043a7a9fc7c91eeaec`.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:699c736da5c750925561e80681e07c690f72b1d453701d89b034b39beec00b55.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:c59d80682ced4dc231ab5a5e2eee49f48e53e6d78753baad047450f99ab1a36b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f7bde3998b13bbb464ee34cba1ecf270b66e6a5fe4a30c001e0b4b0784a4e4b7.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:cf1267c3e93cd30aadb138996e6f53303d50d7bb730c304a24acfd22e2b4d913; certificate sha256:31470cfc67cc99e83e12e9f967814d55ed8aeb6725c114e1be33cd919c8ee3a9; external-validation hash sha256:0232399b0b20900fae5f945a17c32ff14a146310471f3d8e1013d98dc9fdd112.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:9fc8e3d73650812afd6ca2ff83c2faaf5fedd96177422de98734a64c4236f2aa. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html; sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- sft-mat-cryst-translation-001-authority-correspondence: predicted complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-periodic-support__integer-translation-action__local-word-and-adjacency-invariant__observation-depth-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered crystalline translational order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-periodic-support, integer-translation-action, local-word-and-adjacency-invariant and observation-depth-declared.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:82c21d5ad24be00ddd2907d027deee7e0332eef8d95834009a8623769d1e6667; derivation seal `sha256:ed8de31a2f6de9736c9f8dde2f1add2bd74b1cd305d41a3f2252e2f2b3e3f759`; engine receipt `sha256:bf36a3534aab9194516ce002531273c8bd47093f8db5304d723bcb5179c8af3e` at `receipts/engine/model_admitted/SFT-MAT-CRYST-TRANSLATION-001-bf36a3534aab9194.json`; empirical validation `sha256:d5820432483ce1f6db98ac5243feb7d9739305010874fb8312bedf5b1efa5fb2`; measurement receipt `sha256:d863a33e6f7b1f9df5b48c1ca7be3000834fe51921330c7db33edd8a06780fed`; isolation `sha256:4a2fcc8f791a7da72410ba1c5ab79d7ea3b06949abc307bcb34e0032e28236e3`; custody `sha256:31b05b66c269dfbe59d8109ccf02e598b00d239c8876be66387f24d6d6765ad0`.

11. Simple-cubic nearest-neighbour coordination

Claim identity: `SFT-MAT-CRYST-CUBIC-COORDINATION-001`

Question and exact theorem. Three stable spatial generators with two held orientations per generator force six nearest neighbours for the simple-cubic adjacency class.

```
three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__
coordination-count-six__complete-state-transition-boundary-trace__root-bound-for
ward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves three-axis-site-carrier, two-orientations-per-axis, nearest-adjacency-complete and coordination-count-six.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`three-axis-site-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`two-orientations-per-axis`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`nearest-adjacency-complete`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`coordination-count-six`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__coordination-count-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.
- `coordination-six`: three directions times two held orientations generate six distinct neighbours; passed `true`.

Detailed mathematical and physical meaning. The exact product is three independently admitted spatial generators times the two Fold fibre orientations. The six generated neighbours are: axis-one/first-fibre, axis-one/second-fibre, axis-two/first-fibre, axis-two/second-fibre, axis-three/first-fibre, axis-three/second-fibre. No coordinate subtraction is used; opposed directions are held labels.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a557cd3bae435643b540f71a220d86dead2a0af33a4b9db33127a736db1746d4`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ffdd89973296587921dcf25a21228bea827ef70e3557d3b263e39c5b3c5ecf3d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8836eb5156029bb0d0d5e87199dbd03ea5ca0cc32842231958a80aca8dd63da1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:559485317a0696c17c8bee2f0a302354a5738b655be68072aa8f84e6516ff1ea`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:20be4530b1fbc3b79cce20b0c8a94ff0794a8ac42dc8d633c2228dd92dbe0d82`; certificate `sha256:7705717fd0f2eb7831c6a4f828389f697c3b1c0a157da3d9d59907a04de53b48`; external-validation hash `sha256:882c896ef2f8a38ebf582d61bf71dd229923e81adfc46d9474398cb812a2caa3`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:fa547f7fa1727d64e737ac3f8ebc4a041bd00f93f18621aebd87ac98e47da155`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SIX-NEIGHBOUR-MESH-2026-07-24` - National Institute of Standards and Technology; https://math.nist.gov/oommf/doc/userguide20a0/userguide/Standard_Oxs_Ext_Child_Clas.html; snapshot `experiments/external_sources/materials/snapshots/nist-six-neighbour-mesh.html`; `sha256:505ca000d91c5ef134836cbc75307df1584750dfbfb2faf5c4e55756d054f5c2`; scope: six nearest neighbours as forward/backward positions on three coordinate axes.

Comparison and custody records:

- sft-mat-cryst-cubic-coordination-001-authority-correspondence: predicted three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__coordination-count-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived three-axis-site-carrier__two-orientations-per-axis__nearest-adjacency-complete__coordination-count-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered simple-cubic nearest-neighbour coordination observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves three-axis-site-carrier, two-orientations-per-axis, nearest-adjacency-complete and coordination-count-six.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:d9f52d70cc6c34ca39fc40035d128e1c8aa5334910900b03aab1998a85286082`; derivation seal sha256:e42124c8f179997887b6b5e7ab012c111153998f111f7ba98c1ec85a2d82d83b; engine receipt sha256:b63b31c5bf7aea6cde9b7495c3aaed6f34fac4baf513c2824a18d4c42102155a at receipts/engine/model_admitted/SFT-MAT-CRYST-CUBIC-COORDINATION-001-b63b31c5bf7aea6c.json; empirical validation sha256:8a82df98e9284baf366a3492530356dfb56430eafa1cfd56fb362f46856df7e0; measurement receipt sha256:be7fafca4dbef1c845769908689a1b966df05d448dd42a7b6f91a704d1258123; isolation sha256:1b7a43467ab4c4a739ade2cb74f2bb3d1aeda8ad1fbb25ed3adfe8a9cd5c1d10; custody sha256:5608949988ae3687ef60d5f1df5ed7441439153ecac0d84f2fb6d20b19b26046.

12. Crystallographic rotation restriction

Claim identity: SFT-MAT-CRYST-ROTATION-RESTRICTION-001

Question and exact theorem. A finite rotation preserves a periodic lattice only for generated orders 1, 2, 3, 4 and 6; five is the least excluded positive order.

```
integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-STRUCT-GENERATOR-THREE-001
- SFT-PHYS-SPACE-DIMENSION-THREE-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-WAVE-INTERFERENCE-001
- SFT-PHYS-WAVE-DIFFRACTION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves integer-lattice-action-carrier, finite-order-rotation-relation, translation-compatibility-retained and orders-one-two-three-four-six.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`integer-lattice-action-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`finite-order-rotation-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`translation-compatibility-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`orders-one-two-three-four-six`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.
- `rotation-orders`: the depth-independent factor certificate retains exactly orders 1, 2, 3, 4 and 6; passed `true`.

Detailed mathematical and physical meaning. The rank-two integral rotation boundary requires totient degree at most two. If an odd prime p divides the order, p-1 divides the degree, so only odd prime three survives. Every order is therefore 2^a times 3^b with a<=2, b<=1, not(a=2 and b=1). Exhaustion leaves (1, 2, 3, 4, 6); five is the least excluded positive order. This is a depth-independent factor certificate, not a depth-seven lookup and not a trigonometric approximation.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:8e118cd050ba1f91713c4bac85af97b055953558ef2a08ad26c17387a3abde50.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:903699ea736f5b69ac09a10ca848e32b3b7bf4ee50f11d913c7d630147c8f3a7.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a9244da0719ec18d4622cf446c89292a3c2267c7bb08fa457861ad57cf3f2c31.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:f6f523f473b879608080940789f14fa8b6f533bc3ca361aafce15dfffb28763b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:0a7974f67520983c1fdd51fccf1df5b8e5802ebc0562dd9a4e97fc4778af4b0f; certificate sha256:ce3cc18e06f56be003ec62dea474e6214221da3577fe714dc39437d1d192a4e3; external-validation hash sha256:2231c2f7810cf10930fb666398885e031d743432e234512ad4452968d2844c38.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:5bbabc66fa0c30a2122ad479107cf8fdbb52650090cff138262db47417d2bda1. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SHECHTMAN-QUASICRYSTALS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/nist-and-nobel/dan-shechtman; snapshot experiments/external_sources/materials/snapshots/nist-shechtman-quasicrystals.html; sha256:911c06e466a95d58749422c56cd4e33fa1fbece04ec06977af428d7dac23ae0e; scope: periodic rotation orders and fivefold quasicrystal observation.

Comparison and custody records:

- sft-mat-cryst-rotation-restriction-001-authority-correspondence: predicted integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived integer-lattice-action-carrier__finite-order-rotation-relation__translation-compatibility-retained__orders-one-two-three-four-six__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered crystallographic rotation restriction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves integer-lattice-action-carrier, finite-order-rotation-relation, translation-compatibility-retained and orders-one-two-three-four-six.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:14f138cda0fee15f74ec859ff51be2b51c67c810be00b72df700c3d8a14211ca`; derivation seal `sha256:4731ceaa58a721703393d44192a3a9c00d6b8a04efad601d81c99abff6efaab7`; engine receipt `sha256:8fe85fc4bd955efdb5fcc37b159fec38f73a027159ead37660749a04313cd465` at `receipts/engine/model_admitted/SFT-MAT-CRYST-ROTATION-RESTRICTION-001-8fe85fc4bd955efd.json`; empirical validation `sha256:e93f338ce62e1bdbbba29721ad38fa839cd63180536515cf90c50df1a4e6b463`; measurement receipt `sha256:2ff74beb5fbbcbd98e4efb667c9107057fe01c33a592299c0ed2a9b5e4604d9d`; isolation `sha256:e6db0b5eb3ece993947ca064c20b18abdb90df5cc6c021dc2e2281cc38d72220`; custody `sha256:290d1315e496177b5646f2fbc830723be3df434fc37fc87de423ee348b62e35f`.

13. Three-dimensional crystal-system classification

Claim identity: `SFT-MAT-CRYST-SYSTEMS-001`

Question and exact theorem. Complete rank-three metric and crystallographic-rotation equivalence partitions periodic crystals into seven crystal-system classes.

```
rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-metric-carrier, axis-length-angle-equivalence, rotation-compatible-classes and seven-system-partition.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`rank-three-metric-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`axis-length-angle-equivalence`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`rotation-compatible-classes`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`seven-system-partition`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.
- `seven-systems`: the complete rank-three length/angle quotient has seven survivors; passed `true`.

Detailed mathematical and physical meaning. The complete rank-three metric grammar is the product of three length-equivalence classes and four angle-equivalence classes. Quotienting by axis relabelling and rotation compatibility retains seven forms: cubic (all-equal; all-right), trigonal (all-equal; all-equal-nonright), tetragonal (two-equal; all-right), hexagonal (two-equal; two-right), orthorhombic (all-distinct; all-right), monoclinic (all-distinct; two-right), triclinic (all-distinct; all-distinct).

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d6d53473525caba97aac0eef120adbe564356c71e4097f541c0d987a977f95f7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d5ecbdab14d6f4451eabd1972a86974801daa1b3f29fbc0d095e536484148aa9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:89c2d1f3852452f2909a9e5f7bd2121e7df76e5501c07b86b1cb3253df66abf7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ef648732ef475e15f7f44a87514589d7ad95c001bf0beb0fb102ffd1422f6f1e`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:d5514861f898ded8174cc4056d7825fd0e5b938747e48538eaebd3cc84f8f5b8`; certificate `sha256:0e7973bb51c318e20750b3f30a48728851edd1fc9f4b8a4303aae2f3d8841560`; external-validation hash `sha256:e71174f60ede1e2827735dbd37e20c4ab1938398f2be76698070fa1426b66624`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:d7a99045f74e35aa49659bcc6ecb8293ac686764c1a2e6a3f153593a1df6f002`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24` - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot `experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html`; `sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6`; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.

Comparison and custody records:

- sft-mat-cryst-systems-001-authority-correspondence: predicted rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived rank-three-metric-carrier__axis-length-angle-equivalence__rotation-compatible-classes__seven-system-partition__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered three-dimensional crystal-system classification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-metric-carrier, axis-length-angle-equivalence, rotation-compatible-classes and seven-system-partition.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:e2966380d9a9fda51fc07e0b73d5afee4d49e91d5a0af0869a4cb930524697b9`; derivation seal `sha256:05c0736603eda6753dcaed3534e5a3f3e54e3ae4d3a62489c4fcaa0dcf19f335`; engine receipt `sha256:e09270361d9fcad5e3e2e58018dd2c46dace3a54af702d711d8b22b0e34919dd` at `receipts/engine/model_admitted/SFT-MAT-CRYST-SYSTEMS-001-e09270361d9fcad5.json`; empirical validation `sha256:c4b816b9a81da48b1bc8d1a3cc70e41847ad2f22c86313a5106036f69a7322e7`; measurement receipt `sha256:bd488ae3983b3fa598b381a978e307b19aa3263489e578414e7e71d91ec6a9b9`; isolation `sha256:a068a2cff6c89dff9921d705b6a08c6eac4f097a5ba44a238ab72455d3a278be`; custody `sha256:27e2d5486e52bf5f47c1ae8bcad33818e99eea50d8b1efd4791f031a0c6e82c4`.

14. Three-dimensional Bravais classification

Claim identity: `SFT-MAT-CRYST-BRAVAIS-001`

Question and exact theorem. Complete rank-three translation lattices modulo basis and centering equivalence form fourteen Bravais classes.

```
rank-three-translation-carrier__centering-equivalence-relation__primitive-lattic
e-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__r
oot-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-translation-carrier, centering-equivalence-relation, primitive-lattice-classes and fourteen-bravais-classes.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`rank-three-translation-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`centering-equivalence-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`primitive-lattice-classes`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`fourteen-bravais-classes`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `rank-three-translation-carrier__centering-equivalence-relation__primitive-lattice-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.
- `fourteen-bravais`: the complete system/centering quotient has fourteen survivors; passed `true`.

Detailed mathematical and physical meaning. The system/centering grammar contains seven system classes times five centering classes, or thirty-five candidates. Exact basis/centering compatibility retains fourteen: cubic/primitive, cubic/body, cubic/face, trigonal/rhombohedral, tetragonal/primitive, tetragonal/body, hexagonal/primitive, orthorhombic/primitive, orthorhombic/base, orthorhombic/body, orthorhombic/face, monoclinic/primitive, monoclinic/base, triclinic/primitive. The paper claims this quotient only inside the registered rank-three translation and centering grammar.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:78101747be3a1f60452a2a41fef4bbceacc6e5337dcbc7573bf342ffcd82ed15`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1ae5e1a2f713f1df2305360821d929b44dc0ff0a5d58e620083b46859e5a6c77`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a636fbebbfe7ea076b33edcc43949081d4b538dfbf0e6a76d84730ca300815c6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e348fb47b5b11603b3bef45c7ed6a17b63d22d9d6ef59d63964b9ab0dbd010fb`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:5255faf88e98383237d2363fc087370a4d87195e1b2a7960052221bcf3b45d69`; certificate `sha256:2ca709648229e6c57487ee5fa522f233b1084b25fdce360491a2b8656d3f7032`; external-validation hash `sha256:0247320bda7feded0c92b38377a9b25f7c6959cf48051ef443a41a804b8a6237`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `4` claim-specific source discriminators were required; their ordered identity is `sha256:6232a8f37319ed3012b7aa4f9826ecf52c3b38617950fa4facfc4637cfadd968`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24` - National Institute of Standards and Technology; https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot `experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html`; `sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6`; scope: lattice generation, rotation, seven crystal systems and fourteen Bravais classes.
- `NIST-BRAVAIS-CLASSIFICATION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification; snapshot `experiments/external_sources/materials/snapshots/nist-bravais-classification.html`; `sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91`; scope: diffraction classification across all fourteen Bravais lattices.

Comparison and custody records:

- sft-mat-cryst-bravais-001-authority-correspondence: predicted rank-three-translation-carrier__centering-equivalence-relation__primitive-lattice-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived rank-three-translation-carrier__centering-equivalence-relation__primitive-lattice-classes__fourteen-bravais-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered three-dimensional bravais classification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar

- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-translation-carrier, centering-equivalence-relation, primitive-lattice-classes and fourteen-bravais-classes.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:1ba2a2e7694b4e6f545b596b6a15b62501b2a48434945e1c12cd508bc407a61e`; derivation seal `sha256:da91f3783dc2f1e6d6e61900a6e577dc770edb550cbf5e3eef3b8144d95a5a50`; engine receipt `sha256:402977c8ad6256e131a459efdf0762f6f572b919f6a35e60619b4acaf66ad684` at `receipts/engine/model_admitted/SFT-MAT-CRYST-BRAVAIS-001-402977c8ad6256e1.json`; empirical validation `sha256:333e6b1b653c99e953f630f7eb8804adce7dbdd7d3cf7d3f79293700b0f60d16`; measurement receipt `sha256:003021a8d559eb1b26af67c2f11c9ed26cf12f49becff5a2006773efb7a33977`; isolation `sha256:527f1213f3d7c8a54c383d938c9a3d8232e0db8136f64e8aed5995f316676a4a`; custody `sha256:567f6142968e0c21f7265d5e24402a84854192f9c201dae1a7319410381357bb`.

15. Reciprocal support and diffraction

Claim identity: `SFT-MAT-CRYST-RECIPROCAL-001`

Question and exact theorem. Reciprocal support is the exact class of probe path labels whose phase returns coherently over the complete generated material organization; retained classes form diffraction peaks.

```
complete-scatterer-carrier__path-phase-return-relation__translation-dual-support
__peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-
forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-scatterer-carrier, path-phase-return-relation, translation-dual-support and peak-resolution-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-scatterer-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`path-phase-return-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`translation-dual-support`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`peak-resolution-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-scatterer-carrier__path-phase-return-relation__translation-dual-support__peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:dbbd7f07ef61e5278af026e0de60234376a22109306b91f4d1bf17bf3fc92db7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c3cecb0b4bea526fd329b1d5f1ddb4790afef854b36f7c80a3b54d70d679c209`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6e2d17ef89a20df4e99c1242813468012f08aabc1d0f82b92ef260085e73c59b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c03422bc2ec42444b078c5e063341eaaf36057cc69ec87b869a1fba8c8c0f016`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:7fde687d50176b337eb1d74f974b35828f044bea87496e0145f9ed715e7a66c7`; certificate `sha256:b79a9101b1cd1d53c7602769ae73503014124fe572a19913e5e2d32930989b98`; external-validation hash `sha256:d01489f20f1663afdd82a9d4e83200885117dfa5c34feeaa4a85ea9f15449730`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:1f65c1f62878d5c4562fa2d3b4250a9d4266167ee478f1b4b1631a12b10e9daf`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-BRAVAIS-CLASSIFICATION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification; snapshot `experiments/external_sources/materials/snapshots/nist-bravais-classification.html`; `sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91`; scope: diffraction classification across all fourteen Bravais lattices.

Comparison and custody records:

- sft-mat-cryst-reciprocal-001-authority-correspondence: predicted complete-scatterer-carrier__path-phase-return-relation__translation-dual-support__peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-scatterer-carrier__path-phase-return-relation__translation-dual-support__peak-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered reciprocal support and diffraction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-scatterer-carrier, path-phase-return-relation, translation-dual-support and peak-resolution-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:7fe0ba8c722df50c77b1ae34c1574c76db1044d3115cd0de6c04b557de63cfac`; derivation seal `sha256:2cf710a5f2a9502019e2d3816d744823c545bd807f7de28a0f80368e5d6d9856`; engine receipt `sha256:bb69b8443063baf90849c3e3fbb54fe746bf27a39cbb5892e3b7988d38e35d78` at `receipts/engine/model_admitted/SFT-MAT-CRYST-RECIPROCAL-001-bb69b8443063baf9.json`; empirical validation `sha256:71cca974d8629c82315c237652911b0b5577f25513e2fa24cdfec188ef578ea0`; measurement receipt `sha256:b4975788f2336635d491d04088bc4f366ecdc98223a688a48adabce8c40f16f8`; isolation `sha256:7ce6b243c09044f0a801fd8ae4c47c89f24bec45f413bb579ba34fb0b8e017e3`; custody `sha256:2f6fc08ab9a531f9412ae510e8bf4f74b19507eb4c3f421d074c2ff4a0a5b0e8`.

16. Quasicrystalline aperiodic order

Claim identity: `SFT-MAT-CRYST-QUASICRYSTAL-001`

Question and exact theorem. Quasicrystalline order is long-range coherent recurrence with sharp reciprocal support and no finite translation period; fivefold order is thereby admitted only outside periodic-lattice closure.

```
complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden
-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transiti
on-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure
__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-aperiodic-carrier, long-range-recurrence-without-translation, forbidden-periodic-order-retained and sharp-nonperiodic-diffraction.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-aperiodic-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`long-range-recurrence-without-translation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`forbidden-periodic-order-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`sharp-nonperiodic-diffraction`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f482fe7557c1f3d9096abc4c5b6ab86ff984ddeb47b893b09a2df71291ecca34`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:dce768e3d1858fac538cdf97980ba03c7c7a351a5d3243eaeb790fb0d6fe1b82`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b018cbd8640c5d2740f37cd8fb376d36267f8057724af2a3ad80b4bf6ef62ec7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e752ea45c00eb03832377dbbdd9c0bae8c0df9aa99f886ff58cb8cc499898244`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:8713bd0cc130a811d5e81c9332eb037108cbdcecb35a4ea3afd962489cb870f0`; certificate `sha256:6315b1810b5134e92f29441df2c6d737beebb9e3277bcb4032a37e8d06f5c938`; external-validation hash `sha256:25e7cbc99190b490ca1de85a180d406e5f69d1060cc67b8232a4b4d1cdfc488f`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:c5eb79da6aa59b45f7955fda6e77963fcbcc4468da12df970f9ce6c9df522f7c`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal; snapshot `experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html`; `sha256:a1b7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18`; scope: microscope confirmation, nonrepeating order and fivefold symmetry.

Comparison and custody records:

- sft-mat-cryst-quasicrystal-001-authority-correspondence: predicted complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-aperiodic-carrier__long-range-recurrence-without-translation__forbidden-periodic-order-retained__sharp-nonperiodic-diffraction__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered quasicrystalline aperiodic order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves complete-aperiodic-carrier, long-range-recurrence-without-translation, forbidden-periodic-order-retained and sharp-nonperiodic-diffraction.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:7b1915442585a900282440427561c6f7b4df02ce32449a92d2ea47ff898edc5d`; derivation seal `sha256:6a4386e007b87fab581b9006924276970082bced9cd7259a4384243c894a9a07`; engine receipt `sha256:77617aba1e766abd8b64273551afa2ca856c9f67a68f869d86448ef1c5266e21` at `receipts/engine/model_admitted/SFT-MAT-CRYST-QUASICRYSTAL-001-77617aba1e766abd.json`; empirical validation `sha256:977cb5c6105037ae75d97e2312a249489f97a3fa3fe524912be1b7a5fc79bb29`; measurement receipt `sha256:0ad97e77ff4a78474e22cdfc34fa63f25835bd04395e2ce541ef8d2e2fec0bf9`; isolation `sha256:cf3c21b892268b1d10e85c3b0d1f1f505778ea51b5f4939b71f2288730876745`; custody `sha256:d89c32434f6f04e88eb371e35d05e15a90f83cf32e23b31d9f17ec49afb58967`.

17. Acoustic lattice excitation branches

Claim identity: `SFT-MAT-CRYST-PHONON-001`

Question and exact theorem. A rank-three lattice has exactly one longitudinal and two transverse displacement orientations, forcing three acoustic excitation branches.

```
rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-exc
itation-recurrence__three-acoustic-branches__complete-state-transition-boundary-
trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-r
ule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-WAVE-INTERFERENCE-001`
- `SFT-PHYS-WAVE-DIFFRACTION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-displacement-carrier, one-longitudinal-two-transverse, collective-excitation-recurrence and three-acoustic-branches.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`rank-three-displacement-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`one-longitudinal-two-transverse`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`collective-excitation-recurrence`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`three-acoustic-branches`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-excitation-recurrence__three-acoustic-branches__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.
- `three-acoustic-branches`: one longitudinal plus two transverse rank-three orientations are retained; passed `true`.

Detailed mathematical and physical meaning. A propagation direction consumes one rank-three orientation. The remaining two independent cross-axis orientations are transverse. The complete census is: longitudinal/propagation-axis, transverse/first-cross-axis, transverse/second-cross-axis, yielding three acoustic branches without an irrational normalization or continuous-mode premise.

Periodic and aperiodic order are generated from exact site, adjacency, translation, rotation, reciprocal and displacement support rather than assumed from a crystallographic table.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d2e24b132e0f5c972671ff849c506788f98e5f442ca5e40467459c911b8e4636`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:274bdf879f97e05579ec9587adb448fbfa54aaae500b5c950c6b1702e2192c84`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:82599c5b8d2e6423bcd6b667f6a82fc0e740bed1f5baf566351026c7353152f6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1a63a5afef7b1226f407ae481433648a075f0dfe892a946fd213606d40485070`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:32104da4daf1fd380faf112d1ca088d4bffc61a352e61ffa4c246635b881e4d3`; certificate `sha256:e712e3cecd112890f70c47e00d2985ebb8941fb0a3933a188b1522998c457edc`; external-validation hash `sha256:e6f329a8b9200d04c64b33e216bae238ce8bb9fcd662f88aea11ae209aa5a186`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `4` claim-specific source discriminators were required; their ordered identity is `sha256:6913f74d3c5f5fbb3d58697fbb956c68bb97d0cd5240834d73dd97e6109226ff`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TRANSPORT-THERMOELECTRIC-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials; snapshot `experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html`; `sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0`; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-cryst-phonon-001-authority-correspondence: predicted rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-excitation-recurrence__three-acoustic-branches__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived rank-three-displacement-carrier__one-longitudinal-two-transverse__collective-excitation-recurrence__three-acoustic-branches__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered acoustic lattice excitation branches observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered crystal_quasicrystal grammar that preserves rank-three-displacement-carrier, one-longitudinal-two-transverse, collective-excitation-recurrence and three-acoustic-branches.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:d76418e36f165132c2d2757608ccfd33cbf2caabbf7849321efeb8a6ceec504c`; derivation seal `sha256:91b3320ea3273de7f8f7250145371464466d13e21357218c3c7688707b79a25a`; engine receipt `sha256:8c3472817c5da55f7509b6085db046570a7109ea880dd7e38a086f72c07b216c` at `receipts/engine/model_admitted/SFT-MAT-CRYST-PHONON-001-8c3472817c5da55f.json`; empirical validation `sha256:79983d2c6c9ccc400c7d5b9a724ca15387801673a1881287815dbdefe12706ca`; measurement receipt `sha256:ae8c2aac5b764cf95d5996bbecc43548237405912e7bca76b77a1f77161bb877`; isolation `sha256:088998a56bf790b485497a4f0d902cf7c2cb52cba726aac1e09c34716acfbfe5`; custody `sha256:dac4e779449deddb57373829413ebff00af16ad5936d9cf9b3f2b72ec1a778d3`.

13. Defects Microstructure

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

18. Point-defect identity

Claim identity: `SFT-MAT-DEFECT-POINT-001`

Question and exact theorem. A point defect is the least localized retained difference between an actual site record and its declared reference-lattice word.

```
site-and-occupant-carrier__local-reference-difference__defect-provenance-retaine
d__resolution-localized__complete-state-transition-boundary-trace__root-bound-fo
rward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves site-and-occupant-carrier, local-reference-difference, defect-provenance-retained and resolution-localized.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	site-and-occupant-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	local-reference-difference	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	defect-provenance-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	resolution-localized	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organization are separately retained; passed true.

- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d5cca37e5943297a8a5b2d3d85d2a054989f1a133065f6d35ed160baaa65cfd9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f68f961248b9bc315028a34f7cabd89e0b1f40a716840e7fae8f3a0bf451eac3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3f18a95e86b4d63ffc8ae28d1018e1b039db7a8de3cd6e79b9e1cfe87a40d635`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7dd783b6de1acf2c3cf3ee8c73e69fd6b520ec7fb967d22b238900a5d0b23b8c`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:3059cc726018f803ac57d5d892c51d517508bfa3be460c81d502fdc7d2c68244`; certificate `sha256:c759d08df8a75a473cd5e9dec1433b27cd200bb7d4cca6abb72cabb0f4d37eae`; external-validation hash `sha256:3c2b96d2f15fc90e4531999b446db3da556e38c8b51ef1660d68d2855aef2cf3`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:7988e4e687fecb304714c434cb7545ce9d70fa8f823acf23c085c562186f7272`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POINT-DEFECTS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides; snapshot `experiments/external_sources/materials/snapshots/nist-point-defects.html`; `sha256:b885250bd645ed8eaefa55a29d99ea4436b691f77c17c74a457acb6acf88335a`; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-point-001-authority-correspondence: predicted site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived site-and-occupant-carrier__local-reference-difference__defect-provenance-retained__resolution-localized__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered point-defect identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves site-and-occupant-carrier, local-reference-difference, defect-provenance-retained and resolution-localized.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:ce5dba442d917fdfaad1110e0264bcecad74c6a4c9be9ef58ada814cbe2d7cf1`; derivation seal `sha256:25283d53bed00346043921382e190f5b91210459c356fbe8bffeea89ea2aa456`; engine receipt `sha256:b1b3d3707d89ec075a1c23ad71477dc4af6341260e33d7e712aea0d4c35d4b22` at `receipts/engine/model_admitted/SFT-MAT-DEFECT-POINT-001-b1b3d3707d89ec07.json`; empirical validation `sha256:a94db58787562293e8bb51a26af1d3f6966955e3d91a4255ca338c8c4eb7daea`; measurement receipt `sha256:38aecf7fa7d9859f15d89dc3a4baf8b8d0b60043dcc2052a2a76b90835f55a43`; isolation `sha256:a019338b83f6ca0ef9f51f77476ad7d44a4673ef585264cc7173c3bddcde9759`; custody `sha256:d2acebdeb9581146a364ab315d10012ef51ac622920719654b9919208b89890e`.

19. Vacancy as structural absence

Claim identity: `SFT-MAT-DEFECT-VACANCY-001`

Question and exact theorem. A vacancy is a retained reference site carrying structural occupant absence, never deletion of the site or a numerical-zero material amount.

```
reference-site-carrier__occupant-absence-held-label__site-identity-preserved__em
pty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound
-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves reference-site-carrier, occupant-absence-held-label, site-identity-preserved and empty-one-not-numerical-zero.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`reference-site-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`occupant-absence-held-label`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`site-identity-preserved`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`empty-one-not-numerical-zero`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `reference-site-carrier__occupant-absence-held-label__site-identity-preserved__empty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:34bcd6eefbd08e1ae00fee7c919cbaa8d558a45544826b10a57c2c7bd698f9de`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5a8c40e74cfe96c496d6ff7e40fa33f620678a3a3a5afa2f3ba76056e2d6c7e7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0e53cc842ec91c069e757ab5bb956cb974cf58b4cbfe3f5b671dbb3b85de8551`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2a6374a8e168c792adec3263140527f62b65afebcd1c2a85a14fe573562262ee`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:505a9fdcb929e4fe4912e574520857cc16b9843a3599ba905f951ef28e415e1a`; certificate `sha256:072c8a2d131081cd814ed1c6d543f5a27b00b28d1743049df66be684c1a19b1e`; external-validation hash `sha256:1dc1c03612c8e602721414875702b896f302df78952ae98f40bee3f3bb53a80f`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:d66f5170de57eaea5c0a8c4e930eb3e728aaeed6a13417345f946147d5fe3d83`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POINT-DEFECTS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides; snapshot `experiments/external_sources/materials/snapshots/nist-point-defects.html`; `sha256:b885250bd645ed8eaefa55a29d99ea4436b691f77c17c74a457acb6acf88335a`; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-vacancy-001-authority-correspondence: predicted reference-site-carrier__occupant-absence-held-label__site-identity-preserved__empty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived reference-site-carrier__occupant-absence-held-label__site-identity-preserved__empty-one-not-numerical-zero__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered vacancy as structural absence observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves reference-site-carrier, occupant-absence-held-label, site-identity-preserved and empty-one-not-numerical-zero.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:8b6b1466e80a30239147f070f141ead32c10c87ce24d155d975af5d548b78089`; derivation seal `sha256:b4b7599659a0361406a02a2850911db2a54c6f046756a9562f280abd80656e6b`; engine receipt `sha256:e380e7e72cf8fb701cc5cf6bd37e711db7fa7cc9a9fef348aebb81fec9d73c53` at `receipts/engine/model_admitted/SFT-MAT-DEFECT-VACANCY-001-e380e7e72cf8fb70.json`; empirical validation `sha256:6b86a2e46c5ebfa7a9310081ee9972a6f67bc2d21d4684747b684a2a26521959`; measurement receipt `sha256:2d2091fe1ae56f8660db3b54d2cb6b8747cfcef0a10d58857c60c5a4c9cf857c`; isolation `sha256:ae1764141e70fda66147266e60303f406510c93e4f2b9099554fb0c5c4c0ad18`; custody `sha256:2a8ecf59f12ea4aa2c6785dc707dd2b10524954079841556 15a5a2ba719e8082`.

20. Interstitial and substitutional defects

Claim identity: `SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001`

Question and exact theorem. Interstitial and substitutional defects are distinguished respectively by an added occupied site relation and a changed occupant label at a retained reference site.

```
complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus
-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-t
race__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-ru
le
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves complete-local-site-carrier, occupant-to-reference-comparison, extra-site-versus-replaced-label and defect-kind-distinguished.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-local-site-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`occupant-to-reference-comparison`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`extra-site-versus-replaced-label`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`defect-kind-distinguished`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

`sha256:f3948b872639a13d3ee135782f30c8aff9a7773c9fdf26149a4866148efaa330`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9adeb2d2247f93e2869d911066bf4021bd6108567408c919a97b0c473380fe15`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ba6464150d8d66f60708c21cf335227879f90b82d93bd8b62b1ef6011aaea5e1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3b95d8ae638b3a67b12120e44708eadb6d42a6a0db6034c93420958988deb289`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:d26a7f8e2a347a232ba359abd7b605a579728ef6581446703faf206ce7b37f9d`; certificate `sha256:b7a03f2805f3eeab439d48fef49cd06f8670e0d7637638a18a4edd759c7b01ee`; external-validation hash `sha256:337ae07bd1c202cf5affc6fa9cf3f4c6045a8d83a55056f625a351ad6fd99929`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:2b552685e6c97371ba6be2557169b0890be1cc01817fc2a39de49338915614c1`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POINT-DEFECTS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides; snapshot `experiments/external_sources/materials/snapshots/nist-point-defects.html`; `sha256:b885250bd645ed8eaefa55a29d99ea4436b691f77c17c74a457acb6acf88335a`; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-defect-interstitial-substitution-001-authority-correspondence: predicted complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-local-site-carrier__occupant-to-reference-comparison__extra-site-versus-replaced-label__defect-kind-distinguished__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered interstitial and substitutional defects observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves complete-local-site-carrier, occupant-to-reference-comparison, extra-site-versus-replaced-label and defect-kind-distinguished.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:b8b0fce4b02e0890ef18ec6beb51e1064210749c16b5dcf939dee868c30a8a4e`; derivation seal `sha256:28dea2afe3f1fdcf79309764e992de951478be45317f96c14af4e0e839b8640a`; engine receipt `sha256:47a87b557fa200d5e2177b408e83e080eb7d358c52c10f61c3817b60b34707c7` at `receipts/engine/model_admitted/SFT-MAT-DEFECT-INTERSTITIAL-SUBSTITUTION-001-47a87b557fa200d5.json`; empirical validation `sha256:4e6ee278b992402fd86a3894cf2d3ce3b6e29c3cb964073d732f8b1e6f66b673`; measurement receipt `sha256:e78d1384af71ffceb827e8a592a7cd061f1534e806e90b511867da52567835fd`; isolation `sha256:5c8ac345a9fea15a0989b063f23ceb749d6a7e46b80c9b45cc28951c28f4f23c`; custody `sha256:8f9a5e072757dc231f8a117bb38e7f9d6c5dc43a4051c82e0019ab109d7a988b`.

21. Dislocation line and slip

Claim identity: `SFT-MAT-DEFECT-DISLOCATION-001`

Question and exact theorem. A dislocation is a line-supported lattice mismatch whose complete circuit retains a nonclosing held translation and supplies a lawful slip path.

```
line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves line-supported-mismatch-carrier, closure-failure-around-loop, burgers-path-record-retained and slip-system-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	line-supported-mismatch-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	closure-failure-around-loop	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	burgers-path-record-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	slip-system-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d59f3ceee9a9ea4c5fedc027dca3afe12a11b3071f83718e717f5ea4d7962b94`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7d81ee24b84c6e40d34cd4b294fb323ec65c1baf32a075bbba85a363860296e0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6f2b30a9d67d8c00448c0baccd1113371c830b432e882418e1938dc936c54ed0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:870afb6bf3c0fa3e16cb9c36ffaf1ffb021c357a0cb1f7df6080efb52a9d7288`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:ad2fbb72e7f7bfff1236de535b9cc4ddf9618d802e99ba9238b109b1e619136c`; certificate `sha256:eba1b94470fbb42e12dfdbb14fb076926ce197f9965de57ce9e47b36f17a0746`; external-validation hash `sha256:039ab49886efc164de273e4dd73105148fbcc04ed8e98851087a40d5474150fe`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:e7f7cc953536991061b1f57ee711cea8d62059945102b7d6622beb403907358a`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature; snapshot `experiments/external_sources/materials/snapshots/nist-superconducting-grain-boundaries.html`; `sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2`; scope: grain boundaries, strain, dislocations, current flow and superconducting response.

Comparison and custody records:

- sft-mat-defect-dislocation-001-authority-correspondence: predicted line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived line-supported-mismatch-carrier__closure-failure-around-loop__burgers-path-record-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered dislocation line and slip observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves line-supported-mismatch-carrier, closure-failure-around-loop, burgers-path-record-retained and slip-system-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:6188c63c7eca703914838536c9b19141566ce78b61fe8a0b217cd8c483e9dac3`; derivation seal `sha256:956b315813b9db004e985d97806bcf3ad06e798b1c442da065d7a1737b696f2f`; engine receipt `sha256:ff273682b7e9bf3e07a0ebe94391f6a7f6046677fe786751d96667017dbdaa50` at `receipts/engine/model_admitted/SFT-MAT-DEFECT-DISLOCATION-001-ff273682b7e9bf3e.json`; empirical validation `sha256:74c2249f8ad197da4a480dfc1c7f5a336bd21868f2b600eb0388eff6b8d0c0d9`; measurement receipt `sha256:9ddba1e33caf86d459b837718673e37a49246d7fed354968f71ce5b33376e7cd`; isolation `sha256:0e4803cccc3ea43aa1cf21b67ac24af49e54f17f8fa33c9d452ccbf535666406`; custody `sha256:d8faf7892749b1143d64114e17b66476b9329ff8a8919918c02542daa4f249ea`.

22. Grain and grain-boundary identity

Claim identity: `SFT-MAT-MICRO-GRAIN-BOUNDARY-001`

Question and exact theorem. A grain is a maximal connected orientation-coherent crystal region; its boundary is the retained adjacency where two such orientation classes meet.

```
orientation-coherent-region__adjacent-orientation-mismatch__interface-network-re
tained__grain-scale-declared__complete-state-transition-boundary-trace__root-bou
nd-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves orientation-coherent-region, adjacent-orientation-mismatch, interface-network-retained and grain-scale-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`orientation-coherent-region`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	adjacent-orientation-mismatch	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	interface-network-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	grain-scale-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c942c9634762db6520d32ff1ac8e7d7a1c63e39d2c641022dcd7138ef3598216`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:220e22dfdd09c0f9d947632d1b79d338308a1b410141a93266443951748ef628`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:dff2ad7f91bf18ad74cad7113fd23d4b572fc232ae3f3dacd4a2505c0501db95.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:4bed4e2d09f13bdd5c2a7afc570796ce789516387035f49315e09e75991a704b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:349f54df0d5774c84cd33aa94c0bd9445971a549f8d5040319e8c1bb49c31cda; certificate sha256:77a0e2b7f0df63949e54011d392bbc08d9bd34adb66f336af40b02a4ae3368e7; external-validation hash sha256:834416d05874d2011eb1ffd3f172e77f75eb414b9ab7d09635a4ac2362a0ea45.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:17e0d614290974fd2dc557705e24b055f6725241f36b930e4e3fed566c413d49. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction; snapshot experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html; sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; scope: grains, phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-micro-grain-boundary-001-authority-correspondence: predicted orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived orientation-coherent-region__adjacent-orientation-mismatch__interface-network-retained__grain-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered grain and grain-boundary identity observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves orientation-coherent-region, adjacent-orientation-mismatch, interface-network-retained and grain-scale-declared.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:98bcef7ff326b2a30a2c04f4526b617cc2e6b5a08adb067693acf16d4d1d1f67; derivation seal sha256:e514b24f767337ca53e38189e80c4ba8a8acb74613b672b8a5fcfdd042ab9144; engine receipt sha256:ac7a66efd3aedc5341c9fbbd7d72a278c5f2e9e3420f3ac6c5d121b9af1505fb at receipts/engine/model_admitted/SFT-MAT-MICRO-GRAIN-BOUNDARY-001-ac7a66efd3aedc53.json; empirical validation sha256:db61a7d6a0f2d4f932107505d4ce4551d79e5139bf4aabb45252e82265b5f1cf;

measurement receipt `sha256:28bae4d3ddb832f965a34d4a5f50d64ca9a7d2818cc04cfe3ecedd3f53bc3c7d`; isolation `sha256:b23111358cbbd16ddffebadea0f98f4d4710932725c2d69776d218c8afa597b0`; custody `sha256:b7738555c09fa628fe8a367ef171ca1c2e41c56a620fbcd9f304df2afc63d428`.

23. Material interface law

Claim identity: `SFT-MAT-MICRO-INTERFACE-001`

Question and exact theorem. A material interface is the complete boundary relation joining distinct material or phase carriers with explicit structural, content and transfer compatibility records.

```
two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves two-material-boundary-carrier, cross-boundary-compatibility, traction-content-and-structure-ledger and interface-scale-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`two-material-boundary-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`cross-boundary-compatibility`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	traction-content-and-structure-ledger	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	interface-scale-declared	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c1e4dfd258e36706fa84e3cd75aaca7071913c560ebf12f38ff39bbcdf3d222e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:38e2f1f70bfd4f0bc36468142b4370421240bc7e98c6c54d8d0436482af344e6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c1db60d562c2617ef8de5671afd5d54569ef6985af0e15a0014cc202d4747fca`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3439e9c2f088c1bb6c5ce841494806e4d057028abcf2413c50f6c51c1a129bdf`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:057c33aa17088d597f45fd732daa9b8051863699e686069cfa4abb4d15629506`; certificate `sha256:644657ee1c81408bd1464213a5340603387c8ba06f5db9339efcd369df0fdbf0`; external-validation hash `sha256:f722c72b28540d303f2c1be3ed5525483c8f7771260a1132e5fc02a2517180fb`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is `sha256:9f783a523f300809cccd2615a4c90f84a593fd2fbdb5267829fe4312999299b5`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POLYMER-INTERFACE-CONSORTIUM-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/el/mssd/polymer-surfaceinterface-consortium; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-interface-consortium.html`; `sha256:2d48772b1b901e657fa86e4ee28fad15e176dd675911ce8322b54cc718c32635`; scope: surface damage, interfacial adhesion and interface-property measurement.

Comparison and custody records:

- sft-mat-micro-interface-001-authority-correspondence: predicted two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived two-material-boundary-carrier__cross-boundary-compatibility__traction-content-and-structure-ledger__interface-scale-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material interface law observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves two-material-boundary-carrier, cross-boundary-compatibility, traction-content-and-structure-ledger and interface-scale-declared.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:a20fe778e5b1b9c80a7b99792163778ffe132c5828b30a105a4f8dcf48e297e1`; derivation seal `sha256:52261a548994417867de3243a46b64609a8e73134b022c544173fcc2d5b3f129`; engine receipt `sha256:a75d45c8a0bd91e7f206b23cc49ca7c97649ccc12672945d9699ea6e1c8eb1bd` at `receipts/engine/model_admitted/SFT-MAT-MICRO-INTERFACE-001-a75d45c8a0bd91e7.json`; empirical validation `sha256:d7d15758d7c7347d521260dd8c03fdd20873caee2dccd34297ab1ee1994a911b`; measurement receipt `sha256:2789b2f3f5f5d2da7dabda5141971f793ac0c3aa17426e32a28f0a1dd66fe03c`; isolation `sha256:62c4e87fefaa3b164744c4ab25db1267d0676a9fa2507c9733253ee4d2c723e7`; custody sha256:ac3e49780e5c320a0c9b97e62913dbc25e34efa901a369b2f41be3150e774490.

24. Diffusive constituent transport

Claim identity: `SFT-MAT-MICRO-DIFFUSION-001`

Question and exact theorem. Diffusion is counted adjacent-site redistribution of retained constituent labels with complete global content conservation and declared support/time recurrence.

```
labelled-constituent-carrier__adjacent-site-transition-count__global-content-con
served__time-and-support-bounded__complete-state-transition-boundary-trace__root
-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves labelled-constituent-carrier, adjacent-site-transition-count, global-content-conserved and time-and-support-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`labelled-constituent-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`adjacent-site-transition-count`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`global-content-conserved`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	time-and-support-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `labelled-constituent-carrier__adjacent-site-transition-count__global-content-conserved__time-and-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:1d11be3a1ad882f5cd29b311457e62302459d6833393f74d4e942187679fae66`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2ed5233a8868de6e7a8153374b3de47540029249169de8412bcef7af92ebfb5d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0a97789d5adcb49aef646568c5f015237315bb8fa83088af4563c2e6bd4bc54b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:48a8135d5b012c395466cb8e4986b7799d63e98542cd8ca4fa9d311f782ba773`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:9c59f387133a29636589558b2530ee2118991e6afe52a24776db39acffb14f43`; certificate `sha256:5e1381704dc4bb016cca880cdf834a2838db3c214f1ecbd148234faf2c29a052`; external-validation hash `sha256:64cf3e5bd60590e55b421e0fbf9f707a7adb5412ef80884d4445ab68dfd899a6`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:06c38d5113645138ec19c40e94ce2a9cdb3ace8703038b4f73d9197bfbf084c5`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POINT-DEFECTS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides; snapshot `experiments/external_sources/materials/snapshots/nist-point-defects.html`; `sha256:b885250bd645ed8eaefa55a29d99ea4436b691f77c17c74a457acb6acf88335a`; scope: vacancies, interstitials, substitutions, diffusion, composition and defect metrology.

Comparison and custody records:

- sft-mat-micro-diffusion-001-authority-correspondence: predicted labelled-constituent-carrier__adjacent-site-transition-count__global-content-conserved__time-and-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived labelled-constituent-carrier__adjacent-site-transition-count__global-content-conserved__time-and-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered diffusive constituent transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves labelled-constituent-carrier, adjacent-site-transition-count, global-content-conserved and time-and-support-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:40f7cd3e3ea9de556a959e4850c76b1d2b8cb4801a32bced9ebb56d720beeb74`; derivation seal `sha256:fe6aabdae82bc2c12a9539bf9af505c44f86e4fc12a9f8e911c8aea9cf8c2469`; engine receipt `sha256:3e6618450ee819a245f475c45283d761896e350e8c0e1462eb2dabc10c400881` at `receipts/engine/model_admitted/SFT-MAT-MICRO-DIFFUSION-001-3e6618450ee819a2.json`; empirical validation `sha256:3a20a291f5964b0d603cb10e0cb85139d358db3dba65b48bc57dc3ce8996b56e`; measurement receipt `sha256:d5d1645eced54a2feb7096397771785e507c1d1fe702218f2732de6c9e0ad0d4`; isolation `sha256:a3eab68585733f829113f1f4cc7e7fe19431f1984508c8ed8967fde128196956`; custody `sha256:8da37d5bfaf2abcc2106df7c0398aa92c2f6ef38027ea47d4c74c4f90cecd65b`.

25. Nucleation and phase growth

Claim identity: `SFT-MAT-MICRO-NUCLEATION-GROWTH-001`

Question and exact theorem. A nucleus is the least generated cluster whose interior recurrence persists against its complete boundary relation; growth appends compatible cells while retaining phase and transfer provenance.

```
candidate-phase-cluster-carrier__boundary-versus-interior-recurrence__critical-persistence-and-growth__condition-and-size-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves candidate-phase-cluster-carrier, boundary-versus-interior-recurrence, critical-persistence-and-growth and condition-and-size-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`candidate-phase-cluster-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`boundary-versus-interior-recurrence`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`critical-persistence-and-growth`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	condition-and-size-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `candidate-phase-cluster-carrier__boundary-versus-interior-recurrence__critical-persistence-and-growth__condition-and-size-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Defects are retained differences from a declared reference organization; microstructure records their connected mesoscale arrangement, interfaces and transport paths.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:02e1b41356ebd232f4e1ffce226a6e14589a289969615744ce95a85aad756a91`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:85311f60dd07cce5138f6bd6e045ddff85be231164ee115db8a452a521f1fee8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:56ef43ac04bcd811cb45b4c7c6fdbcff6ccdb17e704f75735d8372bd501a556a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dd63ed8dc2853cf4afc78234cae08767527ac748ef137aa0f4a17520975b4ce6`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:9683ac6de970ada2d2d72b0be6c6d861cc8dddd1b21b397d0d9b60b6defbedbc`; certificate `sha256:1dbae78f92c78f583b62df211ba0173ab8f2c89a7c42f608e32c4b285d976cc8`; external-validation hash `sha256:c3b79b0253401ef2d732672ae16b5c5ea73b0c67e363e3f8a7a2f5247d212895`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:1c85d46f9369e6f5de9f63e0cbd6cbb7f02c8739fb3b08ad26d55f7238bbc31e`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SOLIDIFICATION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/publications/solidification-0; snapshot `experiments/external_sources/materials/snapshots/nist-solidification.html`; `sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c`; scope: solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- sft-mat-micro-nucleation-growth-001-authority-correspondence: predicted candidate-phase-cluster-carrier__boundary-versus-interior-recurrence__critical-persistence-and-growth__condition-and-size-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived candidate-phase-cluster-carrier__boundary-versus-interior-recurrence__critical-persistence-and-growth__condition-and-size-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered nucleation and phase growth observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered defects_microstructure grammar that preserves candidate-phase-cluster-carrier, boundary-versus-interior-recurrence, critical-persistence-and-growth and condition-and-size-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:5a3a4319df079e15b89fcc578f2f7ad41536a4d633b5fcea4746de982194b30c`; derivation seal `sha256:00b4f5e631130452dec26095c7e8906936b03dd6d06a591f3a22989b1a66c1df`; engine receipt `sha256:a9245f3d9e9386799ada6b1cc5f501b011bfa74c4b21132e17030eaed3e79bdd` at `receipts/engine/model_admitted/SFT-MAT-MICRO-NUCLEATION-GROWTH-001-a9245f3d9e938679.json`; empirical validation `sha256:2a1889999b40d5869f6b9b7ff919265c7bcfd95f77b61d002f917d8d71b7ba49`; measurement receipt `sha256:c2a6cc6f62a2898a9dfc49b69b928f8aea8f9d17b2110bda7b6eb6eba6342a3d`; isolation `sha256:b8d3766cf94a9c7e8e9e750abb2bf1a0feef8f57dba717ddb3236c425ac6764b`; custody `sha256:90ae992b4e7826a09190aad26ac3d29e2df6487b1e1529c8ef4b21b5f718c1c4`.

14. Electronic Semiconductor

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

26. Electronic bands and structural gaps

Claim identity: `SFT-MAT-ELEC-BAND-GAP-001`

Question and exact theorem. Electronic bands are complete phase-compatible recurrence classes on a material network; a band gap is a declared interval with no compatible generated state path.

```
complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absen
t-support-retained__energy-observation-bounded__complete-state-transition-bounda
ry-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extr
a-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves complete-lattice-state-support, phase-compatible-path-classes, allowed-and-absent-support-retained and energy-observation-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-lattice-state-support`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`relation`	`relation-imported-fitted-or-erased`	`phase-compatible-path-classes`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`allowed-and-absent-support-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`energy-observation-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e7658e1f06537e680b3ac39b7f4cc55681b1f55427f53eac74a237a7914ca6ac`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2430cf6e628f3745292c24ba39678422bf65e31c5bafae7c88e050479ec28fb5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:ccb0fb8846d5f8b815561ff55751d3207480100b14da0039bbbdd6dd480954b2.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:bfde1e54c6b003d0a0606d7d52bfc1046a37dd8f1357f843a5fc77d332188f2a.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:1e5c2bccf261f989160ab67cea576566eb955e4ef4e8e027f5cef5f3b063634c; certificate sha256:21963d4131f0a68880134816c335271db01e439c0b6a68b848658dcbb318d1bd; external-validation hash sha256:c100ff6484b4b8bf70291a8f5b401b7de9a126a6cac9f6a74ad43142ec5a41ae.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:68ff9628a37dce468af12c6384628b91da884cec8991a7cc7e4e45528a7c86d1. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TOPOLOGICAL-INSULATORS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/topological-insulators; snapshot experiments/external_sources/materials/snapshots/nist-topological-insulators.html; sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; scope: topological invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- sft-mat-elec-band-gap-001-authority-correspondence: predicted complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-lattice-state-support__phase-compatible-path-classes__allowed-and-absent-support-retained__energy-observation-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered electronic bands and structural gaps observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves complete-lattice-state-support, phase-compatible-path-classes, allowed-and-absent-support-retained and energy-observation-bounded.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:44addc87284413f5cb47df08df006e5926357c33b694da2ae9da9862470876c6; derivation seal sha256:175a8e789f8b27dd035c876d34d6ad6c7f59c521f797fd73ebde832f3f81d385; engine receipt sha256:7b3d73980e43e5e27c0796cda5840f72f2139cd68462f5de28c32b6fd289f085 at receipts/engine/model_admitted/SFT-MAT-ELEC-BAND-GAP-001-7b3d73980e43e5e2.json; empirical validation `sha256:7940d848f5f262579b0b16624dc6ba450141f83939e11ec7508f2c14b97aea6e`; measurement receipt `sha256:87710ab5dcb4ae5cc803063185f54704445502dbebf37fb7afffafa72b7f0d29`; isolation `sha256:28fc22f923f5c9d892f7e2e01dccdfde00144bae2fced6a676eef828729958f4`; custody `sha256:5d21c6ba0ac6008740250a9ad2c4d5b32112de387241c602fd7c36e703c3128a`.

27. Conductor, semiconductor and insulator distinction

Claim identity: `SFT-MAT-ELEC-CONDUCTOR-CLASS-001`

Question and exact theorem. Conductor, semiconductor and insulator classes are forced by whether occupied carrier support connects to accessible states directly, across a finite activation gap, or not within the declared conditions.

```
carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves carrier-access-support, occupied-to-accessible-state-relation, gap-and-filling-retained and condition-bounded-transport-class.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`carrier-access-support`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`occupied-to-accessible-state-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	gap-and-filling-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	condition-bounded-transport-class	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6b5fc1f6d32f1642235870027f3f6d09f09a59e0640b2c8e0ceed9372a3354a0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8bc23cae5864aba8f8171dd7eabe70270b7240ec4dd08547ac1f140c0eab3df7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6a3880e037cb135ac340b234e00c15070e7aa00ca0d95ab77f580baab73b002d`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:22f9b7b6ae28eef1be31d92d3cc3dea0076e729e6321efcec9f9b1998e18189f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:e51bfd79933191b3db914530ce1a3f2fe977a4e15d0a373a00bfd0620f7dca2b`; certificate `sha256:40be778743ea52098d5752b1130728f7507bc2b71ea29381c51cfba7898bd47c`; external-validation hash `sha256:1caf6cf61b772309acea478f82ffc1af61f3837d67f720225570c7b1955149c2`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:9c9409a2e572bb2d0836dca51f779846076dceabda444c74ff1840b1ae8b95a6`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TOPOLOGICAL-INSULATORS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/topological-insulators; snapshot `experiments/external_sources/materials/snapshots/nist-topological-insulators.html`; `sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449`; scope: topological invariants, insulating bulk, band gap and metallic boundaries.
- `NIST-SEMICONDUCTORS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors; snapshot `experiments/external_sources/materials/snapshots/nist-semiconductors.html`; `sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325`; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-elec-conductor-class-001-authority-correspondence: predicted carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived carrier-access-support__occupied-to-accessible-state-relation__gap-and-filling-retained__condition-bounded-transport-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered conductor, semiconductor and insulator distinction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves carrier-access-support, occupied-to-accessible-state-relation, gap-and-filling-retained and condition-bounded-transport-class.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:3a0672b7756ff9d49ab3b20a85f2c1915685f491d53339920343d42872af33af`; derivation seal sha256:82986bf06b4f76150a9f44e3485710dde93593cb487b90aff3d4518af3f5b29b; engine receipt sha256:e66114ebeeef098723db07db0f8fcf46b242222fa43dfc62cafd7fe664832ca2 at receipts/engine/model_admitted/SFT-MAT-ELEC-CONDUCTOR-CLASS-001-e66114ebeeef0987.json; empirical validation sha256:f4a79e3f2a36290a2a22ba128e7587792a20995bcb71d7eed2ab65c2cbbf0dba; measurement receipt sha256:5cdfb489014fb0be924fde87b453d556866f08a9f42d39b4a2de8460e51b467c; isolation sha256:31ff4726366c6ee968c9e2a86e07d4f4481376fd4c23fc08e32844ea5fd39622; custody sha256:dff32d01f05e8099cb18a076661a0d64c6a7fe2d9bc75d4612b2c046c2a7b273.

28. Electron and hole carrier duality

Claim identity: SFT-MAT-ELEC-CARRIER-DUALITY-001

Question and exact theorem. A binary exclusion-bounded state yields two transport descriptions: retained occupation and its held local absence, corresponding after sealing to electron and hole carriers.

```
binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-CONDENSED-BAND-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves binary-occupation-carrier, occupied-and-held-absence-pair, charge-orientation-retained and two-carrier-classes.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	binary-occupation-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	occupied-and-held-absence-pair	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	charge-orientation-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	two-carrier-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6913b47e4395df4244aac3f1636c31e345f1650091defd63a5a55ad53215427c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a06921b3e66f349f59a1691bdf6941ce10a2bb46d011ff17d496c3e9879cb00b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:e1f4603d1ca5bf94be4382a5ff0171fb92d97f2ed2dac597bdcdcff1969cd9e2.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:8488d36cd152a36b3fca8f7c07b9aee54038f04802540d3e561994b31f723093.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:2c00743cc5db244ed341a55c207f31476829df1719b36131ab05a2e9bdd1d2a7; certificate sha256:d95c160677403b50e44e149e3775ceab1a29413ec3de7c3840d02d669e98325d; external-validation hash sha256:fee3f69c239830d86d49cf5c348de9f74b0ba9301d4365196d7a093b2bcb179c.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:ba6eaeeda55aaefbe09d5d56edd2f15d1f36e1cd04ad57756489bc2a67100d93. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SEMICONDUCTOR-CARRIERS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light; snapshot experiments/external_sources/materials/snapshots/nist-semiconductor-carriers.html; sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5; scope: electron and hole carrier signs, doping, transport and Hall measurement.
- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/theory-optical-properties-materials; snapshot experiments/external_sources/materials/snapshots/nist-optical-materials.html; sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- sft-mat-elec-carrier-duality-001-authority-correspondence: predicted binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived binary-occupation-carrier__occupied-and-held-absence-pair__charge-orientation-retained__two-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered electron and hole carrier duality observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves binary-occupation-carrier, occupied-and-held-absence-pair, charge-orientation-retained and two-carrier-classes.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest `sha256:487f354580f1b942ef88107d8a7e67acaeb612e4800a2d3db21957d7393baa69`; derivation seal `sha256:02e62b0bb16a65e63274071f63abc8695d4dc712151cc16d37c39dea4b57dcee`; engine receipt `sha256:fd598a7661fd3dad3da24311efdcba303cbcee8dad7ddcd14e32c6e69afa7a0a` at `receipts/engine/model_admitted/SFT-MAT-ELEC-CARRIER-DUALITY-001-fd598a7661fd3dad.json`; empirical validation `sha256:32242bbc2ab119d300a72f453ece708ab5c3c3431cf5d19c4d742a0d2a85b81a`; measurement receipt `sha256:fdc881b8019988cb2fa9ffcaf6c97a23f6e9abe4541a24471546cd30626b139c`; isolation `sha256:dbdf98301457a308b3ac52745fe174bc74d6241df9a017d8fe11b0655ffa36b0`; custody `sha256:529bbb4a33e307039b20ec33f0fc389b025fcd8244504553e92cbc392540b299`.

29. Exclusion-bounded state occupation

Claim identity: `SFT-MAT-ELEC-OCCUPATION-001`

Question and exact theorem. An exclusion-labelled canonical material mode admits structural absence or one retained indistinguishable occupation; a duplicate occupation is rejected.

```
canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplicat
ion-rejected__occupation-bound-one__complete-state-transition-boundary-trace__ro
ot-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves canonical-mode-carrier, empty-or-singly-held-support, indistinguishable-duplication-rejected and occupation-bound-one.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`canonical-mode-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`relation`	`relation-imported-fitted-or-erased`	`empty-or-singly-held-support`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`indistinguishable-duplication-rejected`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`occupation-bound-one`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplication-rejected__occupation-bound-one__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9e095d2545b97492889f9c036a377954bc6c30b71ea4df6cf590c33a7ff73e15`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:489cedb9542a3ae1a73c6c0b0cfcae0a35d5b408bb26dcc08cbbac120e976a04`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:450062506d9f4b952b6cfc258ddee57e835465c3ba6069f3fa260372f2bd7be5.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:02cac4eb83231697c6648ffde545483083d1b1beb4f661539dde2c0d7ade8718.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:68927cecb586cf05cfe48019cdd0639a70cf2b46c72e876832380c4b24a11265; certificate sha256:e5f507b0a7e69aa7d8ad60023d63d64a9590dee11e5f7b8667d863fb2f58209a; external-validation hash sha256:c996a7a959521342393af3d9614300fa090a7d658dc386a86baffb1f4268760a.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:586a491bdc820fe537ca3c2684b67f21a9afdd214debf310fdc5f78d2ecc7daf. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-elec-occupation-001-authority-correspondence: predicted canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplication-rejected__occupation-bound-one__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived canonical-mode-carrier__empty-or-singly-held-support__indistinguishable-duplication-rejected__occupation-bound-one__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered exclusion-bounded state occupation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves canonical-mode-carrier, empty-or-singly-held-support, indistinguishable-duplication-rejected and occupation-bound-one.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:2d0fe62702f566ed470e063bb5e5b66dd78ec57a1ecf5952711e588f64e6f0a0; derivation seal sha256:a0a7516941112ac9fc011c492b523eeedf7b8aee7a793fc958fc72c9dacc71d9; engine receipt sha256:506e8ef22d4cde99c31da1531b362820047c16b74619ef23b75785d57ef7cc68 at receipts/engine/model_admitted/SFT-MAT-ELEC-OCCUPATION-001-506e8ef22d4cde99.json; empirical validation sha256:ee35ddf7d203268b4935a82979d01e280f6475b227d378ec8b8eff821b9381df;

measurement receipt `sha256:084ad1d875948d1719f83b9b6a861e91642abbbaa1885ef0ce125aa3a60706a7`; isolation `sha256:938d56f9988c6c5ec659b1c08f398e04a05f1f4b0b848dcf4430a32356ad6967`; custody `sha256:02e2391d12546aa8a4d5e40c33b22264075664191749b1011b94efbf0606c846`.

30. Doping as controlled constituent substitution

Claim identity: `SFT-MAT-SEMI-DOPING-001`

Question and exact theorem. Doping is a source-bound substitution or addition that retains host organization while introducing an exact carrier-support difference.

```
host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves host-lattice-and-dopant-carrier, valence-support-difference, host-structure-and-provenance-retained and concentration-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`host-lattice-and-dopant-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`valence-support-difference`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	host-structure-and-provenance-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	concentration-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:56562612ce8c35623d0f0eecc35d803dedbbb5024e8a9b6ddd9af1ea5273c1dc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e0b7aa75ce3cc58f9f38cccc833e5fffb828ed6a780b76c27e72333ae94cc79d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:cce931a0a9fc2d4bf92c7abfb6695c3c7131c95cf90e190f274acdf42801dd19`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:bb0257ed565ef5277fab5ae273fec802528089e7c14e6b69b5869605228db7ce`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:1b2754be6101b1b3994aa165f56cdc9719922bda78bb8a3e7b49541a130c5cec`; certificate `sha256:bb931bfb8986710e3d9c87afc938b21108e51f8711df610b710f2d2d37447c36`; external-validation hash `sha256:af6af3fceadf777a897ccb954d3a94939fda8a9a940635212ca5f0c7ff90910d`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is `sha256:122bb299288127d263f0e8816b29197cafb3b27fd5ab9450db9e64b49090bfc3`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-semi-doping-001-authority-correspondence: predicted host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived host-lattice-and-dopant-carrier__valence-support-difference__host-structure-and-provenance-retained__concentration-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered doping as controlled constituent substitution observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves host-lattice-and-dopant-carrier, valence-support-difference, host-structure-and-provenance-retained and concentration-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:7d6ba4126f6021c12701e140f9662df0dcac166c04f0411174670731bcaf36f2`; derivation seal `sha256:2c55ece8edc00804dddf74490222c7b077a2b1fb138d08925bacd67c16fa258d`; engine receipt `sha256:2dac934d9906877ee45538a866d4b4e5d01bca7db0fb4b83d1d13f6f3dfe5bc0` at `receipts/engine/model_admitted/SFT-MAT-SEMI-DOPING-001-2dac934d9906877e.json`; empirical validation `sha256:4b2237c7244ac6417cf900786078c5e3e67a6690e0716fa7cd6ce20c2bd1e4b9`; measurement receipt `sha256:76ac92ae9fbcfb18e6d6249bc79a99402af54af309c24eeb36531ec6dc8d07b8`; isolation `sha256:65640601d6884e72972e3ccfd89d29f60ac5ac53b9d2f2642fb10fc37a5041e7`; custody `sha256:75feef131ff8f5a61484f23bc35e133e4d4dcf9aa845ebd919d5e0050a69d51b`.

31. p-type and n-type organization

Claim identity: `SFT-MAT-SEMI-PN-TYPE-001`

Question and exact theorem. The two held orientations of a doped carrier imbalance force occupation-majority and absence-majority semiconductor classes.

```
doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations
__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-b
ound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves doped-support-carrier, excess-occupation-or-absence, opposed-charge-orientations and two-majority-carrier-classes.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`doped-support-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`excess-occupation-or-absence`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`opposed-charge-orientations`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
observation	observation-boundary-unrecorded	two-majority-carrier-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:56ed005fb5cdc0097997f439b277475193ecbc6b55d69b1a477b88ee060926ec`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cb92a6f95ff3cdea45d2e28b701382cb2a7f52dae78b06701b10a88ec5f994ec`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4b980f33219ddfc1ad61861742c6de890fe9d280fdaa67a1f556462fcbeaa528`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d8e94a97ea900fc4dfdec8667a3e360ee94746b72f6b7e344184e234fc506842`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:e452a66255f1a519151033fcce9e6faa6547ddefcecff55bca8aa84762cbfd35`; certificate `sha256:30d82bd025a5aef37135c2532e86294b973a60acc75b57d4c7832a10a1aebe2e`; external-validation hash `sha256:008460b8937e1e40da3576894f4ebb00fdd06ac578102266d2755dfd33919f57`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `3` claim-specific source discriminators were required; their ordered identity is `sha256:c9d08fdf23cf1d1c4a0cbd021530dcec9fe237faec27dcda06c3bc8c6ba06828`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SEMICONDUCTOR-CARRIERS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light; snapshot `experiments/external_sources/materials/snapshots/nist-semiconductor-carriers.html`; `sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5`; scope: electron and hole carrier signs, doping, transport and Hall measurement.

Comparison and custody records:

- sft-mat-semi-pn-type-001-authority-correspondence: predicted doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived doped-support-carrier__excess-occupation-or-absence__opposed-charge-orientations__two-majority-carrier-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered p-type and n-type organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves doped-support-carrier, excess-occupation-or-absence, opposed-charge-orientations and two-majority-carrier-classes.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:c315630088c00248b7a14519816565651cba129871a017b8b196d44835b21999`; derivation seal `sha256:508e9b18d4e217dbe7bb9880439e5ad62146420eedddbd5f4d38ab155c89bbc1`; engine receipt `sha256:09af6c90f13d03c26aa83c7e45157858c3cc873c105d60d24b5d8ecf5416bb56` at `receipts/engine/model_admitted/SFT-MAT-SEMI-PN-TYPE-001-09af6c90f13d03c2.json`; empirical validation `sha256:de8958fcad4ba8ece21a19486d0c82b04ba99b43f1aefa231a80e2a69b25851d`; measurement receipt `sha256:4401c9d604bc206b01f0b0240170a20ab4dbcdbac070de79b10ebfa528e1ffaa`; isolation `sha256:d097f756290910c99b0b5274c7508ea5752523009ebbbae60bc424a248147a40`; custody `sha256:f248e50cffb18f18a9831313e6d9d11fb1281018d9cf6fe04bfe88b3687c4896`.

32. p-n junction and depletion organization

Claim identity: `SFT-MAT-SEMI-JUNCTION-001`

Question and exact theorem. Joining opposed doped regions forces interfacial carrier transfer until a retained depletion/space-charge organization balances transport and field response at declared conditions.

```
opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves opposed-doped-region-carrier, cross-interface-carrier-transfer, space-charge-and-field-retained and equilibrium-and-bias-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`opposed-doped-region-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`cross-interface-carrier-transfer`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`space-charge-and-field-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`observation`	`observation-boundary-unrecorded`	`equilibrium-and-bias-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:04fea8715f50e7048d9416d376c53e3450f90c1c8994797f356f59fc28bb26fd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b60cab4caeac9244a50a12bfbcdba9f6971a485a5c9dca8d0dc056c9e1461973`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:341cc639ba0af26441cfc73b6163439bb85143317cd5e75e78911ae88dc2e7b5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:bed3d1635d1a890677f6efc509da67b859b05cda4243428f4bfa2cb10c603d12`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:e6eb0781d6783a76498a288507518312a16cd86e3b3062787095f1b02f171edd`; certificate `sha256:e9b13a5eb19c146ded679178d0be32f2a73e257475860e846b49a09f9cb13d16`; external-validation hash `sha256:d4b7e5fdc300feba712eba719d137d821e3bf69c7e235835a0281a7832a17eea`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:28acbaad472972858e5363c8410375f0fe98fecce35a163c6af00e8e96a968c8`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-JOSEPHSON-STANDARD-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard; snapshot `experiments/external_sources/materials/snapshots/nist-josephson.html`; `sha256:371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0`; scope: superconductive tunnel junctions and quantized Josephson voltage.
- `NIST-SEMICONDUCTORS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors; snapshot `experiments/external_sources/materials/snapshots/nist-semiconductors.html`; `sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325`; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-semi-junction-001-authority-correspondence: predicted opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived opposed-doped-region-carrier__cross-interface-carrier-transfer__space-charge-and-field-retained__equilibrium-and-bias-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered p-n junction and depletion organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves opposed-doped-region-carrier, cross-interface-carrier-transfer, space-charge-and-field-retained and equilibrium-and-bias-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:4d54dd75b86ef466bb3ae5cbc87a0ca7f701301e43197d0ab078c61c49750fc3`; derivation seal `sha256:555154742ac70136d63a345d0f10ab9c0c161554e1e8014f4b9ba7d52847822c`; engine receipt `sha256:779d6b98fcddc660d82e4349746ac115bf9463663074b27bac1a52fba754d2e0` at `receipts/engine/model_admitted/SFT-MAT-SEMI-JUNCTION-001-779d6b98fcddc660.json`; empirical validation `sha256:bcae9c1e255550e984a5e79458201c89f70b0258eba44b7b1903d1bba3cf78d3`; measurement receipt `sha256:35ceeda6bff041209f8db359a7dfcba8f80dafb726fdce296417f61bf22051e3`; isolation `sha256:76cfb9f4f250904e62748743c6525a455e0f0750a8adcbabd145a21d2c142a99`; custody `sha256:533feb47e28eae5a49bc786c08228949054a2dffe1623d130133575a3ec9d5a6`.

33. Semiconductor transport relation

Claim identity: `SFT-MAT-SEMI-TRANSPORT-001`

Question and exact theorem. Semiconductor transport is the complete counted motion of occupied and held-absence carriers through accessible support under declared field, gradient and scattering conditions.

```
labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves labelled-mobile-carrier-support, field-and-gradient-transition-count, scattering-provenance-retained and mobility-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`labelled-mobile-carrier-support`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	field-and-gradient-transition-count	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	scattering-provenance-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	mobility-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0e7b64dccdebdae3f781a411c669794f0bb3a3e5bdc46971765b794957d212a9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:013b382715fc6ec176e0dc3408ede980c818daf766daf96c476885b998c34ddd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:126ab210a36b80039e752758c90fa83b171da3f4688ecbfab8fb9478a547d6f8.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:bbf482bcded2d0df85a2fcb371710d3e58ddec033cd86e9729e94016097fd96b.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:aedaad9f2cdfe6887baf53d955eae7a0fb8b34bef6cbbb1ef936381ff1b3def2; certificate sha256:1f9ea281dc6bffeca2b178a18d709033fcb47d9f2666f9691f5c20615e0f8693; external-validation hash sha256:07cc521721d714dd3db3311e20236df3c3daf59ff0064523b530398b1e8f4db1.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:416db3b9005e4976355fa7996612862d673d7aced19bdae0fe27a041a619ee80. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials; snapshot experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- sft-mat-semi-transport-001-authority-correspondence: predicted labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived labelled-mobile-carrier-support__field-and-gradient-transition-count__scattering-provenance-retained__mobility-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered semiconductor transport relation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves labelled-mobile-carrier-support, field-and-gradient-transition-count, scattering-provenance-retained and mobility-condition-bounded.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:3af983283357631283d3947def3fd55314859ce664d93c96ae2d9a4aa1d2e69c; derivation seal sha256:fcd13ccc157a4adf9d8bd511d17f75266e4d4514d3c5a8008bb2c4a9fe2e2a2c; engine receipt sha256:7a9e1a097ec9c3c48321750ba1f42b728a9bec278024122acdb49c71d1bfeceb at receipts/engine/model_admitted/SFT-MAT-SEMI-TRANSPORT-001-7a9e1a097ec9c3c4.json; empirical validation `sha256:6b0761d0b8abf30d604d8cd0c0d8fc0bca015f1b61338f43fd2910a8b178aa52`; measurement receipt `sha256:8b454ccd9aee42a77be377617c292db6ea57bd02cbebae033513adf4e802cf3f`; isolation `sha256:9a8ad3fe9172c8975e080fdd5a07d52725f60710ce518c29d8a6ea02fa14784b`; custody `sha256:db2e9f0628622dc8e4a0e9ae541b76556fe17d30ad12204a0b24021289e87f25`.

34. Interband optical transition

Claim identity: `SFT-MAT-SEMI-OPTICAL-001`

Question and exact theorem. An interband optical response exists only when a probe supplies a complete compatible transition between retained occupied and accessible band states under the material selection rules.

```
initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-CONDENSED-BAND-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves initial-final-band-carrier, probe-energy-compatible-transition, occupation-gap-and-momentum-retained and selection-and-resolution-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`initial-final-band-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	probe-energy-compatible-transition	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	occupation-gap-and-momentum-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	selection-and-resolution-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Electronic material classes preserve occupation, held absence, accessible support, gaps, doping, junctions, transport and optical-transition boundaries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c936b0d5bb1ae8aec13da9742921982d562b55f6d4774bd1aab6e334a1e5b753`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8701918a766daef17bc285510ac5721438c7e56272c88217cd085a9d3e145154`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:bc252118ced3be1bd50755e418df34a442529528051ab45a471d57b3cfcfb8f6.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:ceec9c1581794a42484f4f193268ea44aecb938181070c46a7f3f65721a2f46a.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:a3afed5ac8947640ccdafaf35625a2212acb73b39ad06c991752279f3cad2019; certificate sha256:8076ebc7ac57b55ce725e87f69dcce17a303f9329a50b04a951c98f157d8053e; external-validation hash sha256:f68ffc662b3b7c7128a78f5abc04b5a0e0eaa09941e02795cea9ced09ac95ef0.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:278dc1caac24eb6c922c3a6bf6e833a99a93040d869c9a79be2ae4104dbf0bf3. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/theory-optical-properties-materials; snapshot experiments/external_sources/materials/snapshots/nist-optical-materials.html; sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of refraction, absorption, reflectance, dielectric response and photonic crystals.
- NIST-SEMICONDUCTORS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors; snapshot experiments/external_sources/materials/snapshots/nist-semiconductors.html; sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-semi-optical-001-authority-correspondence: predicted initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived initial-final-band-carrier__probe-energy-compatible-transition__occupation-gap-and-momentum-retained__selection-and-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered interband optical transition observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered electronic_semiconductor grammar that preserves initial-final-band-carrier, probe-energy-compatible-transition, occupation-gap-and-momentum-retained and selection-and-resolution-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:8d8279fa3008b12b0b1051877f0af738d00569ad9ba188399d1eba5df11f24f8`; derivation seal `sha256:8e4a5b2c675b5e98ec5e642838d2228f57b6aee5ee1ac95b8b21082fbcf31b8a`; engine receipt `sha256:809590cfd0ac95b5b110d63963478dbeff718642ced0621bb7bd593ce0f09e93` at `receipts/engine/model_admitted/SFT-MAT-SEMI-OPTICAL-001-809590cfd0ac95b5.json`; empirical validation `sha256:d223c4d8de7571cd7987e65eab435a979ca11071a431f928801ba7ff40fc18f7`; measurement receipt `sha256:bbdfdb43fb4478403d0f7fbdebc1396367ef0eb93db04319e389aad5fb1104d3`; isolation `sha256:beff5fea3faf3ab08ccddef2b1a5053453c9a471eee182464a6cd2ce6a342af1`; custody `sha256:2a01435955d2824cb67bf66500b68c1a34f4933eeb7eceee81a2e04ca4db6e43`.

15. Superconducting Superfluid Topological

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

35. Coherent paired-carrier state

Claim identity: `SFT-MAT-SC-PAIR-001`

Question and exact theorem. The least exclusion-compatible collective carrier joins two opposed fermionic labels into one integer-composite recurrence class.

```
two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__
pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forwa
rd-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-fermion-joint-carrier, opposed-label-pairing, integer-composite-recurrence and pair-support-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`two-fermion-joint-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`opposed-label-pairing`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`integer-composite-recurrence`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`pair-support-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:588d928b1e15481455a5bc3aed20e5ecc939285f701ef568ab8854c37a456594`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1fce1ab61ac5cdfd56f21d9a82a0f122803c48d6893b8182a0ec054e44125969`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:06212265fed2d4576c34bb29a7fa1db2397ec20e3ecdda764ca853158bb137eb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d85c54fbbab7de7503b8d8e13076bebd80a9d80d2bf9d3463f2ba5d851c6bf6e`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:ccdabd49d56b90757f60cc9f3d1de2c6b1e70499d23109147029465b0c1a2078`; certificate `sha256:4425dcd748752cd4959cb4846753a73d83912db2c333b36d3295d3212ba1f9a1`; external-validation hash `sha256:cd91799f0836165d952f55222fdc867f9baf72874a1e3c76abb6f573efb6766f`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:6c15a4430a3b63ae8b2d9b46dc34a353ae01acb710cd8481e153960a6e86cc26`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-sc-pair-001-authority-correspondence: predicted two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived two-fermion-joint-carrier__opposed-label-pairing__integer-composite-recurrence__pair-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered coherent paired-carrier state observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-fermion-joint-carrier, opposed-label-pairing, integer-composite-recurrence and pair-support-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:d93b4d5e2b622347f1676960ee60f23945f6a7ced001e0126d43a7c8b8ef3a8e`; derivation seal `sha256:a1338a1e0276c18373474215b722f972da5d845bbbd492ccb324d205c46553b5`; engine receipt `sha256:f65c0f0c9bd7124faa85bdb47c873a6abc42e8e8708b6d1e1a59820b33115855` at `receipts/engine/model_admitted/SFT-MAT-SC-PAIR-001-f65c0f0c9bd7124f.json`; empirical validation `sha256:713c8da110495dfd47aaea6dd319378273a24160916d2c4318d89f2f2fd1d278`; measurement receipt `sha256:61a58faeb48a968867e3b01590511b20540c5a14b184e41518714cf58b250eba`; isolation `sha256:7785d4374ef3eff679275b146a79ab9f5de3cea69341d630072727b1c8d59a9f`; custody `sha256:9c164976a0cf9ee4d1d0855105a2d8a6351a716ee3bee38ae12ec035fb2c8d5c`.

36. Dissipation-closed superconducting transport

Claim identity: `SFT-MAT-SC-ZERO-RESISTANCE-001`

Question and exact theorem. Superconducting zero-resistance transport is phase-locked pair recurrence on complete connected support with no retained label-compatible dissipative transition.

```
phase-locked-pair-network__collective-translation-recurrence__momentum-randomizi
ng-paths-absent__critical-condition-bounded__complete-state-transition-boundary-
trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-r
ule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves phase-locked-pair-network, collective-translation-recurrence, momentum-randomizing-paths-absent and critical-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`phase-locked-pair-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`collective-translation-recurrence`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`momentum-randomizing-paths-absent`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`critical-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `phase-locked-pair-network__collective-translation-recurrence__momentum-randomizing-paths-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:8d318b6cd754552f1635d2d3376fb19a7943dac948ddf64cdb2c469dd2d57981.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2d3c2643f67651e4ca0ce67d52da102053664b47fd07c1aec55f2625f3b5820d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4d3bbf5b42e26656f92e162c71c1b9b62f1636bacdd84ef0eade4dc87e68aa53`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:51e5ed61aacce99183b4b16d20ea877065858d4c167fda8309eca0ac820dd396`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:b1b9adcd5b385285baebd3bcc32081e7bcb07bc9a7423d1d0d8f4b24623b8b64`; certificate `sha256:e767c0f4c1ff932262442a7d1c6b5ae435d89b7adfc2ede2eb66da249e8a5872`; external-validation hash `sha256:bc66886b883b3b04e280b2c157934a163799a8e3abcd37cc54ea9170516dc3b5`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:c5d02ac8fd4e460114d7267f20dc67b2baff4b6cb76be84227c69b692b6557a0`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SUPERCONDUCTING-FLUX-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting; snapshot `experiments/external_sources/materials/snapshots/nist-superconducting-flux.html`; `sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6`; scope: zero resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-sc-zero-resistance-001-authority-correspondence: predicted phase-locked-pair-network__collective-translation-recurrence__momentum-randomizing-paths-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived phase-locked-pair-network__collective-translation-recurrence__momentum-randomizing-paths-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered dissipation-closed superconducting transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves phase-locked-pair-network, collective-translation-recurrence, momentum-randomizing-paths-absent and critical-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:b9e796760dea97e1e9f71baab1a6c8bfd297371b6fbdd5aa16ee8c36edf8c366`; derivation seal `sha256:edbbea4729b78617df2c8f1ecdb55716528ac140a38827276580103223e2b971`; engine receipt `sha256:12eb3dc5c5b6718a083650f79097da45e15e1c0f9a7d6d962fb3afdf672576ea` at `receipts/engine/model_admitted/SFT-MAT-SC-ZERO-RESISTANCE-001-12eb3dc5c5b6718a.json`; empirical validation `sha256:4463105306d4f42745346031c4a24c8910c65ae30aaf2ea125d0108575905655`; measurement receipt `sha256:1bebb8a089facd8820e247e0c2b56def398143bf54d99b01e82b17c475edc90b`; isolation `sha256:3a31a682ebb7914cb6dfe9a572940f32f1a21e3af617a0e474f367180900ffee`; custody `sha256:4b18283de437e6a909184edac099a512c370d092730bd30c5ef62b3927e90677`.

37. Magnetic flux exclusion in the superconducting phase

Claim identity: `SFT-MAT-SC-MEISSNER-001`

Question and exact theorem. A phase-locked charged carrier network generates a boundary response whose complete bulk recurrence closes an incompatible applied magnetic distinction, yielding flux exclusion apart from the declared penetration boundary.

```
coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves coherent-boundary-current-carrier, applied-field-response-relation, bulk-phase-lock-retained and penetration-boundary-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`coherent-boundary-current-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`applied-field-response-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`bulk-phase-lock-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`penetration-boundary-declared`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:41fdd77f83e8d59c2bfcf7bfbd501d37db3c95f5bc0b83d16b44837f40d88f75.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:e4bd038bb4b94d51173bdfb9230bddd24ccc1144277314f85858688c05c4e10d.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:b0b3d381e964f5faaa7119aeaac1005b9b4d579b20faed23038adc1dd012e4cc.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:e1e636a30e4b8e61b27e684806bc21fbf45696298651b7107e96b7f089e0a892.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:928c10c30b33c7559a3dbbc0c913420ed2ab81b7bc1d15273d91edeb97eee1d0; certificate sha256:06e8deb4ea6d9b5aa05a4dfec72f2b735bfa417cc18126dfeb1493310c209e47; external-validation hash sha256:b50c818dc8d34d3a26d99222c94ed78c6d5e46fe14e683dadfd25d9519ca71ce.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:0c547443d909099a887a74f3853eb1e7fc46b03829baa87ede9b6fae8b103170. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-SUPERCONDUCTING-FLUX-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-flux.html; sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; scope: zero resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-sc-meissner-001-authority-correspondence: predicted coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived coherent-boundary-current-carrier__applied-field-response-relation__bulk-phase-lock-retained__penetration-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered magnetic flux exclusion in the superconducting phase observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves coherent-boundary-current-carrier, applied-field-response-relation, bulk-phase-lock-retained and penetration-boundary-declared.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:b59e8e71bf983f7a08fd389c4b3c03789f7d0f5a8bd9b4c5f8864fbfb35608e8`; derivation seal `sha256:f9146fca79784576aa3a1589231dca173d7efe82cd3885b978b60ee3e13ef2bf`; engine receipt `sha256:2cd04f09d3f7371d20ce1b7209cc1dc496ea19835d019733d1fda439b0195fce` at `receipts/engine/model_admitted/SFT-MAT-SC-MEISSNER-001-2cd04f09d3f7371d.json`; empirical validation `sha256:ed1c150af3c3925bcb4915bb464cdcb311c930e5f83b00b8d6175f225bb0f044`; measurement receipt `sha256:3f7d9d6861cb68227b9a4061d8ab60bd253c3e2a7a84b9134c79a13fa58ec14e`; isolation `sha256:f141e62bf73a7f12e615ada65bbff9248644c0f2540ddc848f69ebca25a2b0b4`; custody `sha256:713d272d7326c231a0ba2e21288be842ea24652fdb5f251e11611b0f3113d7ae`.

38. Superconducting flux quantization

Claim identity: `SFT-MAT-SC-FLUX-QUANTIZATION-001`

Question and exact theorem. Single-valued return of a paired-carrier phase around a closed path forces whole winding classes and therefore discrete flux support.

```
closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-re
tained__integer-flux-classes__complete-state-transition-boundary-trace__root-bou
nd-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-phase-loop-carrier, whole-return-winding-relation, paired-charge-label-retained and integer-flux-classes.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	closed-phase-loop-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	whole-return-winding-relation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	paired-charge-label-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	integer-flux-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f0e9a5316c0cbe435499c31ffe40b86071cc48d07b4534c765299a5a35bf0f9d`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d7f3655cbeb3b9850a02dc2d27cfdf0a9c033adbdc9caa63cba999c19e29c974`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c69e26fe4417441d2552b06d56b73199016f8853baa5f0e2a5f34bcac21d8bcd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:64af51fc38ea344b01e2131b58ecba3a1728ed24158136831a4a38b3507f95cb`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:ba1d06e0c591660c25b9d494d140c2eb9c5b5ad7a60667e026d6c7c3f845b47d`; certificate `sha256:7cf210102f6487dc10ff5b4ceb32d3bf816945f961e947c3b153c669c4d4c632`; external-validation hash `sha256:76c78e4fa8a0436139f72942375cfe051a5b9bb6dd6496e8e17cbb720971c2f6`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:a59499d54f66da57c505ea1a9006d13183cf8c79f97b6694d0562a7789a16489`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SUPERCONDUCTING-FLUX-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting; snapshot `experiments/external_sources/materials/snapshots/nist-superconducting-flux.html`; `sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6`; scope: zero resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-sc-flux-quantization-001-authority-correspondence: predicted closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived closed-phase-loop-carrier__whole-return-winding-relation__paired-charge-label-retained__integer-flux-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered superconducting flux quantization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-phase-loop-carrier, whole-return-winding-relation, paired-charge-label-retained and integer-flux-classes.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:c8b02fef0c388c57e46670d576003802b43b86270c1afaff3f410a87423f6911`; derivation seal `sha256:85f5457eddebd585da75c677d2b205089e95a4558c7cc6f9af8783fe82ea8e0e`; engine receipt `sha256:5222d9301d5077db8f7fe1adccb876b76af5caeec23508a322fc8393ac62e22b` at `receipts/engine/model_admitted/SFT-MAT-SC-FLUX-QUANTIZATION-001-5222d9301d5077db.json`; empirical validation `sha256:0b4b3abdbac20b0e33a7ee4a688a80a5a4090133d5ad8bb7d756508ea41e47aa`; measurement receipt `sha256:ee23932a5a16f9a89e0b3dfd7839be86e66580e95a8c9a7621fef766dd025324`; isolation `sha256:b80a2d163e7842712cce6b99df6f2c0337bcf60b91f002180b285ac4d1ab8ed6`; custody `sha256:9d61fe50609999ec7322ca722ebf3962578a79550c05102d4c8a2a06e82e31dc`.

39. Weak-link coherent transfer

Claim identity: `SFT-MAT-SC-JOSEPHSON-001`

Question and exact theorem. A weak link between phase-locked pair supports carries a coherent transfer recurrence determined by their retained phase distinction without dissolving pair identity.

```
two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-condensate-and-link-carrier, phase-difference-transfer-recurrence, pair-identity-retained-across-link and junction-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`two-condensate-and-link-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`phase-difference-transfer-recurrence`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`pair-identity-retained-across-link`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`junction-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:399845f807dcd160de6e469aad258bfa25f83f72dc4570e597b8b67651d0aab2`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c0e22c5b1eff9f4e8d5756445e63ce5b47873c214a9e6225659f30250ae45e44`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:7b8e05babc5f8a0f80750a7c96e67151bc76e63513889500c048513a184222c3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f9a41df391a6e3a93cb1c58b912920fb74a9f3d3cfe4d8231ca7cded8f9f5816`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:4d646761ef86e203d4dc92fc307815d351f2f955293517f5549e5dcc1e949b39`; certificate `sha256:28ea20d0b7536a61f4df9e90490a956ebd4d4d57e33dbf816d7a7011b0260a1b`; external-validation hash `sha256:74274c0b5926ea435009d31d46f20c137898814c025f01bac6ccec42b1905e3c`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:ee4e8e99314f54e4c03074f8570f0e1abe2959be6265b7bcf80807a92d298924`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-JOSEPHSON-STANDARD-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard; snapshot `experiments/external_sources/materials/snapshots/nist-josephson.html`; `sha256:371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0`; scope: superconductive tunnel junctions and quantized Josephson voltage.

Comparison and custody records:

- sft-mat-sc-josephson-001-authority-correspondence: predicted two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived two-condensate-and-link-carrier__phase-difference-transfer-recurrence__pair-identity-retained-across-link__junction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered weak-link coherent transfer observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves two-condensate-and-link-carrier, phase-difference-transfer-recurrence, pair-identity-retained-across-link and junction-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

`sha256:de9469a89d2867722b9eaed6fa3760649daaa5553d70c469eb7cf83e15e2525a`; derivation seal `sha256:3bd854b2b132fd641e4e5f5955b5ffce6518e0c7ac28bcdd65660d450b12d489`; engine receipt `sha256:7081c6da685356122c147a5935e3b767d81a0c0c6ce6d5e0a7b75f6c0b4da6d0` at `receipts/engine/model_admitted/SFT-MAT-SC-JOSEPHSON-001-7081c6da68535612.json`; empirical validation `sha256:d871b3b487789842475feacb4d5f08c14ca3faa00faab4518a71d1f3cf2ceb03`; measurement receipt `sha256:2faf6ca3e067d3c131f05245b0149df1d19104e157da47b7accab873cd704cda`; isolation `sha256:7bc6387d84c77a4c1527b40b42bd1d5b808abd5603f45727497e2621e7d90a08`; custody `sha256:48ac2f8f2ee7cacdc2cdd908a0e4e30408ceced8311b194c717c481488f74827`.

40. Dissipation-closed superfluid transport

Claim identity: `SFT-MAT-SF-SUPERFLUID-001`

Question and exact theorem. Superfluid flow is shared-phase recurrence of a neutral collective carrier when the complete accessible transition census contains no allowed dissipative path below its first retained excitation boundary.

```
neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves neutral-coherent-fluid-carrier, shared-phase-flow-recurrence, loss-paths-below-gap-absent and critical-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	neutral-coherent-fluid-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	shared-phase-flow-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	loss-paths-below-gap-absent	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	critical-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:53aeb9bd83d18de3cd8d06065a5ef322fc212de21e769e8ca85c254f3a235c1e`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0dd29512db07cf4c534254d6059cf520fc5edc37285f2abc39c5a4c037549c9b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f6e09bc043b7b7342f350da3404ad33d5781dccd6d6a9ee2adb0e1d3c39d5b21`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:502b285ce7563d3d21f1c79789e6ae682a6022b3e36db0a107366d25c3c7fa2b`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:f56b139525ec8db1263aebf85b98f0d1417c8c2ef70d4f4ecef2abb94bc22a8e`; certificate `sha256:737b7188dd1647b964cd6655764a2a2db0aea236180922a145ffbddfb1ffa9ad`; external-validation hash `sha256:dbdd999b36e350be4f7fc5b44e9ef260cea53c782eccf9d0cb8e356c5eb0cc93`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:00b0b7c16330fdcbb90a5ea51473a63afe04ee29fbb606a9c345ba91717e8eee`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SUPERFLUID-CIRCULATION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid; snapshot `experiments/external_sources/materials/snapshots/nist-superfluid-circulation.html`; `sha256:49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0`; scope: superfluid flow, vortices and discrete quantized circulation.

Comparison and custody records:

- sft-mat-sf-superfluid-001-authority-correspondence: predicted neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived neutral-coherent-fluid-carrier__shared-phase-flow-recurrence__loss-paths-below-gap-absent__critical-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered dissipation-closed superfluid transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves neutral-coherent-fluid-carrier, shared-phase-flow-recurrence, loss-paths-below-gap-absent and critical-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:cb2ec54d212e954c015f00c0dac3b0b91ad8433dddd8dfeb5e842f81a20a3158`; derivation seal `sha256:c4fa97a72987df4fb8e8db6b3e67e94202e5bd15d51c09701501a72e26228d5f`; engine receipt `sha256:f97e7d21474da7f1f117d9fdf57a2a714f0844f88ece7bf4d90daf91e8e79b3b` at `receipts/engine/model_admitted/SFT-MAT-SF-SUPERFLUID-001-f97e7d21474da7f1.json`; empirical validation `sha256:e10012064a7012eaad97c30443e19ed8db341b8c01a039ad34cec73910fc9ebd`; measurement receipt `sha256:f5e2e939e5d8bbed493cbca552c915b723dbef99a7a0602b8a960eac7d1a8b2d`; isolation `sha256:5c0d5f2cea3b7d6c7242bf7c842f4a0713086806e9d89c4909cff5b976545779`; custody `sha256:65f0b4b610b2e75226f3cae0b0a7eb10f9d9093837f3fc6cb940dde6bbfb33e8`.

41. Quantized superfluid circulation

Claim identity: `SFT-MAT-SF-CIRCULATION-001`

Question and exact theorem. Single-valued return of a superfluid phase around a closed path forces whole winding and discrete circulation classes with a retained vortex-core boundary.

```
closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-superflow-loop-carrier, whole-phase-return-winding, vortex-core-boundary-retained and integer-circulation-classes.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	closed-superflow-loop-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	whole-phase-return-winding	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	vortex-core-boundary-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	integer-circulation-classes	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f02eec7391d7631590d2bdf547034d8deb9704bd1216a21b9f245de6f8279715`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:47e3e962a411f836a67dcfcfb25ec672026bec43ec887acbfb46cdd0e5d2fd37`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4e454943e4f2ef6f47e060c907ae345a5b14d1aa029de2e0f12f423693052ff7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:65529a60c0bbbd22624bcc4ef2744b5a19a59898aaea44440259859ce3e8d166`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:8309220cfb4f71ac91bb6c78e1b8e93d5fd7d982a212003a218b11b52185129b`; certificate `sha256:c2182c823e85f1379edf6bbcae692fb0a0597df0bf835c2b4a567d343189c797`; external-validation hash `sha256:c4741e9aec860d4696db44b536e7ed688067581da702492261813d3450aa487d`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:3b4dda8db9ff0cf748bd3069c4aa37b649f64bf399b70947c75841856eaf0257`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SUPERFLUID-CIRCULATION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid; snapshot `experiments/external_sources/materials/snapshots/nist-superfluid-circulation.html`; `sha256:49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0`; scope: superfluid flow, vortices and discrete quantized circulation.

Comparison and custody records:

- sft-mat-sf-circulation-001-authority-correspondence: predicted closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived closed-superflow-loop-carrier__whole-phase-return-winding__vortex-core-boundary-retained__integer-circulation-classes__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered quantized superfluid circulation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves closed-superflow-loop-carrier, whole-phase-return-winding, vortex-core-boundary-retained and integer-circulation-classes.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:f16ac71b88734483350d41ca16b2b40556ee06e0987c89a426fb1985a4b5907a`; derivation seal sha256:b372c91d91c4134850d6c8fb14dca1272e0daaeaacce8ca48d23bd0f66a3498b; engine receipt sha256:d7416c15ecaa36ba63cd90357f14a677eba102cc7a50158fc5fda860d820aebd at receipts/engine/model_admitted/SFT-MAT-SF-CIRCULATION-001-d7416c15ecaa36ba.json; empirical validation sha256:d0a3ab6803a908c6d5278dadb5d3099c6ef00e2f4b1c721818568eaa604bcb2d; measurement receipt sha256:01272ae44e4f4e7d9ba40480b6afd4158ecdb7a6bdd572fceb256005fe603f75; isolation sha256:a6bf12743e23eb7e27f83abfac2fab464763d8dcc4f16491218c03255d3b3e3a; custody sha256:b9112c2aaa8190490bcec24ceccf35ccc1ef0b1a0a82050c5612ebe0472af434.

42. Topological material invariant

Claim identity: SFT-MAT-TOPO-INVARIANT-001

Question and exact theorem. A material invariant is topological when its complete global path class is unchanged by every generated local deformation that retains connectivity and boundary records.

```
complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-MATTER-FERMION-BOSON-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001
- SFT-PHYS-CONDENSED-TOPOLOGICAL-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves complete-global-path-carrier, connectivity-preserving-equivalence, local-deformations-quotiented and integer-path-class.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	complete-global-path-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	connectivity-preserving-equivalence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	local-deformations-quotiented	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	integer-path-class	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:13efa656a19aecf32a12b6813904ce4d2fcde0814a104238b0f2940d99e2c78e`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e181bd02e17925831b4a6b22c0d3ae5c3578c1a6ce53367eeabced88df33d8a6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8905e525f2d1831fae398bb4d42ebe70cd9d0eb4243660c738c48c791372c83e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2b0c618640e78c9e6a61d0cd591b39a31111d3a4bcb515fa4230310307ad5b0f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:52be2a7f7455b7444b99c8ee1fb48683ea99fdce6dd35ca5f94546db602cac5a`; certificate `sha256:324729b6e2e4e99286bacf7e230b6054ad0584c770944fde8bac4f7d7e3ca46f`; external-validation hash `sha256:d5da1e69cff9663a4438aa6a46b03c0d80ee306d6917c10d13e544d6c476d28d`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:e25423817b22188b5ad5603572fe72556d6d623de47ea8d79cac1a22ac17e8e8`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TOPOLOGICAL-INSULATORS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/topological-insulators; snapshot `experiments/external_sources/materials/snapshots/nist-topological-insulators.html`; `sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449`; scope: topological invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- sft-mat-topo-invariant-001-authority-correspondence: predicted complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-global-path-carrier__connectivity-preserving-equivalence__local-deformations-quotiented__integer-path-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered topological material invariant observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves complete-global-path-carrier, connectivity-preserving-equivalence, local-deformations-quotiented and integer-path-class.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:39b15b0d045f1b1d1b7ff8ad2ee76ac387e0f1a555cdf04c6cfe0a7e5c93e16a`; derivation seal `sha256:355e1bda262e641217018e5bfd12142fb515f53642ce68873f49b667d47e9dbb`; engine receipt `sha256:0aa06ea813c11acea4d063c83c11e606519be491423b0806a196e2876fe9391d` at `receipts/engine/model_admitted/SFT-MAT-TOPO-INVARIANT-001-0aa06ea813c11ace.json`; empirical validation `sha256:e2546e5a5c2cffbad153836b434eee2a7d90ffd41f42024f1d71c7fbfa5e9c84`; measurement receipt `sha256:c2ed694d7588b3efc388b89c3d8225bce6a6b223050cfa18c3666f6752c8d69b`; isolation `sha256:d877347a8eb8de659a662a2d14c4f0f00357e99386626f2e72ebd2da2b6a733e`; custody `sha256:1ec350459cdf675e1b52403b1065a89816d482f69231144349ef769376bfac96`.

43. Bulk-boundary protected transport

Claim identity: `SFT-MAT-TOPO-BULK-BOUNDARY-001`

Question and exact theorem. A nontrivial bulk path class forces a boundary recurrence whose removal would require closing the retained bulk distinction; compatible local backscattering paths are therefore absent within the declared gap.

```
bulk-and-boundary-joint-carrier__global-class-boundary-obstruction__local-backsc
atter-paths-excluded__gap-and-boundary-bounded__complete-state-transition-bounda
ry-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extr
a-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-FERMION-BOSON-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-PHYS-CONDENSED-SUPERCONDUCTIVITY-001`
- `SFT-PHYS-CONDENSED-TOPOLOGICAL-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves bulk-and-boundary-joint-carrier, global-class-boundary-obstruction, local-backscatter-paths-excluded and gap-and-boundary-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	bulk-and-boundary-joint-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	global-class-boundary-obstruction	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	local-backscatter-paths-excluded	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	gap-and-boundary-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `bulk-and-boundary-joint-carrier__global-class-boundary-obstruction__local-backscatter-paths-excluded__gap-and-boundary-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Collective recurrence, whole return classes and global path obstruction organize superconducting, superfluid and topological material response.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f204c5e7146eb04130ad4807231ad0637e358eeb604715a0c22917bb8e16da4a`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8ed623d8b4faf8ed6f5d8eabc89a41c340ddf240b0b45ecc9e15bd39089bc68f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ad1aad827f8955b0b0041cf25546c6841d11c1b396331884019eeb73aab28e00`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4d467a33e09e676142c88ff0e26ce9d516bfa0201a789ecef92a812414280b8f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:c1ad057b49304b4368ba4fad93e4f0ee05abf12ee690c16fd5c5d7c3a8eda48b`; certificate `sha256:b0899dd406db35b3d8d9d3b00a5254964967f26d618d1a345bb4f7a4f3537c19`; external-validation hash `sha256:5b8e12fedd650e5769d332dbaf6afda85a21d1f6ab5c33aede7b21fedad07d32`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:1d4a59568502972f6927fa8cb2944be5ea47b884c616a7a56c4d71c0d8fede16`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TOPOLOGICAL-INSULATORS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/topological-insulators; snapshot `experiments/external_sources/materials/snapshots/nist-topological-insulators.html`; `sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449`; scope: topological invariants, insulating bulk, band gap and metallic boundaries.

Comparison and custody records:

- sft-mat-topo-bulk-boundary-001-authority-correspondence: predicted bulk-and-boundary-joint-carrier__global-class-boundary-obstruction__local-backscatter-paths-excluded__gap-and-boundary-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bulk-and-boundary-joint-carrier__global-class-boundary-obstruction__local-backscatter-paths-excluded__gap-and-boundary-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered bulk-boundary protected transport observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered superconducting_superfluid_topological grammar that preserves bulk-and-boundary-joint-carrier, global-class-boundary-obstruction, local-backscatter-paths-excluded and gap-and-boundary-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

`sha256:624e50eb6c37a7cb6aa302bb729524f97d404589da3d4dacfb4b391c2f0fd9ee`; derivation seal `sha256:786171e535a2997aaaa15642857733ac66028d16eab119c2349d6501ea056489`; engine receipt `sha256:abb28732eccdc748940dce9e3629db011dc41a205ef3b968b06d5cf1c0fb4f1b` at `receipts/engine/model_admitted/SFT-MAT-TOPO-BULK-BOUNDARY-001-abb28732eccdc748.json`; empirical validation `sha256:74b655977a85543c9d7d4722f055436f75559358e69d248ace5caaa8253b1b88`; measurement receipt `sha256:6fe03baefebf6c5f8b5194955b9d1132490c993a55fc531f5407d89ae26235b7`; isolation `sha256:6b2e7114ed5081ebd7faf0314e4d1e0aa595766da8b28fd6ea5bbb6f8dc75bc0`; custody `sha256:03b7784db09f06251f28e2779209ce357bee47b308dce3fabf2cb5625b6e2dc5`.

16. Mechanical

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

44. Stress and strain records

Claim identity: `SFT-MAT-MECH-STRESS-STRAIN-001`

Question and exact theorem. Stress is oriented load transfer per generated boundary support; strain is the exact retained change of internal separation relative to the same reference material state.

```
material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-region-and-boundary-carrier, load-transfer-and-shape-change, direction-support-and-reference-retained and method-and-scale-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`material-region-and-boundary-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`load-transfer-and-shape-change`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`direction-support-and-reference-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`method-and-scale-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:adb5d05a21d454b6bf5a2017f8be3f371d55fbd9a0b7ad25b947caa373ad534f.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:109a21a2e46dcd23f5466f23be5156d57b500eff7391011d89ecc112be5e6872.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:6391dbcda50225bace528b7afbb330ee156b47e635d52cdd610f75541e72b83b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:04e5725ac444cc83fe68338efe05502691abd15b8808ea95150d1c8bb7212fe6.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:60cf9be4f23f11d4bef081dbc622511a6b00ba9a95d7c50d982b096248f1af5e; certificate sha256:166b798626def8349847d2bbf1f68dca63d749c0a498661c151eb032cc0da216; external-validation hash sha256:64538778c885dec7c2c0b9e159dc79e019cbcc17cace487879a2ca3a86e5b28f.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:58cbbf4e31983da152d2e9df9813867504aea0339fd3dd88255134648592ea34. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-FATIGUE-FRACTURE-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture; snapshot experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html; sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; scope: processing-structure-property-performance, defects, fatigue and fracture.
- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-grain-boundaries.html; sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2; scope: grain boundaries, strain, dislocations, current flow and superconducting response.

Comparison and custody records:

- sft-mat-mech-stress-strain-001-authority-correspondence: predicted material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-region-and-boundary-carrier__load-transfer-and-shape-change__direction-support-and-reference-retained__method-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered stress and strain records observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-region-and-boundary-carrier, load-transfer-and-shape-change, direction-support-and-reference-retained and method-and-scale-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:72ec1a5c263bd16dcc14f88e80cde6ddbce93c49574494ea6d506ef427d69a6b`; derivation seal `sha256:d3b9e2aa529d399c2335b68a60fb2429b910b89d6a5c9463d99af63692002ffa`; engine receipt `sha256:85482dc44ea7d4480cdd7474370679a5ef92c2b374ade6acd5604e5da740e766` at `receipts/engine/model_admitted/SFT-MAT-MECH-STRESS-STRAIN-001-85482dc44ea7d448.json`; empirical validation `sha256:b255306584d1626058d63a03e6c05d19c2039983e3939726f8e7e9d1feae2f65`; measurement receipt `sha256:905fe24fd20a32060c8e829ca181725b21b382c21b9580ce56ac405699ed9f02`; isolation `sha256:4eb94e72da4ebae40c69edca01de9e83b273d60729611f3ac44fa71099fd27e2`; custody `sha256:f680002c5dbcfd1e4fa1ab346fedf28362a1c9de3b8c2c7e067d99fe7112d62b`.

45. Elastic recovery

Claim identity: `SFT-MAT-MECH-ELASTICITY-001`

Question and exact theorem. Elastic response is a material transition path whose load release returns every retained constituent and adjacency relation to the original canonical state.

```
bond-network-state-carrier__reversible-displacement-path__original-adjacency-cla
ss-restored__elastic-boundary-declared__complete-state-transition-boundary-trace
__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-network-state-carrier, reversible-displacement-path, original-adjacency-class-restored and elastic-boundary-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`bond-network-state-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`reversible-displacement-path`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`original-adjacency-class-restored`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`elastic-boundary-declared`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `bond-network-state-carrier__reversible-displacement-path__original-adjacency-class-restored__elastic-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:d965c15a37557dcfc3605cc273e1a1880c9e41083dbcfc9d3a708f979636fd61.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cd6eb0e5456c7e6e8a8e5f7fd98dfc5b0a7eae9a32d7ecce593b09f29c908bae`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ac6d6673b36ac8725d7bfe8f9d12e9924b9024d0edd255e90883210b8efb8fe9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:47aaed4f4a0ea0724465aceb9bf4aba1ef44ce469cad8d3d06fa3e0f47f21236`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:b0a0f914de0a4b46b3d7efd5c82bdd05cd890bec50d642f36863a2e98494e644`; certificate `sha256:9d1e0777b17da1c0589983ef1da7f0c3ec0ba76e129d86df0a1b1a58f37c3c93`; external-validation hash `sha256:a0374dd3a632fe0a266ad5bf3f615aac82e7a7b413a508f0b7c44bcdb0e433ad`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:1f5231592213d1cb22ac6eee1db2e48d66131790c95c6401a6712d5e12f4bc25`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SEMICONDUCTORS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors; snapshot `experiments/external_sources/materials/snapshots/nist-semiconductors.html`; `sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325`; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-mech-elasticity-001-authority-correspondence: predicted bond-network-state-carrier__reversible-displacement-path__original-adjacency-class-restored__elastic-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bond-network-state-carrier__reversible-displacement-path__original-adjacency-class-restored__elastic-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered elastic recovery observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-network-state-carrier, reversible-displacement-path, original-adjacency-class-restored and elastic-boundary-declared.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:b346b8fee12d428022c4ae51969a24edb3f509f9de381374c5735ea1eb0d92a5`; derivation seal `sha256:260626b207e5763c1058d0fc4ec4281cb5ce31d71314b505f4027525115ab722`; engine receipt `sha256:b490d2bc4771b5f91d7458764359748fcb9fd73ecb665b5f66637be92c9d9b17` at `receipts/engine/model_admitted/SFT-MAT-MECH-ELASTICITY-001-b490d2bc4771b5f9.json`; empirical validation `sha256:91e76ec99100ab2b4bcb9de89117654abe30c261b652a123188a14eccb7d5e2e`; measurement receipt `sha256:0fc70106063eb948741b12ccaf1d72685502d7185847683b38e43c272ba1b467`; isolation `sha256:5f44ba6f53e33a525c41ec5a037db2da522c7f9ab958f32f1cc61771aa19cae6`; custody `sha256:4746d3544ccf6e014f821317b7525878c79c6de8fa91be7cb75968f039c58fc0`.

46. Plastic deformation

Claim identity: `SFT-MAT-MECH-PLASTICITY-001`

Question and exact theorem. Plastic response retains material content and phase identity while a defect-mediated transition changes the canonical adjacency organization after load release.

```
bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-and-defect-network-carrier, irreversible-same-material-reorganization, content-retained-adjacency-changed and yield-boundary-declared.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`bond-and-defect-network-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`irreversible-same-material-reorganization`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`content-retained-adjacency-changed`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`yield-boundary-declared`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9150c7c34bf6ef1d0009e84afa667c913cb1142a022c9425f3fc6a140ffd3b9d`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a7d6901fc35b585477d6b688a690b3c162e2e621559cad8b4d96464a4ec0170f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:844e5f851097100443c4f3d0dfa284fbae50e59dff142d5d5c9b8fdddff629d3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:316c01f0fe9eb2fb72cde97a9c3e7af4199b41650b03f8a77be1d286fd0b766f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:f15e6957edff21a3f20f016b86d2c1eec8d49f37c03d565312ea934715ed36d1`; certificate `sha256:db298f3a0e0fc3ab181511dd1bea67a8856c29679fb939576d2814ae6dc44876`; external-validation hash `sha256:20e178859f18007683e55801f7a6cb2240a269f32c4aa11dd20fa58e04218056`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:c36087458657bcd08909aee6600aa4bcabf26dce1f9f96112864f0e05ec356d9`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TEXTURE-PHASE-FRACTION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction; snapshot `experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html`; `sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc`; scope: grains, phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-mech-plasticity-001-authority-correspondence: predicted bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bond-and-defect-network-carrier__irreversible-same-material-reorganization__content-retained-adjacency-changed__yield-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered plastic deformation observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves bond-and-defect-network-carrier, irreversible-same-material-reorganization, content-retained-adjacency-changed and yield-boundary-declared.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

sha256:3006a582b76abd466eb5e052f89c2465497a3c270e20fe6db6dbb339179a8ac2; derivation seal sha256:12ebaca61ddae97d7c9eb477672cc0de736f0ab3c7f9af57c6d2e80f45d61158; engine receipt sha256:77ce9abbdc078fd565e3a48d5a5b2f65b128d91f104012cc32b4c4a02eb7d2f1 at receipts/engine/model_admitted/SFT-MAT-MECH-PLASTICITY-001-77ce9abbdc078fd5.json; empirical validation sha256:b3532d1895dd1296d276e1735e26099cf264926c90c65d74200d5e625debdfe7; measurement receipt sha256:391ccc840a880b49ecdd2940ed67b3596c473564b0d566fe1b63c106bffed26a; isolation sha256:d873591d0a8d9bb028422f4b911183a711f9c215eb9344d67d2f95798c1a8f7e; custody sha256:efd5e346f7ffd5349aa7bb765688da67e26c84fda13093741e87732a585a073b.

47. Dislocation-mediated slip

Claim identity: SFT-MAT-MECH-SLIP-001

Question and exact theorem. Slip is a counted whole-lattice translation propagated by a retained dislocation along a compatible plane/direction without requiring simultaneous rupture of the complete plane.

```
dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and
-content-retained__slip-system-bounded__complete-state-transition-boundary-trace
__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-FLUID-PRESSURE-STRESS-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves dislocation-and-slip-plane-carrier, whole-translation-advance, burgers-label-and-content-retained and slip-system-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	dislocation-and-slip-plane-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	relation-imported-fitted-or-erased	whole-translation-advance	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	burgers-label-and-content-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	slip-system-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:89538877c541c0e3b80b081a953e40d057a01f631b47a0e8282b46c7a892f43f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0151dd250ba88c10559598d8778c71c3c11e36e9edcd1034f8ee2fc45cbfb0c5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:f077732e53b69bf50948fe5266a30e63f5f80a7293c8bc4ae7d2a46e0d248e76.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:16203aa2aa450696e0421d92c1b3cc0a003c3fd86ca1413a987ca923e90c1af7.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:ce30006112583d5a105cd9e64a8deec6bdef9c9a58e2ccafa5b65005bd351415; certificate sha256:c9a1f12156a69f5368449a3cc781bbff9923903bd31e5d7f38c8d32b9beeec75; external-validation hash sha256:1ca21fd13495c4d6ec9d83b742a4d8006a197f875c48dccdd1140cb435248696.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:a5c1c011368a43bfcccc04cf8c412ee846c22c570ed3a97aa228d2280274c1f1. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal; snapshot experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html; sha256:a1b7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18; scope: microscope confirmation, nonrepeating order and fivefold symmetry.

Comparison and custody records:

- sft-mat-mech-slip-001-authority-correspondence: predicted dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived dislocation-and-slip-plane-carrier__whole-translation-advance__burgers-label-and-content-retained__slip-system-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered dislocation-mediated slip observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves dislocation-and-slip-plane-carrier, whole-translation-advance, burgers-label-and-content-retained and slip-system-bounded.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:1e7c4abd9525c4f5aa9107b87bff628f69fee285b2d2a20cf4c4b5ddb071f60a; derivation seal sha256:5c9bc421d35c9d3b9ed2d15aacce9dda69982b549984f8c43b1d769f9d16501e; engine receipt sha256:ca7d3c3104a0dfdf5dde9127e49e575edfbba784c1c2d58859ab2f22267eb1e3 at receipts/engine/model_admitted/SFT-MAT-MECH-SLIP-001-ca7d3c3104a0dfdf.json; empirical validation sha256:262f49da288b06c0eadc6b88cf734f56ef39fe5c5f9ca0093b53805735c49a9b;

measurement receipt `sha256:612c261a80046a405268cd495f42234bf4b68f6b2b4e82c49826f0939b1e8b52`; isolation `sha256:7274e98371ac622e94f1b2247c5c951a7650f0e5270ff34fc798957594909bbd`; custody `sha256:d3ef1d91e8259ecd6710a62a7ebf7ee7c3741be9f8d9e6b5668b3591fa03c339`.

48. Elastic modulus as response ratio

Claim identity: `SFT-MAT-MECH-MODULUS-001`

Question and exact theorem. An elastic modulus is the exact stress/strain response relation for one declared material direction, condition, reference state and reversible response window.

```
elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition
-and-range-retained__linear-window-bounded__complete-state-transition-boundary-t
race__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-ru
le
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves elastic-state-pair-carrier, stress-to-strain-exact-relation, direction-condition-and-range-retained and linear-window-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`elastic-state-pair-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`stress-to-strain-exact-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	direction-condition-and-range-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	linear-window-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7d6248f498d15a2edbf0aea112d99e5a2f1daf19a804222723337b55c86c4ace`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:06222eed10c6b068ebb6c92d7456fde86013ee12119f30ebb92fe895b8f0fffc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2bbd140e1af9ff6e2ba9427d90da1fbe2b21c9c8a379fecc1902d3f131edf130`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9d740d421a442846bdb6849544d4930fc9837d3a2f4ef6f6a838b16d1b9541f1`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:18bf6748d138471a11dcd6ba96c75fe25842d895dd65d2f2ced937d8eca68784`; certificate `sha256:36744a94fc531a579c5f99ec93300f4892e6f6e9dd73cd52fe56f5d70fba9de6`; external-validation hash `sha256:111c32e35dd4f098864a80c781a12cc6cf9e944f89cbae2640eb53a44f1a4854`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:85c03f12333c98d0238b67dfce07d56b00dbd561755af223f44fbc04919ff7e9`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SEMICONDUCTORS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors; snapshot `experiments/external_sources/materials/snapshots/nist-semiconductors.html`; `sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325`; scope: semiconductor structural, nanomechanical, thermal and spectroscopic measurement.

Comparison and custody records:

- sft-mat-mech-modulus-001-authority-correspondence: predicted elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived elastic-state-pair-carrier__stress-to-strain-exact-relation__direction-condition-and-range-retained__linear-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered elastic modulus as response ratio observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves elastic-state-pair-carrier, stress-to-strain-exact-relation, direction-condition-and-range-retained and linear-window-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:1153be6043acfefa94d2197d88eb7ba687bea8d2e7db45b379d2b884ec2fe603`; derivation seal `sha256:e9f4cc9b2d4c4888f1326180b6143f911b195fa9076791fc8bb4e403fb286e78`; engine receipt `sha256:fd39c697c448c56d56a6d63557ec19a7b249f2ca0ff72bae1f895ee7fc77fd88` at `receipts/engine/model_admitted/SFT-MAT-MECH-MODULUS-001-fd39c697c448c56d.json`; empirical validation `sha256:841d85ef828079b201e69f2e3cb0bfc8c17992ebc430516174765f5a3641725a`; measurement receipt `sha256:e195b8c1f1a29cab72fdd2ba5198b76a7c3eaf63c8afce760acfd8ab09ba87bf`; isolation `sha256:9281062f09a9ad8c389d3cc4cd86a231f35c1720f0bb32a9799f953a5285a3da`; custody `sha256:0269158f503aae4190601227607fce846f38422cd5356c8b20326c355ff2d890`.

49. Strength and hardness boundaries

Claim identity: `SFT-MAT-MECH-STRENGTH-HARDNESS-001`

Question and exact theorem. Strength and hardness are method-bounded thresholds at which a declared loading path first produces retained plastic, indentation or failure organization.

```
material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-and-test-carrier, load-to-persistent-change-threshold, method-geometry-and-scale-retained and threshold-not-universal-constant.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`material-and-test-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`load-to-persistent-change-threshold`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
organization	organization-collapsed	method-geometry-and-scale-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	threshold-not-universal-constant	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:862336e016548ff4c2c82fd3ac77ca0323be9c77556e971a6aebebacd0c76fb5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bc14a47bca29bd87794372fa028e2771a41e0b2d2e2b3f1ec19666f2e362e0c7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4be80645c59da9b639daaaf753ec6ce5c2df9dec4c06899ac073ceb7572efc38`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c2c68a86de634604b832653e2398fde3c925a952212ee0b01e72d81390035d1d`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:bd81bef0333505d69e1bb6b71f2ea590586d3910f93d18e8c9770aa2d979ea68`; certificate `sha256:6e74e10acadf20f87e2b884dbb28242db27c0b03c2917387319e000cb73953a8`; external-validation hash `sha256:05602a8fb8c3d7a23827d52a2847b910135f7aa50c9622b0198f2e704299548c`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:306bd49533d46cb19fcf0627564636eda8ee8cc1a15790255742306b9d9a8c67`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POLYMER-COMPOSITES-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/polymer-composites; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-composites.html`; `sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5`; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-mech-strength-hardness-001-authority-correspondence: predicted material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-and-test-carrier__load-to-persistent-change-threshold__method-geometry-and-scale-retained__threshold-not-universal-constant__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered strength and hardness boundaries observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves material-and-test-carrier, load-to-persistent-change-threshold, method-geometry-and-scale-retained and threshold-not-universal-constant.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:8c586386b9a6b3354402a0c4f30ff03c2f27d866530dcb39d08757be0caad2e2`; derivation seal `sha256:74b8b045099cf7851df7063a11da3c2a0e7870c7148bf04fb703f638f213e800`; engine receipt `sha256:2db59637af7814e2711eba819d183bafc63ff7b259573bed972188dd7aaf2a08` at `receipts/engine/model_admitted/SFT-MAT-MECH-STRENGTH-HARDNESS-001-2db59637af7814e2.json`; empirical validation `sha256:0e7d03af5c7af07d3e7714193bd5d51bb707adabef48ef5c6d82ed7100c6176d`; measurement receipt `sha256:43ad4c5ed82a361d710e614767541948b7402e476eac6a6b773b321d8ee0ec09`; isolation `sha256:cb858f2adbe1786193d85075804549f15206839b715fad6d7a87296ed263bf6c`; custody `sha256:7605813ca3176eda76d4d8ce6a2512d1a68bb9f25cf017efef4d138523c4197f`.

50. Fracture and toughness

Claim identity: `SFT-MAT-MECH-FRACTURE-001`

Question and exact theorem. Fracture is connected bond-path separation creating new retained boundary support; toughness is the declared resistance relation for crack initiation or advance with a complete work ledger.

```
crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves crack-and-bond-network-carrier, advance-versus-resistance-relation, new-boundary-and-energy-ledger-retained and geometry-and-rate-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`crack-and-bond-network-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`advance-versus-resistance-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`organization`	`organization-collapsed`	`new-boundary-and-energy-ledger-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`geometry-and-rate-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a5733f9eb6f9903bd7d45570055f974ecaedde3f68a6a62ed63ebde4607095d2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6a5d0a7bb0a8e2f66788209faf7630ad1686264d0c376bc6aa1c3c9c934ce1e4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f27d462d208ffce1fd4f85a280d48e599aefab736bdf059196e25dc0714e5118`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6509c61f547b936474888b2417bcd4a5c72cfd3a0ae057432717257c1443a7e0`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:d6bc4e4588e8d5185f39670215ad2f0e681c18f43d64af3415da20f97af3e995`; certificate `sha256:f13fcb4efc42df89e9e318ffd78f831ed7652201114029d301e13d7ace8e69e6`; external-validation hash `sha256:a99696668e62b4f2d3dce17f8752af6a911915de7dffa7eb432db42293f12515`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:01b75c79f54df1e6372093d29ee1c45470f56ff393dbed86e238d8a2c2e91a74`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FATIGUE-FRACTURE-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture; snapshot `experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html`; `sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11`; scope: processing-structure-property-performance, defects, fatigue and fracture.
- `NIST-POLYMER-COMPOSITES-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/polymer-composites; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-composites.html`; `sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5`; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-mech-fracture-001-authority-correspondence: predicted crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived crack-and-bond-network-carrier__advance-versus-resistance-relation__new-boundary-and-energy-ledger-retained__geometry-and-rate-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered fracture and toughness observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves crack-and-bond-network-carrier, advance-versus-resistance-relation, new-boundary-and-energy-ledger-retained and geometry-and-rate-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

`sha256:bbc88ba37928e3ac9514e4b72395806d1d0a2cc60f604090a4fabd73f7b3bdbc`; derivation seal `sha256:8b5df3b57c3581998968ea6f3337fb3d1ade12a540eaae72c4dc0b7e3bf1a88b`; engine receipt `sha256:66cd7e1bb5e969fb2cf9920411459685ec4b19cce6ddc54415e5bf1e449497cf` at `receipts/engine/model_admitted/SFT-MAT-MECH-FRACTURE-001-66cd7e1bb5e969fb.json`; empirical validation `sha256:31ff5bbedcc5575ff326542e7720d80181ab36505299b0828dd244e41ab283b0`; measurement receipt `sha256:f5fcd156b6b0806c04a181af2e91fb90172cfe31ca081e8d9fb7b3f2e123eabb`; isolation `sha256:90c60d61e5bac21634efcd59ce8c04fb5e939f5118bbfad4c168a6032a6678e1`; custody `sha256:1f63d814bc73955153dabccf74d08cbe10b6657f10c76bb37a2872dff423ae46`.

51. Fatigue and creep

Claim identity: `SFT-MAT-MECH-FATIGUE-CREEP-001`

Question and exact theorem. Fatigue and creep are distinct history-dependent material processes: recurrence-counted damage under cyclic load and retained deformation under sustained load and time/temperature conditions.

```
history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-FLUID-PRESSURE-STRESS-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered mechanical grammar that preserves history-labelled-material-carrier, cyclic-or-sustained-transition-accumulation, damage-defect-and-time-provenance and load-temperature-time-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`history-labelled-material-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`cyclic-or-sustained-transition-accumulation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`damage-defect-and-time-provenance`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`load-temperature-time-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Mechanical laws keep reference state, direction, load path, deformation history, defects, crack boundaries and method geometry in the same auditable record.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:138e417ec271e2d16689255fb807b76d4e11a863a53f75a4be29e4664f40edaa`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f4a38be94309bdd54a09b8c098de0bf328e971aaf99abb5afda17dce0565aaa3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5964c78e400b2cccbe8f604adf24d54cb4afaadca4cf3524164fcc0b10c9811e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:96533e94cd35ecc95cede0615590e5f0915afcc9ae1e73b589bc9ad5feb6ff99`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:50718c3f7afe9f2f21ff7e49da32cb9694d31dceb34873026cf5ab117835fb04`; certificate `sha256:b9dc76d661417daa718f8f2c9614ab248424d3786dd7fb08cfcd03b9d7680ae6`; external-validation hash `sha256:22152c9fcd29cc509c7d11b289060e20c3b52f613b2a4721bd6678fd3f58976a`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:30bae96d9d87e88765e2e13ea6f741387b1141dca008910c29f8318ec151b536`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FATIGUE-FRACTURE-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture; snapshot `experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html`; `sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11`; scope: processing-structure-property-performance, defects, fatigue and fracture.
- `NIST-POLYMER-COMPOSITES-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/polymer-composites; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-composites.html`; `sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5`; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-mech-fatigue-creep-001-authority-correspondence: predicted history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived history-labelled-material-carrier__cyclic-or-sustained-transition-accumulation__damage-defect-and-time-provenance__load-temperature-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered fatigue and creep observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered mechanical grammar that preserves history-labelled-material-carrier, cyclic-or-sustained-transition-accumulation, damage-defect-and-time-provenance and load-temperature-time-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:6259261c2f433280ac5b338213ad30a639c165bf120e029689cd7cc6b0f61e4b`; derivation seal `sha256:161e752c1fbcc5111fa1dca42857a94f4d8918ae581e3e23f81e30df3059a745`; engine receipt `sha256:0533e9af7333170230d829ce66828f6bc1982defb9c8fb311cf01e53196f13e8` at `receipts/engine/model_admitted/SFT-MAT-MECH-FATIGUE-CREEP-001-0533e9af73331702.json`; empirical validation `sha256:b2728846e5e1df985715934a704a0d3e072542786645908f030c00d21547ca7a`; measurement receipt `sha256:645d29b7f2bb33fcad245033ea69bd2534dba99f4df7b9a74d6b153c77f1df80`; isolation `sha256:fc65a26867d6ef55044a2f9e33afc6460d9a90381f191f27c46fe0a05893a6f1`; custody `sha256:a2f03701274180403fafd17c220b8ad21dad330cab01ff5cb46bd025c103928a`.

17. Thermal Magnetic Optical

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

52. Heat capacity and excitation storage

Claim identity: `SFT-MAT-THERM-HEAT-CAPACITY-001`

Question and exact theorem. Material heat capacity is the exact relation between supplied thermal carrier change and the resulting accessible excitation-support change at declared conditions.

```
material-excitation-support__energy-to-state-count-change__condition-and-phase-r
etained__temperature-window-bounded__complete-state-transition-boundary-trace__r
oot-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves material-excitation-support, energy-to-state-count-change, condition-and-phase-retained and temperature-window-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`material-excitation-support`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`energy-to-state-count-change`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`condition-and-phase-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`temperature-window-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `material-excitation-support__energy-to-state-count-change__condition-and-phase-retained__temperature-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9b1c92f07930722df9def1146b48afb9053a331fe13649f8dbc811937f62bda5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f5a5463e58ef4db286448b3ccf1df2c4a0f2f2c5934e286ad3c5c109878ac038`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a6b3bf01252235db636981ba4ca9b966662f720f8b2f46ea7bfc07d9beb3bdc9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:09b10d18c04b2ebd6f0139afdc0e36b99291ceb2381d3b1485aa6dbff83644ee`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:c85a958591bc27b8bea1709497f13281fc169c363ae08e5f0d3c2d231bc34257`; certificate `sha256:3b8c30953cea0c023f330da888aa7ee92a4162a93e01cd00fa210aa29c8a15ad`; external-validation hash `sha256:609c83bbdd41d06080381f919f9aac4e58eb32bf48a496e68e78b41ed213737c`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:962425c0c7e89286623b46cb21a73a7bfd40a4e3549f825ec5937dd421beca5d`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TRANSPORT-THERMOELECTRIC-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials; snapshot `experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html`; `sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0`; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- sft-mat-therm-heat-capacity-001-authority-correspondence: predicted material-excitation-support__energy-to-state-count-change__condition-and-phase-retained__temperature-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-excitation-support__energy-to-state-count-change__condition-and-phase-retained__temperature-window-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered heat capacity and excitation storage observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves material-excitation-support, energy-to-state-count-change, condition-and-phase-retained and temperature-window-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:e42ae574ea210e58e7db4db2b4dea63b4c2a952a2c3003d1494486b831a51cce`; derivation seal `sha256:7abc09c00bae8245182e239280d27c4e0fcd5752a0f60ddf6adc1531bfebc332`; engine receipt `sha256:9a919059d9d03fcdc366ad97dd95eed602d6df50b8db4d49e3acb1d492dcf1d8` at `receipts/engine/model_admitted/SFT-MAT-THERM-HEAT-CAPACITY-001-9a919059d9d03fcd.json`; empirical validation `sha256:6e1bf64a4d135d263d2f6e1a0a7ce530df19e721915617a6ed4e4cc56060a319`; measurement receipt `sha256:5c348ebe8e6ee46f9ac074932fb49e4af0501d044da5826744228e4d7d8afc9e`; isolation `sha256:e3b1e98bbbb1fa295e5694748d0a260d5e9516397fdad151680d860c0d743008`; custody `sha256:cb2428cec3ca8bfac3b1cd7e5da284e0d4f80482444e02eb6f6b01292f22deac`.

53. Thermal conduction

Claim identity: `SFT-MAT-THERM-CONDUCTION-001`

Question and exact theorem. Thermal conduction is counted adjacent transfer of retained energy labels through connected material support with complete source, boundary and scattering provenance.

```
energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledge
r-retained__gradient-time-support-bounded__complete-state-transition-boundary-tr
ace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rul
e
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves energy-labelled-cell-network, adjacent-energy-transfer-count, total-energy-ledger-retained and gradient-time-support-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`energy-labelled-cell-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`adjacent-energy-transfer-count`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`total-energy-ledger-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`gradient-time-support-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledger-retained__gradient-time-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:19069236c3c4b3a7dfaaab7b12bbef6c542b7fa84585b15712bd64dabb87e580`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e830330106bc942c9651b68e4685a201d4266513d442e71e66fd2321a9463b3e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:14fe4fe4b473610bb862eb6beff2dfd07c036ad39ee6a12848a26d4e24a8af58`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7a817c1177ec039fa173c7e2f4b2458c3bc9a478948cec1e1e7b45539ae54486`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:e6979253f0608812554abf41f81ef0099f77c5cc4c214da12304a1b55437d891`; certificate `sha256:8e9fd289da0b25f75129f0b35c2ace40504a81b64a66962a0b25cfa6e7386d8e`; external-validation hash `sha256:31ba40a4f94cd387d02921b23d30d4c580558c96bdbfe0ef2e03fce22fcc583a`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:f0aa7c18ee1acf7349c5163388a2ca04b1c881e3486769bc42c642df190e9803`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TRANSPORT-THERMOELECTRIC-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials; snapshot `experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html`; `sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0`; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- sft-mat-therm-conduction-001-authority-correspondence: predicted energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledger-retained__gradient-time-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived energy-labelled-cell-network__adjacent-energy-transfer-count__total-energy-ledger-retained__gradient-time-support-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered thermal conduction observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves energy-labelled-cell-network, adjacent-energy-transfer-count, total-energy-ledger-retained and gradient-time-support-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:e897e584b3dd80bc4168883855f14430c86cd1a0d62bda4ad2814b4c6ebf8554`; derivation seal `sha256:1fdc4a1db8b717fa6573da8c1c79cf3d21145d4cc07736eb12ed7fd13e1e50c5`; engine receipt `sha256:1bc2b97d196c9750d41d454a0aa47fa7b36d142e108da02aeba0e7fed8134e33` at `receipts/engine/model_admitted/SFT-MAT-THERM-CONDUCTION-001-1bc2b97d196c9750.json`; empirical validation `sha256:e0c3e7778a79d6e1256434b13cb64fcc8c903bf2cadef8c62fc60890f47bd1d1`; measurement receipt `sha256:afa71e5c2000da0945d1c1cc12bcaa435c411f13ac286cfadcfb847b0cc71a2a`; isolation `sha256:e0ae991f27d27091ef5c4f2d234f76b510a90c4d02cd87451b0cc9a7af342af7`; custody `sha256:be4f7ae87af0abe604d3a65bbcb6ace9caf68fc7941833d5e2d5b4d44144779d`.

54. Thermal expansion

Claim identity: `SFT-MAT-THERM-EXPANSION-001`

Question and exact theorem. Thermal expansion is the condition-bounded change of recurrent constituent separation with excitation while material identity and complete adjacency provenance remain retained.

```
bond-separation-network__excitation-conditioned-separation-change__material-iden
tity-retained__direction-and-temperature-bounded__complete-state-transition-boun
dary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-ex
tra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bond-separation-network, excitation-conditioned-separation-change, material-identity-retained and direction-and-temperature-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`bond-separation-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`excitation-conditioned-separation-change`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`material-identity-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`direction-and-temperature-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `bond-separation-network__excitation-conditioned-separation-change__material-identity-retained__direction-and-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:0e53cee45fef353af8c1fc3a048db16697e59a4d1945eea47b3e9a9ae4ae9c0c.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:28f81c0bd83923a6393f7cf96abef9aa601ace13fbe9c2e13d8a95e762b6775e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:35b43f42468a1c1e771a8aa75e61ee96f3241e1249bfb8bfde861a74df429ab0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:016edca3decdf384605400b1fcfa0e140add94b17145adf5269e163906f9311f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:2430b2dfd7b2daa237185f98bc1c08842018b04eedfc432b4aca14a108bc9c25`; certificate `sha256:12cf1d124a03f9181174b282315f745c3854f4129d0540c209fd60966356acc3`; external-validation hash `sha256:f43b131aa11d8a1e83be60efe50ca5486c5336a52fb45526c6d32755c4bf3a72`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:51a71476779069d408c1bdc3413b8c879c8a17f2e40b35bbd6caca235d49ee36`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-therm-expansion-001-authority-correspondence: predicted bond-separation-network__excitation-conditioned-separation-change__material-identity-retained__direction-and-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bond-separation-network__excitation-conditioned-separation-change__material-identity-retained__direction-and-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered thermal expansion observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bond-separation-network, excitation-conditioned-separation-change, material-identity-retained and direction-and-temperature-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:d08e166df831402c873d497a2b6ebe99c4557b02f7a6686f6bbcf42d229df9c8`; derivation seal `sha256:0214f5934c9a967a9aa0ac4ddcde7068fc7560b1c26f9ea683d9c5f23a5cc25a`; engine receipt `sha256:e43221596aae9f61e140436942ee5dbb1625f0bfa5a520f34f217d780558cbdb` at `receipts/engine/model_admitted/SFT-MAT-THERM-EXPANSION-001-e43221596aae9f61.json`; empirical validation `sha256:be6cdeb54ff17afaa198f0414e8ea3404907e36e1b79a041e23f13f399cec260`; measurement receipt `sha256:b871d56ca42a2531ba7e55209ef6306482b6aeb06c6825d06320164217ab5345`; isolation `sha256:a299c268357a8d8afccf08b9285f38ec3440b0a33f6dc2497ec2a36ad2b1d44f`; custody `sha256:3f767b087bfbcc3375d204a69b2fba54d2c14a7282e893c1d63f95f35b76e021`.

55. Ferromagnetic order

Claim identity: `SFT-MAT-MAG-FERROMAGNETISM-001`

Question and exact theorem. Ferromagnetism is connected recurrence of aligned local held-moment orientations producing a retained macroscopic orientation class.

```
local-moment-network__aligned-shared-orientation-recurrence__connected-net-momen
t-retained__ordering-condition-bounded__complete-state-transition-boundary-trace
__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves local-moment-network, aligned-shared-orientation-recurrence, connected-net-moment-retained and ordering-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	local-moment-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	aligned-shared-orientation-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	connected-net-moment-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	ordering-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `local-moment-network__aligned-shared-orientation-recurrence__connected-net-moment-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:18c6e7d85f2e9146578d2645a8213fdbf355c75e79e05273dcfc9a89bdc6c8f7`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:09dd1edb95f74fb641872c23fef4e458b8b03d180b58c83bcf96284e7c2e84d9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:22995d5074cfb5ecda427c7205df471e65beb5216cf1f63a3073a4d46746ab5b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:290a1b4e2cb6f280907d51338d128896ce8ed6e11c6cd49d332f1bbbccb0340c`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:d0db4c43c76befd8e81f178bb7d60c32918904c030aec387054fd7cac0d279d3`; certificate `sha256:b6ccb098a72046791e3b63c766f6baafd7006b7b10f8a30543b374f3f2a1b383`; external-validation hash `sha256:dd32b24dd4b45270d8a255cfbbcbd691505465416487c457ea4a0d8c47875be8`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:8f81d89cf2d73a6809d674ceca2c208912e0f9998eaa5252837c8d3ce4fd2a3e`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-mag-ferromagnetism-001-authority-correspondence: predicted local-moment-network__aligned-shared-orientation-recurrence__connected-net-moment-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived local-moment-network__aligned-shared-orientation-recurrence__connected-net-moment-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered ferromagnetic order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves local-moment-network, aligned-shared-orientation-recurrence, connected-net-moment-retained and ordering-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:874d335c33a53a6df0256d9f5e149acc452d0743a4f57132fbbad96958d83b63`; derivation seal `sha256:ef659703b1b78755c811fadd47b2d4d01767e0be0db9aa4ed6357fb2e6738878`; engine receipt `sha256:8702b789cbad4cb166e5003ed111ac105b9908c3b921c7694876a7542a47c9ab` at `receipts/engine/model_admitted/SFT-MAT-MAG-FERROMAGNETISM-001-8702b789cbad4cb1.json`; empirical validation `sha256:81ddf6d5cd8f35488a8fd9a766f2aa59bd53d2cfce1aef09c382fea4d32521be`; measurement receipt `sha256:711bb821fc4020f451a199fa3d1e50db79292aec3b2096e80b7f84bc7be5ba0c`; isolation `sha256:706c62f32b92289aebeaf28bb5c38f54efd56a53938d566f43efca10e89f4062`; custody `sha256:e06f5ee0a0adc7afe327d4983e262f49654c4374c76cc69517f35ffb41a20a5f`.

56. Antiferromagnetic order

Claim identity: `SFT-MAT-MAG-ANTIFERROMAGNETISM-001`

Question and exact theorem. Antiferromagnetism is stable opposed recurrence on distinguishable sublattices that retains local order while the complete macroscopic moment distinction closes.

```
bipartite-moment-network__opposed-sublattice-recurrence__local-order-global-clos
ure-retained__ordering-condition-bounded__complete-state-transition-boundary-tra
ce__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bipartite-moment-network, opposed-sublattice-recurrence, local-order-global-closure-retained and ordering-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	bipartite-moment-network	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	opposed-sublattice-recurrence	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	local-order-global-closure-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	ordering-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `bipartite-moment-network__opposed-sublattice-recurrence__local-order-global-closure-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3a97cd3e05edbee53ee54b4478bab16cf055ce518e721868e07f6c11502ba8cd`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d91fbf826b29dfc56d95eaf525c93c5a097a496f66fd54bdc07270e4d97c1709`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:21fc3784752313c95d471cc7f4c37eba8dc481651af93a44b97ae34e2574b7f9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:00d0d75dd210b3b851ee3af7c6b1b5e9382df37025806f700f9e3c4a914ab1b2`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:554a049abd839fa53066a0d078721f9d32d1e2da2ab50d6bad079166de236149`; certificate `sha256:5bcf84137ee43fbe7eaa1e6c5cb69622b45205f73a6e2c53f144f0c95bcab8ce`; external-validation hash `sha256:a46609ffce4544872f8f33cad521d02fd03fb08c46966d098291c136698c7eca`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `3` claim-specific source discriminators were required; their ordered identity is `sha256:e52715234c5d0275b83377bae9ec6c31e571fc23fe49d47cd26aac633767439c`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-ANTIFERROMAGNETIC-COUPLING-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/ncnr/acns-2020-tutorial-ii-practical-approach-fitting-neutron-reflectometry-data/understanding; snapshot `experiments/external_sources/materials/snapshots/nist-antiferromagnetic-coupling.html`; `sha256:b8ce5105622c02b84bffe643693c05c199df57beae79bd9c5e29f26c4360e85e`; scope: polarized-neutron measurement of magnetic structure and antiferromagnetic coupling.

Comparison and custody records:

- sft-mat-mag-antiferromagnetism-001-authority-correspondence: predicted bipartite-moment-network__opposed-sublattice-recurrence__local-order-global-closure-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bipartite-moment-network__opposed-sublattice-recurrence__local-order-global-closure-retained__ordering-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered antiferromagnetic order observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bipartite-moment-network, opposed-sublattice-recurrence, local-order-global-closure-retained and ordering-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:3df466cadcac208382ade4c4777ac52b418e0c3e01286abe21cd23a00899994d`; derivation seal sha256:66b2d9f5998a96ae9e7a4b0f607b4b3701336de1c18415dd2cfa78eaff6de448; engine receipt sha256:4e024776e6572a09e72fd47738e804b731ccffff0685efb0993023aecafc15c7 at receipts/engine/model_admitted/SFT-MAT-MAG-ANTIFERROMAGNETISM-001-4e024776e6572a09.json; empirical validation sha256:9c06b2a0f59c2933efdaa85f3a3f7107cdfb23da035952b54ebbb1571c31b065; measurement receipt sha256:1144769f267e47cca17f0dc4ddd226220357394daaece0f0da693ac5622311f1; isolation sha256:324097c4049c4005726db64e2a93937542c96b9be63d8c95bea908389c63d4e8; custody sha256:c52d305b11f8dbf80db49ca64d8ce900ce41725947b43d8721e896ad58ea6163.

57. Dielectric polarization

Claim identity: SFT-MAT-DIEL-POLARIZATION-001

Question and exact theorem. Dielectric polarization is a retained field-conditioned redistribution or orientation of bound held-charge labels with complete conservation and relaxation records.

```
bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-RESPONSE-001
- SFT-PHYS-THERMO-KINETIC-TRANSPORT-001
- SFT-PHYS-WAVE-POLARIZATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bound-charge-orientation-carrier, applied-field-displacement-relation, charge-conservation-and-relaxation-retained and frequency-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`bound-charge-orientation-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`applied-field-displacement-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`charge-conservation-and-relaxation-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`frequency-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ab4c971eddd4a0592c52799f300cbb68ccdd19e0d17929704c030e932e3147ef`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f811d9b97206cb817d188a47fc41c6ee4857a456316be6a1d8a144544071bcd3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e5b70b12ddb119fd8aab8971c8adb87e5830aadd0fa9b644bfc648aa356cbdf8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:419fd084389f321aa44b601dd00e318fb0647694ec700f5298800692c0e802c1`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:00a9ba701d96966b56fd31c7b4d206d442ec6d652fd7453700c985196d1c92b6`; certificate `sha256:59d7757274ab8ecbe48f4c00bc9b612f3d32ff9fa583f3cd2803d0fed7eb7482`; external-validation hash `sha256:53450986c1b2ebc93c03ac06621f5d9b2598b3048bfeec6730e0dda3f02dcadf`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:a96625233799c6676585da9b9389b00e18e4f90a9ec7b3b86d06d2b1c40ced6b`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-diel-polarization-001-authority-correspondence: predicted bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived bound-charge-orientation-carrier__applied-field-displacement-relation__charge-conservation-and-relaxation-retained__frequency-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered dielectric polarization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves bound-charge-orientation-carrier, applied-field-displacement-relation, charge-conservation-and-relaxation-retained and frequency-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:1498859c348dca8e370798bb01a8832aea7e4e8d376b158d788bd6156378dc0e`; derivation seal `sha256:0a118061761c83630046d49ac13b24f8877554e4f646101aa11e5dba5b8f1420`; engine receipt `sha256:f432cb7b60b503a89a22333e6942725b659572fd12da81a606267858f3ac637d` at `receipts/engine/model_admitted/SFT-MAT-DIEL-POLARIZATION-001-f432cb7b60b503a8.json`; empirical validation `sha256:606adb7ecb391af5f1f37d0122a6ca840f6a4b532ea334cd0fb1c4205cadc2aa`; measurement receipt `sha256:e912dcb76211979f7aca50f6f3348c37518499fc73f6cd86ec3c8aadc480477c`; isolation `sha256:3abe9bbad740c8ab4999df5dbec74fe762361aee816a6c135941212a29e44b2c`; custody `sha256:8344d6447b2b2cb16be34b7690434539f57962edbde04bd804c4b811d17e8539`.

58. Optical propagation and refractive response

Claim identity: `SFT-MAT-OPT-REFRACTIVE-001`

Question and exact theorem. Material refractive response is the source-bound phase/delay relation produced by coherent probe coupling to accessible material excitations while propagation, loss and dispersion records remain separate.

```
probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__fre
quency-phase-and-loss-retained__spectral-condition-bounded__complete-state-trans
ition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-clos
ure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves probe-and-material-excitation-carrier, coherent-delay-and-transfer-relation, frequency-phase-and-loss-retained and spectral-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`probe-and-material-excitation-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`coherent-delay-and-transfer-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`frequency-phase-and-loss-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`spectral-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__frequency-phase-and-loss-retained__spectral-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:9c630b0915ecb27ecd4a8e9e3a7467474e98d990d0618c7a85c74363f19b6bc7.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9ae669c1738af532adaf47a7e7ad4f3d7043304c6043f5d358a4a2940af80005.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:73d971daac95bf3c58df6f424c3181e678ef094a14cfca717ca19c769d266212.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:b5e103cfd88598d9b7655978391197ab5224bcc9d005792573d8a058c49d84c4.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:65dc955c5bc0b935e97be9bc2733c36b012a32a1a1e56f9c7c9d67b631ab63ef; certificate sha256:191167b5a78855912ddc9e723c619c26dce388f41e3e0573f98e36c081b53714; external-validation hash sha256:7338f1b3d4505446488721394e707d1a2d078e07f0c0fd0dcf8a817f8ba2905d.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:d8b7289d06e63b982d20adbe30509ea26d9351d73e84dbb6f49daa777954288e. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/theory-optical-properties-materials; snapshot experiments/external_sources/materials/snapshots/nist-optical-materials.html; sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- sft-mat-opt-refractive-001-authority-correspondence: predicted probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__frequency-phase-and-loss-retained__spectral-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived probe-and-material-excitation-carrier__coherent-delay-and-transfer-relation__frequency-phase-and-loss-retained__spectral-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered optical propagation and refractive response observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves probe-and-material-excitation-carrier, coherent-delay-and-transfer-relation, frequency-phase-and-loss-retained and spectral-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:6338ee48ea6cf26874c7b32eb792944b85ed28d91cbcf78412f4f07738f8a2f3`; derivation seal `sha256:1b93339c2b96508f0faecd14fa3af55d3369e93d5a2cf3410c3b8cffecf3243a`; engine receipt `sha256:82335b4d53a29fa158d4484e54572994b56b5dda246c8654034cbdc01f485d99` at `receipts/engine/model_admitted/SFT-MAT-OPT-REFRACTIVE-001-82335b4d53a29fa1.json`; empirical validation `sha256:a09d090deef51116d29b164006bf17e5a792eac1e4af65e9ca63528a3ab350e1`; measurement receipt `sha256:f2305f1463ef3636441aa52e0a33cc956f137b10e879102d6f956a3ca8c464ec`; isolation `sha256:029f56e5c4af77cb4977858abb7ad0a734f216f21e388c340a7c094da910c654`; custody `sha256:ddcc328651656989f385b50503615b950a835d1186cc1970f2953b7db45a12e7`.

59. Material phase transition

Claim identity: `SFT-MAT-PHASE-TRANSITION-001`

Question and exact theorem. A material phase transition is a provenance-retaining change between stable macro-recurrence and symmetry classes with explicit conditions, path and hysteresis boundary.

```
initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-THERMO-RESPONSE-001`
- `SFT-PHYS-THERMO-KINETIC-TRANSPORT-001`
- `SFT-PHYS-WAVE-POLARIZATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves initial-final-phase-carrier, stable-macro-fibre-change, symmetry-content-and-path-ledger and condition-and-hysteresis-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	initial-final-phase-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	stable-macro-fibre-change	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	symmetry-content-and-path-ledger	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	condition-and-hysteresis-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Thermal, magnetic, dielectric and optical response are condition-bounded transition relations with complete carrier, phase, loss and history ledgers.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:699bb0538a254745ef67cc361ac8f4844e8c1fa2538837ac5938e52053255541`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3192cf970f56809f280e80c2c9820e5498273e7e42e1999057f39d8970014427`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e183813b92b1fe15c3e6eccee915107ea2ab934872dcf80d50f64eea683505b9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:406f3613f24df373d9f299a2659d9ad0e2bcf07a282d0ba2cebccb1230a0620b`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:a5971eb324e6f0a7959b7a16ebaee120a132038c909c3c221b6c9a779eef1c7c`; certificate `sha256:c7ca23959f208aede92422515466358ab2b11bb5ecbefd73d808513ca2d5dc7a`; external-validation hash `sha256:cafe572aaadaa8b0658384852fee24d1d662945aa2b0d6c7931694fe40e7de38`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `4` claim-specific source discriminators were required; their ordered identity is `sha256:e20ae266b0ec1dd474ba078d282cf54d525ff318993fa9787c5b9fd426c2c8a6`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.
- `NIST-SUPERCONDUCTING-FLUX-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting; snapshot `experiments/external_sources/materials/snapshots/nist-superconducting-flux.html`; `sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6`; scope: zero resistance, Meissner expulsion, quantized flux and vortices.

Comparison and custody records:

- sft-mat-phase-transition-001-authority-correspondence: predicted initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived initial-final-phase-carrier__stable-macro-fibre-change__symmetry-content-and-path-ledger__condition-and-hysteresis-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material phase transition observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed

- Every positive finite generated Materials carrier in the registered thermal_magnetic_optical grammar that preserves initial-final-phase-carrier, stable-macro-fibre-change, symmetry-content-and-path-ledger and condition-and-hysteresis-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:0297cd0782de90adb0d271e4db03d88413a6636799f16e3fbdcfc3b5d84617c4`; derivation seal `sha256:c219d7cd41090ebb065ba94181b6f6cdea11e507feac6b44e0424b3af3e26e8b`; engine receipt `sha256:c40ad4320e96682f79e7b338df47728fd53e8df1455276c8a6673294e60ea745` at `receipts/engine/model_admitted/SFT-MAT-PHASE-TRANSITION-001-c40ad4320e96682f.json`; empirical validation `sha256:5cefdebc7a29e14018d3791b7023149b6671cfa8c81121bee0a4e4667f743cd9`; measurement receipt `sha256:f97aa41aceeec4f79873517d7940d2804059220eee142435cde997ab2b1fe951`; isolation `sha256:fca5b15e153357ae3fcc8f912e9b78e0da3ec9e98cfeb1944b5528738c27536c`; custody `sha256:fbfbbaa82bf8ce93b146cb1a97a3e85e00bc21ab4c4f20e2516ee0ccdf94433a`.

18. Material Classes Bulk

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

60. Metallic material organization

Claim identity: `SFT-MAT-CLASS-METAL-001`

Question and exact theorem. A metallic material is a connected atomic organization with delocalized carrier access across multiple constituent cells and no declared transport gap at its operating conditions.

```
connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-atomic-network, delocalized-carrier-access, translation-and-bonding-retained and condition-bounded-metal-class.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`connected-atomic-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`delocalized-carrier-access`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`translation-and-bonding-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`condition-bounded-metal-class`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:75257e1c7d9eaa9bff969b6d2330e1265d2a0f9bba87c9aece8e402510ab3988`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1b001bf09b7d2091d3d1b8bbeb596db998ff93fd4e7203badeebe99ab867c5b4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c7b74d8a2285a0aab500e235bd3898e00a6c65155c152646c59586906e97f402`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:cc6ff66be0902ac0ba61c87beee88d6283725c77049ca60aecc1a9358568fb3b`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:bb4ef02ce0fb08afdd526d8b4a3fc086373575480832e1d9cfb5b462bb5f6bff`; certificate `sha256:2458dd1b4f8560378cf749959b1476d3cc8b0e8a8e003582c2fbb109136fbb6e`; external-validation hash `sha256:61ea3121a344907c6519355dde4ec7510dcf6362f3a0fda5897da2741b33cb77`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:87fd1a3ef3fb7ef8871ddb7ecd0d074dda38732b6f59dc8a0acfa4dafb814bd1`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SOLIDIFICATION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/publications/solidification-0; snapshot `experiments/external_sources/materials/snapshots/nist-solidification.html`; `sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c`; scope: solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- sft-mat-class-metal-001-authority-correspondence: predicted connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived connected-atomic-network__delocalized-carrier-access__translation-and-bonding-retained__condition-bounded-metal-class__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered metallic material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-atomic-network, delocalized-carrier-access, translation-and-bonding-retained and condition-bounded-metal-class.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:5f02bd1b0f32f5693c1791781165d3bcb38fc8c49ff0ec01254dc518ced91df6`; derivation seal `sha256:af475c9901883694ec27d4a4693e3ca1eb56373d75c42e6c0b073c66e977bced`; engine receipt `sha256:ff88492d77ed22ed6106d1f981e4d70142431b1bfe9aeaa7cc5a6f26966951e2` at `receipts/engine/model_admitted/SFT-MAT-CLASS-METAL-001-ff88492d77ed22ed.json`; empirical validation `sha256:c5422a5c61e616c84db792a559545f816ea6a3d6dfee8456af9a560e2d6c8588`; measurement receipt `sha256:84f09dc3af6ef8df70c69226be6eec901246c4e203c597a51b981db60890fc1f`; isolation `sha256:3ec6b3346deb1532d82db336b639a7be66c91b7b1c4b4146bcf4d40108b37540`; custody `sha256:eb8356b4e42e5e256d654f54b378fdeed0f9fb5059c09c23e394e2e3d4102391`.

61. Ceramic material organization

Claim identity: `SFT-MAT-CLASS-CERAMIC-001`

Question and exact theorem. A ceramic is an inorganic nonmetallic material whose bulk carrier is a localized ionic/covalent network with retained phase, porosity and grain/interface organization.

```
inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves inorganic-network-carrier, localized-ionic-covalent-joining, phase-and-grain-structure-retained and processing-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`inorganic-network-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`localized-ionic-covalent-joining`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`phase-and-grain-structure-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`processing-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:76deb008a987f16ea7cf2d23813da1ce5b61f9665f182459ad93958acd125655`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:62d814937e216c438b3031aa1287c27a1c099fc17fb38a779a32a4c0abda3867`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:52134d6b9c5c728526c88b12b66102baf449f59f6b8deab50eda64a5d8430d18`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:b1df93ebe9661b7a1504c2c444af7ff8ce24fd932d309719a4a077d0effcf610`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:48db99ea691a8154b6bd4b946ecd20cd7d61481745e9e73e611241b91eb35d41`; certificate `sha256:9872a8af3e2b80be8d45b13f6f0d4175c23baf4f58eacdedd1afdee08f3fa44e`; external-validation hash `sha256:fdb0f31db398bd99134453418f3830811ffc0ef4924dd1e4b3e2d9d5d24cb566`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:01eb6862c745c23944528b05975766210e00b897c0ef858c5cccb000421b24cf`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-CERAMIC-ADDITIVE-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-ceramics; snapshot `experiments/external_sources/materials/snapshots/nist-ceramic-additive.html`; `sha256:540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9`; scope: ceramic feedstock, defects, sintering, densification, phases and microstructure.

Comparison and custody records:

- sft-mat-class-ceramic-001-authority-correspondence: predicted inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived inorganic-network-carrier__localized-ionic-covalent-joining__phase-and-grain-structure-retained__processing-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered ceramic material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves inorganic-network-carrier, localized-ionic-covalent-joining, phase-and-grain-structure-retained and processing-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:36cccb1dd2e5e1628e3512f5fa6df179b27bb82e1499c77dcc1baca99f0d6238`; derivation seal `sha256:369e843e6bc60b72015a21594789a04f78935ae133eed27afa02dad43c23c0ea`; engine receipt `sha256:c8d2a975402210330273a38078b9cb98953aa151f947aef809e2a6877aa82749` at `receipts/engine/model_admitted/SFT-MAT-CLASS-CERAMIC-001-c8d2a97540221033.json`; empirical validation `sha256:05b56e875a7c653dbdeb13e30878e702946a5d6c85e3e47da657abf31287437e`; measurement receipt `sha256:d5391cef95ca7f7c7ab19f64f4b6eab3ddcd8ade6de2050843457b42c1f1bb85`; isolation `sha256:34ae11265cd7a19fde0a5f163f473131a9d88c056516a17c01e0a67aa9493a6f`; custody `sha256:c65e9e11483fa4575b83fe65d209fea3569ede6cbad5ccf6048ed0aed39d9905`.

62. Polymeric material organization

Claim identity: `SFT-MAT-CLASS-POLYMER-001`

Question and exact theorem. A polymeric material is a population of covalently recurrent chain carriers with explicit repeat identity, sequence, length/branching distribution and interchain organization.

```
repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves repeated-unit-chain-carrier, covalent-backbone-recurrence, sequence-length-branching-and-entanglement and distribution-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`repeated-unit-chain-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`covalent-backbone-recurrence`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`sequence-length-branching-and-entanglement`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`distribution-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:19eea086be5305ab4d9ffbe6bbf3f5a9286aec315bf29b7d4403a02f9aeb76c3`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7e11f2ef9579ffc07192430aeb6c40fa225c2ac06d7e114724ecc9219834280a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:11aaf3801d8105f70582cb88872a4c36298334d504540dafcbb43a38041c1ed7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:881a7d9a75b566364a003b319abaeca54bdbf2cd3263ffbd9ab1aeb7813f0df7`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:c1061dd2fce337b40a2e45f157414d832614f3037b6f908a09deb0ef7f1e04d4`; certificate `sha256:e2dc53b388ebdf459e8929547810cf4c3d778209fea737eb1bf27af6795d1b08`; external-validation hash `sha256:7c8e5ef81f03d00e04ef56779653dd0d5cffed973bca994f80b6ae1c9ed47974`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:b4d03886dc24480e527ac4522ef8eb3213f4940617f016588852655c95ef6d8a`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POLYMER-PROCESSING-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/polymer-advanced-manufacturing-and-rheology; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-processing.html`; `sha256:11279429040fb5daff9108435f0c510fc12d89e09710e799b1a6182f63a63a18`; scope: polymer processing, crystallization, rheology, kinetics and recycling.

Comparison and custody records:

- sft-mat-class-polymer-001-authority-correspondence: predicted repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived repeated-unit-chain-carrier__covalent-backbone-recurrence__sequence-length-branching-and-entanglement__distribution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered polymeric material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves repeated-unit-chain-carrier, covalent-backbone-recurrence, sequence-length-branching-and-entanglement and distribution-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:6a1b61cf7b27fd57917e294a49338c0d42f55298366417e4c1c0e6ee8680e27e`; derivation seal `sha256:f3bf4d70c0a28bed53287d8770575102bf8e2585e68ae7310e414b9dc218ef3c`; engine receipt `sha256:a99eafa3c8ace458d61210eebad617b1826e865d7025839889d21e69e28f6ff6` at `receipts/engine/model_admitted/SFT-MAT-CLASS-POLYMER-001-a99eafa3c8ace458.json`; empirical validation `sha256:5fd119728138b9711ab1e4dc0020f406fcd5489ae5d9dda98813de8936b3983e`; measurement receipt `sha256:cd90b19a87eb545697cc05bd0370d1b1ac3f616e0baa870bf5a41fe44481e356`; isolation `sha256:11825e09dd0ac943dd33d6d22bdbbd9ff7ba42a8c0c68a92d5c72c8e5ccd07e6`; custody `sha256:ffa4c921c9ac95448b5471c3dfe8126bbce6d40721a51c4ba3744e339a384af7`.

63. Composite material organization

Claim identity: `SFT-MAT-CLASS-COMPOSITE-001`

Question and exact theorem. A composite is a joint material whose distinguishable constituent carriers persist while interfaces and architecture compose their load, transport or barrier responses.

```
multiple-constituent-carriers__interface-mediated-load-transfer__constituent-ide
ntities-retained__architecture-and-scale-bounded__complete-state-transition-boun
dary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-ex
tra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves multiple-constituent-carriers, interface-mediated-load-transfer, constituent-identities-retained and architecture-and-scale-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`multiple-constituent-carriers`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`interface-mediated-load-transfer`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`constituent-identities-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`architecture-and-scale-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `multiple-constituent-carriers__interface-mediated-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c6c46419af214daaf146350e487c95de5bc30d082ae54dad12b5579ad28b1633`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ecccf75b15e102f438af9f7c385a5cc558ed848ca7c8313848451f843f4856a9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ce4a29905dae4aa34196f4664e2f919f02c6b3eab2d8cd11401b744b38a79e1b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e02674fffcbbf54bd9a50f41cef4217f5351d525fb98fc7931dae2a22496d021`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:8d4c7ec3138919dd7154707c09abc73b7d0fcecefb5de1ef5330929a19b6ef56`; certificate `sha256:e0737335b366ab6f6be9f854b63f51a38c01a6727f89f1e92a8958ab93e87def`; external-validation hash `sha256:ab91a5e87b1f7ead61352f386d150be5fec82f2d722d41a666c2f3e18d595300`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:99c767633d82ee90ffa49c7f507f599fa6fd42c297c565e2ce1dc9e4a339472c`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POLYMER-COMPOSITES-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/polymer-composites; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-composites.html`; `sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5`; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-class-composite-001-authority-correspondence: predicted multiple-constituent-carriers__interface-mediated-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived multiple-constituent-carriers__interface-mediated-load-transfer__constituent-identities-retained__architecture-and-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered composite material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves multiple-constituent-carriers, interface-mediated-load-transfer, constituent-identities-retained and architecture-and-scale-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:0c4d6765e00200da4b3797781c97c267cd941b131622a573de77c323c27824ab`; derivation seal `sha256:c8d209ef1e580b1bae19f6f4a3c8c816ecac58ba3747d66810463b448cf61240`; engine receipt `sha256:e068e79f1b5490cd352d10a8669b04741ce9fedbf15e89115fbfa514487b6983` at `receipts/engine/model_admitted/SFT-MAT-CLASS-COMPOSITE-001-e068e79f1b5490cd.json`; empirical validation `sha256:a8e4192ff987d1ec55606bf440fe850195cf076253bef45c4287a44b7dd1ceec`; measurement receipt `sha256:04a5264efc4594b50729f4788e24b5b702585fd07b9127b6840eb7df565b6d49`; isolation `sha256:1451cd65a42e6b1336107ce0653b3d3a2c9742c625bcc8d8af0e498b61c5cfc0`; custody `sha256:b55a8ef809423e1b68ca5143fc757b398f522cd4b4ec94fdad9342c02fa767b5`.

64. Glass and amorphous organization

Claim identity: `SFT-MAT-CLASS-GLASS-001`

Question and exact theorem. A glass is a connected material with retained local structural recurrence but no crystalline translation class, whose macrostate depends on preparation and observation-time relaxation.

```
connected-nonperiodic-network__local-order-without-translation__history-and-rela
xation-retained__observation-time-bounded__complete-state-transition-boundary-tr
ace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rul
e
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-nonperiodic-network, local-order-without-translation, history-and-relaxation-retained and observation-time-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`connected-nonperiodic-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`local-order-without-translation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`history-and-relaxation-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`observation-time-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c8117ff6f4fb77e532becbb51883f31b96b64fc2032238b47ee0761bfd73ce3c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:48ccd96f2cef517058dcc908ea88079c276754135e112e1ce69ebf05da1070a2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:dc7537443e2a29e6ec64d48fdd218123e16753cbd276ee7cbcdb8adee543f54a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e4c5829e88d7ee7cf3bd6e46e6305425f6c0aae67dfa329036fbe4f8ab143be4`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:7e5fd03d9ff4c3789544066770b9bce294e351f8cc31f47b82515957acbebd5e`; certificate `sha256:633a1eca778acd3c503f297e2512b44ae93ce6dde346cdcb88dde7ef694d8f51`; external-validation hash `sha256:bcd2bad747a845796be107bb29cfd82be8b8a5f595131d1dadaaf578261e267d`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `3` claim-specific source discriminators were required; their ordered identity is `sha256:bca2ffd249200181edd6ffefb138fc5f42e3029ab07afc7b74f34735ef7970d2`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-GLASS-TRANSITION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/publications/glass-transition-its-measurement-and-underlying-physics; snapshot `experiments/external_sources/materials/snapshots/nist-glass-transition.html`; `sha256:2d6b99e46aae9d3c6a503494eedd17db45d4b48a633bb254cd3a6a5195440777`; scope: thermodynamic and kinetic glass-transition measurement in glass-forming materials.

Comparison and custody records:

- sft-mat-class-glass-001-authority-correspondence: predicted connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived connected-nonperiodic-network__local-order-without-translation__history-and-relaxation-retained__observation-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered glass and amorphous organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves connected-nonperiodic-network, local-order-without-translation, history-and-relaxation-retained and observation-time-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:ead873ba4e541f4086b9cf10d55335b648a49936c705d212d34a67042618ba67`; derivation seal `sha256:20b63fcf87d6da5eff04eb20dda0fe29cf66a0f8502357ec603f104e95066877`; engine receipt `sha256:48121416b6e96a7ca43dfe1cd3cd81cb6122dca865c2d5757249708911c2d8bb` at `receipts/engine/model_admitted/SFT-MAT-CLASS-GLASS-001-48121416b6e96a7c.json`; empirical validation `sha256:82fa9de65ba0566f78de7820f4837b4fa446ff6d8564270c79fd9abe6b2d82b6`; measurement receipt `sha256:963696e40f2dd738d98875acbeb98ee72b964edfd075b4ea88f9ca04de50fa1a`; isolation `sha256:2fb7bfab1a75e763ff1da7ffbc65554ff91371b787f01d199b07624b7c7f2dac`; custody `sha256:5b5891f1bf93f5e9ce4ef68c1d14aafbd3e29547eebf393d707bd525b16a298d`.

65. Porous material organization

Claim identity: `SFT-MAT-CLASS-POROUS-001`

Question and exact theorem. A porous material is a joint solid/structural-absence support whose pore connectivity, interface and accessibility are retained at a declared resolution.

```
solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-top
ology-retained__resolution-and-access-bounded__complete-state-transition-boundar
y-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra
-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves solid-and-empty-support-carrier, connected-phase-and-pore-network, interface-topology-retained and resolution-and-access-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`solid-and-empty-support-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`connected-phase-and-pore-network`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`interface-topology-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`resolution-and-access-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:86474f877d49101e7e3ace7b1ea74f2d9a093c01fe7680f3b1271e4489adc258`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:aa09a3db065810ef015d5e9c1c6c88deee9b27a1f27fe6dbf4b0cf92e5801cc6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:de7c2b045d20ac78f59d8ec13cc4ff932cc6c3c6f7cd7c236358dbf0836e8d40`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:79a409094422197b89e8c3e50fdf69c57e508575d8db62334202a4d2f429b0c0`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:9df78b11cbc6e6c95b2d9a9e0a9f99e1c79ff7e0c979633cebef497f96e33a74`; certificate `sha256:a3e148d5b2218edb7365c024ec306f7d09058931911d8cc7a88ee26c66aa1af0`; external-validation hash `sha256:7d669197ff1fddc9a4452d46a59fde18bc7e56e1d41e894b145b83fd6603ac8c`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:13583e6afba8522a3fd7dd9d3c8fd22796763215d55db5dfbd11a552b2b30528`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-class-porous-001-authority-correspondence: predicted solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived solid-and-empty-support-carrier__connected-phase-and-pore-network__interface-topology-retained__resolution-and-access-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered porous material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary

- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves solid-and-empty-support-carrier, connected-phase-and-pore-network, interface-topology-retained and resolution-and-access-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:69f8356c8af5397580d2a11cf32c7e72fd277e6e8bed3270680e722abc3cefbe`; derivation seal `sha256:c54ad9ff0af549841d8deaf72c13a3262cecde1f9c6babf627c99aa57579d918`; engine receipt `sha256:38df22075765cbfdde541b7c7c5909a5657a9a5d88c17badc55cf9e1691d896f` at `receipts/engine/model_admitted/SFT-MAT-CLASS-POROUS-001-38df22075765cbfd.json`; empirical validation `sha256:450e94e6f317469bb6b703f57926d750ce630510f52b2625b57fb1b86b746636`; measurement receipt `sha256:7ba33b9a0f10949d4026a021623572caf081489251cb9fd806df2a9ca191e427`; isolation `sha256:21dd1f085996b65e7f1dd705fab7ba3a6e648ad01ebaa1cfb389d1ecdf6e2366`; custody `sha256:6b400b1a8dfc8c669425cc3c4d17624d8621bf1ca1828ad820b17ab91ea2d375`.

66. Molecular-to-bulk correspondence

Claim identity: `SFT-MAT-BULK-CORRESPONDENCE-001`

Question and exact theorem. A bulk material law is admitted only when the complete constituent/interaction network composes to a scale-stable observation while fluctuations, interfaces and preparation history remain explicitly bounded.

```
complete-constituent-ensemble__local-relation-to-network-composition__fluctuatio
n-interface-and-history-retained__representative-scale-bounded__complete-state-t
ransition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-
closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves complete-constituent-ensemble, local-relation-to-network-composition, fluctuation-interface-and-history-retained and representative-scale-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`complete-constituent-ensemble`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`local-relation-to-network-composition`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`fluctuation-interface-and-history-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`representative-scale-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `complete-constituent-ensemble__local-relation-to-network-composition__fluctuation-interface-and-history-retained__representative-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8aaa2be2cb1aaf75093de967bfa9bb883f60e3bfb38d10c1501e9e67043ca4e0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:af1ae25c467d916ea85f9b6610c5ba82923fef5eece84c75ea61a4f185806e16`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1aa726163603796dc22e356d791678eb295c11adc115abd3c4435a34c6960e8e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:64c386dbbe1c3aa910b83a1516e7b297cbebdb40d4ca072c8985bbf628c260b1`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:2730e1897d8309a7942b0a6250869b93142c515f1222b012fab05c66195db660`; certificate `sha256:dc8f66c5f350939ed51ffd3201e28c851664e6533f6eefa0bb451050bad8ee9b`; external-validation hash `sha256:d0944a5281e0ad757098205af86c358371b8c15f418d733c9961798ef1fafa56`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:95c6dd46934ae777e83fddc8889a1df0d2b9d0e0f586a7c703736741414c412f`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-POLYMER-COMPOSITES-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/polymer-composites; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-composites.html`; `sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5`; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.
- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-bulk-correspondence-001-authority-correspondence: predicted complete-constituent-ensemble__local-relation-to-network-composition__fluctuation-interface-and-history-retained__representative-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived complete-constituent-ensemble__local-relation-to-network-composition__fluctuation-interface-and-history-retained__representative-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered molecular-to-bulk correspondence observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves complete-constituent-ensemble, local-relation-to-network-composition, fluctuation-interface-and-history-retained and representative-scale-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:b7c45e293722535b1fee18b96470c7d74e777249add1635ab885cfb643012bce`; derivation seal `sha256:18006e7469ea4d53ae3e2432398c8677dce5f5cd10dd2be3fa2658c714da22ad`; engine receipt `sha256:447912e5ddb8853c1ee41231f0d106962ca3940a118a3fd420384a77f06509b4` at `receipts/engine/model_admitted/SFT-MAT-BULK-CORRESPONDENCE-001-447912e5ddb8853c.json`; empirical validation `sha256:37e602662e5ef301a345ca2dcaa89ced48957308154af015e5d5a8e23fb71221`; measurement receipt `sha256:3844092558e29c9ff4835b0c2c4df9bc85fdb6b9418f73032b67837012d39920`; isolation `sha256:3953777fe17d106465d44bfe5c9158304296a968e2186bf3fbe0ba5891ce252d`; custody `sha256:feadd5dd045f6ba5e2e5f2c2688ab882456e1efeb3630d53f2e0ca0ab2066b0b`.

67. Material anisotropy

Claim identity: `SFT-MAT-BULK-ANISOTROPY-001`

Question and exact theorem. Anisotropy is inequality of response relations across retained material directions after specimen, condition and method identities are held fixed.

```
direction-labelled-structure-carrier__orientation-dependent-response__same-mater
ial-identity-retained__coordinate-and-symmetry-bounded__complete-state-transitio
n-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure_
_no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`

- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-PHYS-THERMO-MICRO-MACRO-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves direction-labelled-structure-carrier, orientation-dependent-response, same-material-identity-retained and coordinate-and-symmetry-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	direction-labelled-structure-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	orientation-dependent-response	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	same-material-identity-retained	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	coordinate-and-symmetry-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.

- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6cbc7c78de53a91c49c276e589ee1970b58c34859a374b9e536116f635df6977`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:32e5a3fe10672f05fcc0c334670205b39f340f5e57431dd46f3b50d3742f3fa3`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:175cd6c6266b529caa8f9d2671fa7ae2b95dd96739c241f9fbdef8a23d8e613a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d1c1e14e228849b8d8b6e38554430a79cd3247aadccdb0a8de858714adf77819`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:b984029abd34332b72aa4acbe09bba4fa14c8c7e328e04abe4a2d287d0d8ec73`; certificate `sha256:d3277213f7cc1a63985c32223dc1176a3a60269bd1d12589f6b5becef88e0d77`; external-validation hash `sha256:f31046c84f6ca5ab88a49c834753bfd3a8c8fb142ffb4cb2b1148a7cd13517c1`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:5415b2701a13d019225ae29acb353bfe745f8505d009fe5b4176d0a86c279069`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TEXTURE-PHASE-FRACTION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction; snapshot `experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html`; `sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc`; scope: grains, phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-bulk-anisotropy-001-authority-correspondence: predicted direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived direction-labelled-structure-carrier__orientation-dependent-response__same-material-identity-retained__coordinate-and-symmetry-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material anisotropy observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves direction-labelled-structure-carrier, orientation-dependent-response, same-material-identity-retained and coordinate-and-symmetry-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:72f09f5534c17c27099b1d5257c3786dbac85ef798492155e0d32d5a6a10e3e3`; derivation seal `sha256:fbdd70413189e72b586102c3e9366e8bcb48fc378e7e134073135a4e0529a77b`; engine receipt `sha256:c3c22fd6b9e4de8ba6af34698e7d7dba28cf604ee5d62cef540b8459446182b6` at `receipts/engine/model_admitted/SFT-MAT-BULK-ANISOTROPY-001-c3c22fd6b9e4de8b.json`; empirical validation `sha256:551c9e104169aae809fc0394df484bdcb23a543b85a8346628b6209e3723d3b3`; measurement receipt `sha256:1a455d7914018cf69f07f94afb5e7a4abf16bf467f77e237738d2008f9cff2c2`; isolation `sha256:e984687ace695a4b2491c37d9f3cfe69d4dbb7cbb9d5f60cb8b992d004c67869`; custody `sha256:61e237fb7d23a7c17168cc1fb16bc841b1020e9a1d5e5df1860c43e029becd5a`.

68. Size and surface effects

Claim identity: `SFT-MAT-BULK-SIZE-SURFACE-001`

Question and exact theorem. A material size effect occurs when the exact boundary/interior support relation changes the accessible state, transfer or mechanical census; it is never a free scale correction.

```
finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-st
ate-and-curvature-retained__size-shape-bounded__complete-state-transition-bounda
ry-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extr
a-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`

- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-POLYMER-CHAIN-001`
- `SFT-PHYS-THERMO-MICRO-MACRO-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves finite-bulk-and-boundary-carrier, boundary-to-interior-support-ratio, surface-state-and-curvature-retained and size-shape-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`finite-bulk-and-boundary-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`boundary-to-interior-support-ratio`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`surface-state-and-curvature-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`size-shape-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-state-and-curvature-retained__size-shape-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.

- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Material classes arise from retained constituent and network organization; molecular-to-bulk claims require scale-stable composition without erasing interfaces, fluctuations or preparation history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a71adddee9b5e490d396e46c2d4ce8e7d6df9fd103a368f317d5404c794f5962`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1cb16132da27181a2ea269de5aeccca7fd5c6fb9308e530c9eb4762bdd7c6cd9`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:16e259f649e3a6a6d3f8aaf73de1ae125511d669c7b238cab1230047fa53e70b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0abbadc72e2bb18b61d548347a7fc809891b92f314caf46fd012efe82f96adb7`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:8b01df9a6447d154451b82d24d6fbe0a6cd3d71941e037c9c9c056cc28fdf109`; certificate `sha256:71bf3b0b6332824a4a93e13e69b343a7f22245cac791167a29f3c097e5d0f4c3`; external-validation hash `sha256:5b57f4f341beaf218e7fdd9becd318ff208a58060bf09221cbfdda95b8aea683`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:96c7a8b7a44f93e71147fa23ce577cc4434028fcf92b7d9f39cab20f952bc8b5`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SURFACE-DAMAGE-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project; snapshot `experiments/external_sources/materials/snapshots/nist-surface-damage.html`; `sha256:7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d`; scope: surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- `NIST-NANO-PROTOCOLS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/nano-measurement-protocols; snapshot `experiments/external_sources/materials/snapshots/nist-nano-protocols.html`; `sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51`; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- sft-mat-bulk-size-surface-001-authority-correspondence: predicted finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-state-and-curvature-retained__size-shape-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived finite-bulk-and-boundary-carrier__boundary-to-interior-support-ratio__surface-state-and-curvature-retained__size-shape-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered size and surface effects observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered material_classes_bulk grammar that preserves finite-bulk-and-boundary-carrier, boundary-to-interior-support-ratio, surface-state-and-curvature-retained and size-shape-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:95bea428b8eefb5e8f23690a77f8b094becf3ba585242b61cb740f778c8cc4f8`; derivation seal `sha256:5f43cb2e35c18fe49c584059308b7af0afc7c6a72bad42a13c824446b7e64489`; engine receipt `sha256:d42d7f0a8dfb8cd74cb9b7857420beef13a926f5ec874c61e613976f557dac14` at `receipts/engine/model_admitted/SFT-MAT-BULK-SIZE-SURFACE-001-d42d7f0a8dfb8cd7.json`; empirical validation `sha256:46a2c11f07c97d5bca30736ccaac4a8677cea56d58998eaf25795921cd21e80c`; measurement receipt `sha256:22c019851ed0dcc75e91525362b462fb42429f0006f626a2460bac869fe59b66`; isolation `sha256:6a00639920ccb0d0be88becd640b378050d2bd32f0cd668d7a8022d2e2aab7fb`; custody `sha256:96a1bc3e068eea0c455024d13b01d9ecf1d2b794de18c5c4783991da5e4fc51c`.

19. Processing Degradation

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

69. Processing–structure–property path

Claim identity: `SFT-MAT-PROC-PATH-001`

Question and exact theorem. A materials process is a complete ordered path from input carrier through retained structural states to property relations; endpoint labels alone cannot erase the path.

```
history-labelled-material-carrier__process-to-structure-to-response-map__interme
diate-states-retained__route-condition-bounded__complete-state-transition-bounda
ry-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extr
a-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`

- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-material-carrier, process-to-structure-to-response-map, intermediate-states-retained and route-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`history-labelled-material-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`process-to-structure-to-response-map`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`intermediate-states-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`route-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `history-labelled-material-carrier__process-to-structure-to-response-map__intermediate-states-retained__route-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:de1ccf6eb7be4b98104a537ca43370f1260822868e975e19263742b742042635`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2a2242612fcf027a838edcc8079ef2b4a0a8103741fd3d4dac0fb3faa9783e70`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:361024627d19d9bae6cc5a148da978c4f72f58f46563b8d250daba4ecff56ba3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c570df606d3d54c7acc82474a18b688a269c52278fa3ed9e40cf3fb952620731`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:3de27f586e0a54a133982ba8f398e3da38fac53b3510ae159a9d1aa2f07a33c0`; certificate `sha256:03e7e4f1fc0dc184d8177651acc4a903420c58e4b0d458d84f3f307ecb17b308`; external-validation hash `sha256:7f4acaefbfb04449e58fbb65dc23f59ec79c48a4162681365f5b4d931ef7faaa`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:7cbdc5dc38d614e32c43f7f765797cd184d0e4187360667ac81f809ede32d647`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FATIGUE-FRACTURE-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture; snapshot `experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html`; `sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11`; scope: processing-structure-property-performance, defects, fatigue and fracture.

Comparison and custody records:

- sft-mat-proc-path-001-authority-correspondence: predicted history-labelled-material-carrier__process-to-structure-to-response-map__intermediate-states-retained__route-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived history-labelled-material-carrier__process-to-structure-to-response-map__intermediate-states-retained__route-condition-bounded__complete-state-transition-boundary-trace__root-bou nd-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered processing-structure-property path observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-material-carrier, process-to-structure-to-response-map, intermediate-states-retained and route-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:1f883e2d57a7b61755f14fc762c6739e9ce7beb81863b37d8e95f4ad26909eeb`; derivation seal `sha256:702060f13eedb527da17fb13321451beb4f372bf20b7eba39441e125afe297d1`; engine receipt `sha256:44a5d1efe1f1541917ea77e5eed4af2fdadbf277107194470e0cd69a279a2bf6` at `receipts/engine/model_admitted/SFT-MAT-PROC-PATH-001-44a5d1efe1f15419.json`; empirical validation `sha256:96ece9f7a71e3deb2748bec2f31385040cbee3d68883d1f5bcf6a94d67a8cd6d`; measurement receipt `sha256:cbc82b899d3bc7f483ec6881a345df54813c7cbdd9a4ab034da56209bb2f053d`; isolation `sha256:87b63a192292d43bf9591f2f3dfa1354236cc5b24b75811f1ba2e0eacb225d5d`; custody `sha256:c441dd3a1ce057baa9ad7680adcd01a5ba977076a692f9633121dd6ae9017865`.

70. Phase stability and coexistence organization

Claim identity: `SFT-MAT-PROC-PHASE-DIAGRAM-001`

Question and exact theorem. A phase map partitions generated composition/condition support by the stable material recurrence class, retaining coexistence, fraction and transition boundaries without fitted extrapolation beyond its domain.

```
composition-condition-support__stable-phase-minimality-relation__coexisting-frac
tions-and-boundaries__declared-domain-bounded__complete-state-transition-boundar
y-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra
-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves composition-condition-support, stable-phase-minimality-relation, coexisting-fractions-and-boundaries and declared-domain-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`composition-condition-support`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`stable-phase-minimality-relation`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`coexisting-fractions-and-boundaries`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`declared-domain-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `composition-condition-support__stable-phase-minimality-relation__coexisting-fractions-and-boundaries__declared-domain-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9054bd12994d5f41bcdf6177b6fed1a34c64e07f8bc3fa509b475f602c4ea5c2`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:40353a579820b968d584b0a61eba404cd3123713f03e571bd52c017083c2f5cc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b3bf6511512a363058e595b8843d98098ef829b121614b2c706fac58032f2bf3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2f7e1698d0a482289fd10c54b7e271b8255002aff2c60c9580d49d52be03b00b`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:5b0d3302fca9faacc388d07b983abfce483d21355e3e376bafddfaeb2611318b`; certificate `sha256:f57fb9ac93816e6dfe7d21e87d4a975ff683877da72cf233739c2ef07f175a5a`; external-validation hash `sha256:e19d6dc3ca0588280f74e04f9517cf7ac7566243ff1d1a70e74bb3323873b877`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:cac50917ffb4e7db7eaf36ef805abfc1b25c482a863f8a04e987b8d380a19de2`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-TRANSPORT-THERMOELECTRIC-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials; snapshot `experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html`; `sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0`; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- `NIST-TEXTURE-PHASE-FRACTION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction; snapshot `experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html`; `sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc`; scope: grains, phase fractions, texture, deformation and uncertainty.

Comparison and custody records:

- sft-mat-proc-phase-diagram-001-authority-correspondence: predicted composition-condition-support__stable-phase-minimality-relation__coexisting-fractions-and-boundaries__declared-domain-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived composition-condition-support__stable-phase-minimality-relation__coexisting-fractions-and-boundaries__declared-domain-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered phase stability and coexistence organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves composition-condition-support, stable-phase-minimality-relation, coexisting-fractions-and-boundaries and declared-domain-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:2697defe1bb570f0e0a431bf86071a61097ef77fa6a094e7bed3496b89580dd0`; derivation seal `sha256:e0c1577c109e318217fbd8353609550171447bf5838f1c229e48b62946ff50ac`; engine receipt `sha256:b2ab10b5f18cb90daa694ebe0ed304d3a153b06b007290c11a4ef97f4e157d0d` at `receipts/engine/model_admitted/SFT-MAT-PROC-PHASE-DIAGRAM-001-b2ab10b5f18cb90d.json`; empirical validation `sha256:ef30582b4adcc27a061c4c1f53dafc306fd144b7f3d0da2fdda718fbee0ef391`; measurement receipt `sha256:b6121ca9be3efcfdea534009debcb7982d58154d474e27907ba17eac057b4615`; isolation `sha256:9b3a1a5224c4dacf767cf438ecbae4065682fefe679f87ef9cdda13a118c6e3d`; custody `sha256:e89d44de4b24180fffe9afd082483ad17c54feee960bc30fad6ece7eda5a639a`.

71. Solidification

Claim identity: `SFT-MAT-PROC-SOLIDIFICATION-001`

Question and exact theorem. Solidification is the ordered nucleation and growth path from mobile to position-retaining organization with complete energy, constituent rejection and interface provenance.

```
liquid-and-solid-phase-carrier__nucleation-growth-and-rejection-path__latent-transfer-and-composition-ledger__rate-gradient-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`

- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves liquid-and-solid-phase-carrier, nucleation-growth-and-rejection-path, latent-transfer-and-composition-ledger and rate-gradient-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`liquid-and-solid-phase-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`nucleation-growth-and-rejection-path`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`latent-transfer-and-composition-ledger`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`rate-gradient-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `liquid-and-solid-phase-carrier__nucleation-growth-and-rejection-path__latent-transfer-and-composition-ledger__rate-gradient-bounded__comp` `lete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:28b0e07297d86f968e940b1b2ac43f3094e57aeb51ca1cf3957be002d4d3c900`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1d9abadcefe83f3d19ed07b99fe3c194097758fd2c19ca09640c2d36d329b587`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:25a0e1d82af8173155367a44fdfda6c03a8a457875f6d6237beb39a838f16497`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:97e19a857b778351e4607a1e2b02197c20cd922c00b6ab7a535f726ea471fec4`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:9008ceb7d860deb437d0278efb1db506b4282e8994c079cb8310927d30804be8`; certificate `sha256:ca3d3ad1afbbf9780ad9132a1260095507c75475a0390dfe859deddde190e2b3`; external-validation hash `sha256:8442dec51ae104811a14c830ef71370b48550b273d30be09822d7dcd9ec6d08f`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:0c9f870f7f230ffeb3890f9e71ae58c89c7463037ae121df10cc17d648146081`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SOLIDIFICATION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/publications/solidification-0; snapshot `experiments/external_sources/materials/snapshots/nist-solidification.html`; `sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c`; scope: solidification transport, nucleation, growth, interfaces, segregation and processing.

Comparison and custody records:

- sft-mat-proc-solidification-001-authority-correspondence: predicted liquid-and-solid-phase-carrier__nucleation-growth-and-rejection-path__latent-transfer-and-composition-ledger__rate-gradient-bounded__complete-state-transition-boundary-trace__root-bo und-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived liquid-and-solid-phase-carrier__nucleation-growth-and-rejection-path__latent-transfer-and-composition-ledger__rate-gradient-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered solidification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves liquid-and-solid-phase-carrier, nucleation-growth-and-rejection-path, latent-transfer-and-composition-ledger and rate-gradient-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:cd4d0b64eac45c1d3a28447a50aa223d74e3c5531439f94b4aafb0ad44ec43ff`; derivation seal `sha256:44d972ac53f7700b71e42f3b296ec7072c34c45c9f864c2c029cbe4af795bad3`; engine receipt `sha256:882454d5441763ff1ecfdc81b6298141741d84056879c62ad3fa9b1cd7c3bf85` at `receipts/engine/model_admitted/SFT-MAT-PROC-SOLIDIFICATION-001-882454d5441763ff.json`; empirical validation `sha256:8f82f3fcd57f8f8f6dcbd0b69e42bc940cd8c94021d24df8dafcd8f016a00f4f`; measurement receipt `sha256:e07ec032b3d672f10f84002045b8112a885dc28567393f3f27097c22abaa2d6c`; isolation `sha256:d2bc013d9aadeadd5d8593a793caf61908d3490425424619a807233e8dfa3da7`; custody `sha256:549b964885d9add4a9fb9c1b65671c42f0e813321766e22d95b1f4d9e922ea1e`.

72. Sintering and densification

Claim identity: `SFT-MAT-PROC-SINTERING-001`

Question and exact theorem. Sintering is interface-mediated constituent transport that joins particles and reorganizes structural absence while conserving material content and retaining the processing path.

```
particle-and-pore-network__interface-driven-material-transfer__content-conserved
-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boun
dary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-ex
tra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves particle-and-pore-network, interface-driven-material-transfer, content-conserved-pore-reorganized and temperature-time-load-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`particle-and-pore-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`interface-driven-material-transfer`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`content-conserved-pore-reorganized`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`temperature-time-load-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `particle-and-pore-network__interface-driven-material-transfer__content-conserved-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successo` r-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organization are separately retained; passed true.
- observation: the observation boundary is distinct from the material organization; passed true.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:427fba8bb3e13d9e946485513932e9addb8efa06915387bb6f29c16955dd431c.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:a95765c073bde4df0ddee77cbd0bba9d791e2da4a9df0f3b85ffba97e4c9a869.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:a86e1fcd6ea91d3d19e8c4ba54fe00c6aed81ef80e74c371cb01a3cf5d220e80.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:973a0bc896c8d4274acedad03ad2cbdb968478e6872d029074dcbe70d0885cb8.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:6b45355ff97dda505fb391570767e63172d3e98a3db5bc1a18561bc19b1ee990; certificate sha256:e171069695aad4a42995390f5a4d33efa475bcf7ab4de7d4e49caa9013ac2cfa; external-validation hash sha256:8ad24f2106da658a5eaf4ae15e3958ee5bbdccfd4fe1e6c0753e5c4770761a6a.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:7892b6461c84d32bd9487a05f441b0e38c9d4640a478edd10b49c14a6cb2ace7. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-CERAMIC-ADDITIVE-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-ceramics; snapshot experiments/external_sources/materials/snapshots/nist-ceramic-additive.html; sha256:540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; scope: ceramic feedstock, defects, sintering, densification, phases and microstructure.

Comparison and custody records:

- sft-mat-proc-sintering-001-authority-correspondence: predicted particle-and-pore-network__interface-driven-material-transfer__content-conserved-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived particle-and-pore-network__interface-driven-material-transfer__content-conserved-pore-reorganized__temperature-time-load-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered sintering and densification observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves particle-and-pore-network, interface-driven-material-transfer, content-conserved-pore-reorganized and temperature-time-load-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:3a48aa91150d2edb674a0ea59f30e9d6dcff091178031b3568c20f1bfe7608f4`; derivation seal `sha256:6dcdbf2eb9b8b5012bad74a050454b19eca9f1a608ddeffc30e364e667083445`; engine receipt `sha256:2405176e959c34190b77cd5b5acfb77dedd92245de91133d652c24ae73e097da` at `receipts/engine/model_admitted/SFT-MAT-PROC-SINTERING-001-2405176e959c3419.json`; empirical validation `sha256:5954fe3b5f12ec2660cfb6021c0d0517f40add8351bdeb0d8a388492b1e1028c`; measurement receipt `sha256:6db1b0cc6f2ba17d3ce2852251728a164746c4d146429002f978129e6e66979a`; isolation `sha256:a13f7d3ad965c5aa3dcaaf81c675099da3d320214db342f785bf416fcf7506eb`; custody `sha256:06ec2311f8604a307fcde247724ac7551f9fa3cb88f4b52cd156ca063fe0f795`.

73. Heat treatment

Claim identity: `SFT-MAT-PROC-HEAT-TREATMENT-001`

Question and exact theorem. Heat treatment is a declared thermal/time path whose phase, defect and microstructure transitions are retained and whose property consequence is path-bound.

```
history-labelled-phase-network__thermal-path-to-microstructure-map__time-tempera
ture-transition-record__cooling-and-specimen-bounded__complete-state-transition-
boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__n
o-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`

- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-phase-network, thermal-path-to-microstructure-map, time-temperature-transition-record and cooling-and-specimen-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`history-labelled-phase-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`thermal-path-to-microstructure-map`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`time-temperature-transition-record`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`cooling-and-specimen-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `history-labelled-phase-network__thermal-path-to-microstructure-map__time-temperature-transition-record__cooling-and-specimen-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b9bccf53662d1f29124adbd3570acc88fe4777fd0a652a1e1919e532d8ed6c2e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:54c522c6df12bc9faa22bd4f2e7cb3e5466453da6c0e58ea1133735b6fbcb9b0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:955e631bff387114e5252f19d68542f07a446c13563573d355fc803804d8d9d4`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e69374df1be210f621e0f3e134dc23dc86ba34914fdd90ad287d39530861f637`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:d5c8b9fd2403519162e8c42d68270b6519c09b147cb35d04e29d194cbd536948`; certificate `sha256:2009f709ca9587b3ee9090d9fe6638b2040f5762ac7c979215d8cb8ec2821dce`; external-validation hash `sha256:18935d07883ce48e19d9973db302e753092903a41dd892d13d8a9f6384a3fe4b`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:f4f5c6df4ece53d108f79157d9e39adee7ab3f6e995d38609ad491640f1c285b`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.
- `NIST-METAL-ADDITIVE-CORROSION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-metals; snapshot `experiments/external_sources/materials/snapshots/nist-metal-additive-corrosion.html`; `sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322`; scope: phase transformation, heat treatment, corrosion and environmental cracking.

Comparison and custody records:

- sft-mat-proc-heat-treatment-001-authority-correspondence: predicted history-labelled-phase-network__thermal-path-to-microstructure-map__time-temperature-transition-record__cooling-and-specimen-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived history-labelled-phase-network__thermal-path-to-microstructure-map__time-temperature-transition-record__cooling-and-specimen-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered heat treatment observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves history-labelled-phase-network, thermal-path-to-microstructure-map, time-temperature-transition-record and cooling-and-specimen-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:137c5dcf16efb098c8ba33efc12ca790c51997d030f971ed5ee21538263c3e0e`; derivation seal `sha256:b48749c0b7e8c1220f3d810e0fe1c7315f214c907330a1c10b73fa00ced47eab`; engine receipt `sha256:71c2de533e261b753919a23d2dda9c99717dc6b0893a99806067e734e9ef5081` at `receipts/engine/model_admitted/SFT-MAT-PROC-HEAT-TREATMENT-001-71c2de533e261b75.json`; empirical validation `sha256:48550300b7b1bc18d7b5a89db1d438340b5f3332fa47c775af37b4a7e93a9529`; measurement receipt `sha256:ea6182376624969f7747697ca792534ee7b5b7583dc9a85812fd0f0bcfeb5341`; isolation `sha256:3990d3430a57e033ce1931a8557cb3f20880d2e069e3da1e186471eebee159f6`; custody `sha256:5136a32f8fdbfbbdbe1df064160bc81eb12efd63dcdc487f609789aaaeccff8d`.

74. Corrosion

Claim identity: `SFT-MAT-DEGR-CORROSION-001`

Question and exact theorem. Corrosion is an interfacial reaction/transport process that transforms a material and environment jointly while retaining complete constituent, charge, product and boundary provenance.

```
material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-environment-joint-carrier, interfacial-redox-and-transport-path, mass-charge-and-product-ledger and environment-time-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`material-environment-joint-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`interfacial-redox-and-transport-path`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`mass-charge-and-product-ledger`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`environment-time-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9d15174b2bb1a9c07c26c8ba5711f9cdd9bb2f8a11a30aa320ce3f46bf1fe60d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:66247b21764c92c8fd72734d44cdcdba2d4db4e84b4faaa730823ee7a5db3327`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6c13de46c871b855dc4deb0ad522cb2e7c51962bf71905b9717e9d62e1decd18`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7cf442abda319bc3a9c74bd4d226b43396730c350c7f6f90566a15f15c27c7db`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:bc99ee0b1e7dd4c872735e6e261e18d1b1a6c045e6165dba52feb87aba82c3b0`; certificate `sha256:322b3fc3c7a431f278d9a3f9d46a2322315693e645c3c955a86a8a9358fa0a8e`; external-validation hash `sha256:c1b0a450d3020e0da5e0c078254ae93c3e1f4b760c4df000c6e1da4857be82d7`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:12b98b8c7184be61af10b2461cf587f811865f7d0f89a232f61fe44e983ae817`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-METAL-ADDITIVE-CORROSION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/additive-manufacturing-metals; snapshot `experiments/external_sources/materials/snapshots/nist-metal-additive-corrosion.html`; `sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322`; scope: phase transformation, heat treatment, corrosion and environmental cracking.

Comparison and custody records:

- sft-mat-degr-corrosion-001-authority-correspondence: predicted material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-environment-joint-carrier__interfacial-redox-and-transport-path__mass-charge-and-product-ledger__environment-time-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered corrosion observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-environment-joint-carrier, interfacial-redox-and-transport-path, mass-charge-and-product-ledger and environment-time-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:cbe464ce8433a734a64bd9b7432682d3246b2a27c2e1dfdf8fa429a58d8d2513`; derivation seal `sha256:dd05f442eff002b7acbae6f66aed2befa009ad9fc2ef5adada01cdaa20bf3838`; engine receipt `sha256:433102762a4f8027717c9429dddf7428947bf0bb850b88acb4929dd5d6b5078c` at `receipts/engine/model_admitted/SFT-MAT-DEGR-CORROSION-001-433102762a4f8027.json`; empirical validation `sha256:7397c246db07236a4bebe94e39a5390cbe1a75c43d58addd67e2e87aca429bff`; measurement receipt `sha256:bdb80356d1b616ae5ea6fb262d58658987549ab20ca590623ae3d8c4b2906183`; isolation `sha256:e79ebce2c1fddc766f90acbd9028cfed588fe6beb296c5bfc4cd88d717a42323`; custody `sha256:72c2ef2583c1c8b4bbe902480e18bf0cb56cca88d710ecb969a1f8347cf9f995`.

75. Wear and tribological material transfer

Claim identity: `SFT-MAT-DEGR-WEAR-001`

Question and exact theorem. Wear is retained material transfer or surface reorganization produced by a declared contact/load/motion path; frictional response and material loss remain separately recorded.

```
contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`

- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-PHYS-THERMO-IRREVERSIBILITY-001`
- `SFT-PHYS-MATTER-SCATTERING-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves contacting-surface-carriers, load-motion-interface-transition, debris-transfer-and-surface-ledger and contact-history-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`contacting-surface-carriers`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`load-motion-interface-transition`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`debris-transfer-and-surface-ledger`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`contact-history-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-clos` ure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: depth_independent. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- carrier: the generated carrier label is held and nonempty; passed true.
- relation: the generated relation and organization are separately retained; passed true.
- observation: the observation boundary is distinct from the material organization; passed true.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:0efa487d64b1eee6cd7ed3a8527673f988a9ec16ecd45ae8220ccc0757354b1f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:8229d0af59b1f56d1ea82fa4cdbbbd8e7ede783433fe74e9419d3f2bb8f61fca.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4347b34ae82985dfadd12d02f667736bf31f2ea92d6347f510d65e4f5162297f.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:4b02a5f8e1ee1b658b888159069c30bfcca22de1c25516ba7278faf493cbb6d3.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash sha256:247df125f5c11760b072521e72106849cde1ca0ddf036aed5c47f6b5100eb78f; certificate sha256:46c5627398f978342d464c3c65e7c094a0b903406e797f56d2b374c14effe281; external-validation hash sha256:a2426bbe0d4a4c3e36b80076f60ad0181b8c4e09889767de1e1dbd2bc78b985f.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 2 claim-specific source discriminators were required; their ordered identity is sha256:6b5d358984148115f3731a4f470093239e7a3bde5e29e0180f773b33385efa39. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-NANOTRIBOLOGY-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/nanotribology-nanomanufacturing-archived; snapshot experiments/external_sources/materials/snapshots/nist-nanotribology.html; sha256:3823f313343779c2a4a74a3e251a620319c89fef2142f2b1e58fa03250fec523; scope: interacting surfaces, relative motion, friction, adhesion, lubrication and wear.

Comparison and custody records:

- sft-mat-degr-wear-001-authority-correspondence: predicted contacting-surface-carriers__load-motion-interface-transition__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived contacting-surface-carriers__load-motion-interface-transitio n__debris-transfer-and-surface-ledger__contact-history-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True

- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered wear and tribological material transfer observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves contacting-surface-carriers, load-motion-interface-transition, debris-transfer-and-surface-ledger and contact-history-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:99ac90f50feb5f2af8a5030ff952e87e3438f02da297fb6bee8c61f4489fb672`; derivation seal `sha256:2bc4d5b3e790c3d5ea3c51b06600f9aa2470632ff440c74aacd6b3be89fa1f11`; engine receipt `sha256:92c996a0f5bfa825783a55376275b47e3635f67bc54552647a07899572b75f73` at `receipts/engine/model_admitted/SFT-MAT-DEGR-WEAR-001-92c996a0f5bfa825.json`; empirical validation `sha256:b834c07064aee6699c3fcdc85825d0c731978cca680844c88aa92ea00f0d4ab5`; measurement receipt `sha256:6a0798c98fc0d49a3d347d1f9652267352ca1c9c64617f6609f52c626d2f9f82`; isolation `sha256:857edb5a43adcea4878ef787f307a6ffd5010ed00e07ec12712474e3c5f8c7a1`; custody `sha256:48e96ae55cb515efe98c53c5d32c53794e7a66ffa1bb72dd0a6e1b02e1f06021`.

76. Radiation damage

Claim identity: `SFT-MAT-DEGR-RADIATION-001`

Question and exact theorem. Radiation damage is the complete source-bound path from probe energy transfer to retained displacement, excitation or transmutation defects and their subsequent migration/recovery.

```
material-and-probe-carrier__energy-transfer-defect-generation__displacement-tran
smutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-bo
undary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-
extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`

- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-PHYS-THERMO-IRREVERSIBILITY-001
- SFT-PHYS-MATTER-SCATTERING-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-and-probe-carrier, energy-transfer-defect-generation, displacement-transmutation-and-recovery and dose-rate-spectrum-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	material-and-probe-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	energy-transfer-defect-generation	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	displacement-transmutation-and-recovery	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	dose-rate-spectrum-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: material-and-probe-carrier__energy-transfer-defect-generation__displacement-transmutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Processing and degradation are ordered paths, never endpoint-only labels: every intermediate structure, transfer, product, defect and environmental boundary remains held.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9038baa8e438b1ac2c472866e42bf6bf72685d5b5fcccafff338d1022d68b1fc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:41a6bac5d899d6b3dc3443ea2f8ad03f200a69dc058c1c064bb1692905e50d94`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c5a8b28f1713b2c745128747d78b44c51d07371449023bddd94d0cb6f48d0518`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0eae8804b4960020b1078c44aeb79a2bcf0b3b24bd45b92fee471ed6121b6014`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:4f95022daaa462fc02c4d5629e2dce03074ed2a6402f7db3fe88c45f732ebe14`; certificate `sha256:2a3ea836c86f60a0cd147b73185a1047da7c534efca61fa7978d850f566f80b3`; external-validation hash `sha256:8e49be83d0c7ef29a09413543caed1843c57a721e1441a22efdf106c9a749ebb`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `4` claim-specific source discriminators were required; their ordered identity is `sha256:f6da78675cb2d1530cb7a3007c0e23d3a8e7ad00b3a3bd09478b36b62572f8ae`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-SURFACE-DAMAGE-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project; snapshot `experiments/external_sources/materials/snapshots/nist-surface-damage.html`; `sha256:7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d`; scope: surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- `NIST-MULTISCALE-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot `experiments/external_sources/materials/snapshots/nist-multiscale-materials.html`; `sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a`; scope: multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.

Comparison and custody records:

- sft-mat-degr-radiation-001-authority-correspondence: predicted material-and-probe-carrier__energy-transfer-defect-generation__displacement-transmutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-and-probe-carrier__energy-transfer-defect-generation__displacement-transmutation-and-recovery__dose-rate-spectrum-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered radiation damage observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered processing_degradation grammar that preserves material-and-probe-carrier, energy-transfer-defect-generation, displacement-transmutation-and-recovery and dose-rate-spectrum-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:6cd407968904d1edcba930dfcd606074c06ccc12d24464f820a04bf357931e8d`; derivation seal `sha256:698efffcc699b74190a413a43a06ba91aa6b1d1dd5c490766e0a6d22b501ca42`; engine receipt `sha256:e6e36bc645b877f36b9c969dd7bcbbda391a9f6e765e806c80d5fc11e214de42` at `receipts/engine/model_admitted/SFT-MAT-DEGR-RADIATION-001-e6e36bc645b877f3.json`; empirical validation `sha256:ee06521307cfbf5c880200c3b4c53e271fa83da81341ee32e8eb79fba8b1e43c`; measurement receipt `sha256:8b764f65f5eb032dc6e38b45a153ac5030f775a4900fc2c5164e8c19bcc87faa`; isolation `sha256:72088d354ffef387f18fefddef742e2a4eae7b9cac2a1b92dd65903dd7638b17`; custody `sha256:e03d344139946255498886f2ef0be18c7ccf242d053080f8aea0062802a9381a`.

20. Advanced Functional Sustainable

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

77. Piezoelectric coupling

Claim identity: `SFT-MAT-FUNC-PIEZOELECTRIC-001`

Question and exact theorem. Piezoelectricity is a reversible cross-relation between mechanical deformation and held electric polarization permitted only when the retained material symmetry does not close the orientation distinction.

```
noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orient
ation-and-charge-ledger__field-stress-condition-bounded__complete-state-transiti
on-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure
__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves noncentrosymmetric-structure-carrier, mechanical-electric-cross-response, orientation-and-charge-ledger and field-stress-condition-bounded.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	noncentrosymmetric-structure-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	mechanical-electric-cross-response	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
organization	organization-collapsed	orientation-and-charge-ledger	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
observation	observation-boundary-unrecorded	field-stress-condition-bounded	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
record	result-without-state-transition-record	complete-state-transition-boundary-trace	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientation-and-charge-ledger__field-stress-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8ab926031b7ddd6f9ae6723610d89598c05082b6a6aae33124e1485e2155a550`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:46b8080073955bdaff587c54bb623525c892ff9137218f8adc2ce6d36b3c9278`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:769c539ede54297da90196ce94b20c57bbbfa5cea6148c8d4b022abc7c36e8a9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:178a33c9bd0f4b6d3e6f7842ba574a87781b2a3174b4f3c77ea99f725c82198f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:0edb98b42db8eb35abf9129bf24b204bde6a7b1f7723f398d61319711f842fec`; certificate `sha256:0eb2407afb1deb0d42da9134ce6101bc3eb518446e460f33ed4948cea367c142`; external-validation hash `sha256:7ea355f4b6500f3743e60bfa36f6f943be05e13ae2c759ce168cc4085e9b94ae`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:0199fe3b4329ace3bce2d717af5ca80589160d0648df85bf0b5b56d1a2e622b0`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-func-piezoelectric-001-authority-correspondence: predicted noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientation-and-charge-ledger__field-stress-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived noncentrosymmetric-structure-carrier__mechanical-electric-cross-response__orientation-and-charge-ledger__field-stress-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered piezoelectric coupling observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves noncentrosymmetric-structure-carrier, mechanical-electric-cross-response, orientation-and-charge-ledger and field-stress-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:cdfe88675f774917ec9e902154853bc89336ea93db2feda54d87a9ac4e61fb8d`; derivation seal `sha256:4f680e305121b953aa576a0d8655400d3fff5a059be9a6c72c1b8035da333bc0`; engine receipt `sha256:7f640ec82c5c73c47f5749e38393747843704417ad8bd9279f671aab42365d26` at `receipts/engine/model_admitted/SFT-MAT-FUNC-PIEZOELECTRIC-001-7f640ec82c5c73c4.json`; empirical validation `sha256:37594ac98d49f7968d62437760964d340829aacbfaae3c9f628d17f905f8c83d`; measurement receipt `sha256:87d6d92549f3c144716efa44e1286f3c5f188352f6208e71998bf895a634c516`; isolation `sha256:ef1082ab87a9d58062773893911c04c5a836d5e6cc67300969b03531e0478173`; custody `sha256:e55d04e7bca4b730de4be5419299976028216ef5ec38114ceadd743681279539`.

78. Ferroelectric order and switching

Claim identity: `SFT-MAT-FUNC-FERROELECTRIC-001`

Question and exact theorem. Ferroelectricity is a material phase with stable opposed polarization recurrence classes and a retained field-driven switching path between them.

```
polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves polar-domain-network, stable-opposed-polarization-classes, field-driven-domain-path-retained and hysteresis-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`polar-domain-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`stable-opposed-polarization-classes`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`field-driven-domain-path-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`hysteresis-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c8c27f9a90d8b0fdf120aa63794dad5be1b2554f0f0e6cd36b647d26c59abc87`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8162c61883e1b3fe67fef308ebacf2e9ed54670ae52ee7067edf8d80ca467aa7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d63ee927baa6c8404ff75eb6407622cdfafe2ad7d894e0556ae9b7572cd4bab7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f47c545d9bf91517601315bcd5136409a1a228df3c7edab9920e71bc1afb39d`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:5467315283f5f64b807d962309ddcf27cd5f1c65a26683e6d316faf565008499`; certificate `sha256:e8f91344b1dc05fb3c8e81712d23b08b06cb1612c30f64a3a83a49a66b4e595f`; external-validation hash `sha256:f6cc2ab5195e78512f9c6a49d2fc79945b2b79a484890d38ade80e7a93576c82`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is `sha256:950537b9a096411ed993b95bfaeb1b77308569ce1b70e1834e2c82c09f6a60c2`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-func-ferroelectric-001-authority-correspondence: predicted polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived polar-domain-network__stable-opposed-polarization-classes__field-driven-domain-path-retained__hysteresis-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered ferroelectric order and switching observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves polar-domain-network, stable-opposed-polarization-classes, field-driven-domain-path-retained and hysteresis-condition-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:da506701f72bb70612bca1f2478c1b91e30dbc5c121e71ccb91e601b080acb0d`; derivation seal `sha256:8611bbfd8a418797896a462e54d5cfd398aa65d0e3bd39effa88daa38add9c80`; engine receipt `sha256:36580c9520ef5c2fbe57824b854c8e16716aacb4b2165fe37a23e6a3d14fbd55` at `receipts/engine/model_admitted/SFT-MAT-FUNC-FERROELECTRIC-001-36580c9520ef5c2f.json`; empirical validation `sha256:74498fd9ace1ce0b625e2b44cc82a6c1ebd9faec3e735e2f19fca20778d3a743`; measurement receipt `sha256:d6c8aaadf1d41547c53d2d042b751edc04af4f75c71c14f1bb8dd0b08fed2817`; isolation `sha256:1d266587e8d483f234e7469691b423578536738ae724f1795770a9a17601a8cf`; custody `sha256:4ae27b63ebc319dcd721ece0a9d989885ce280368273a53f8e1a402036bc0ad9`.

79. Thermoelectric coupling

Claim identity: `SFT-MAT-FUNC-THERMOELECTRIC-001`

Question and exact theorem. Thermoelectric response is the joint conserved relation between energy-gradient transport and held-charge transport on the same material network.

```
charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves charge-and-energy-carrier-network, temperature-gradient-charge-transfer, joint-energy-charge-ledger and direction-condition-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`charge-and-energy-carrier-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`temperature-gradient-charge-transfer`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`joint-energy-charge-ledger`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`direction-condition-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5b2e498cf9e75a44a5a5fbcb0345af47f4deb173f28fe5dc65f0ae665ee2fe21`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d7fd528b32ee51d1a154d5a48b34ed9216d7534090f6e967ecc4f8a632ad9f69`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1b3c38b099cd9b9ec5998f66e220e1435bbe784b5c2e30a5f0d2835f7280ed36`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7c8fad072572679744edac6081724e4c31ee8f824747cf90d31041aa70856b65`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:52d94c68bc0c817a04d1992ddf14e2f8e7371008376f88e58b6b2212c7c9659c`; certificate `sha256:e2af0809eb20acbef176b000e81b81f08d1d24274e74136cc94a9af4ceb83c5d`; external-validation hash `sha256:3a69e07d1138a3073f22b46120a349145fdd7034830be55e85a4b4271c1f9a61`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is

sha256:49468d618ab6361c0f75798af8c71ae5516ac5325bf6b6c287bd154616fb8059. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials; snapshot experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; scope: thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.

Comparison and custody records:

- sft-mat-func-thermoelectric-001-authority-correspondence: predicted charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived charge-and-energy-carrier-network__temperature-gradient-charge-transfer__joint-energy-charge-ledger__direction-condition-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered thermoelectric coupling observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves charge-and-energy-carrier-network, temperature-gradient-charge-transfer, joint-energy-charge-ledger and direction-condition-bounded.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:95f83d012fde1ad0ca6942fe7be129412ba9e0e36dbeeb433f5c4d778a9daec4; derivation seal sha256:1c5893f35485b76b913f5de741de5cf87349e4cb7eb89f62b4687a7902a62dc7; engine receipt sha256:c861a777de27b860ffce46411542047fe44e88ddd68fcae21cc9d48d658184cd at receipts/engine/model_admitted/SFT-MAT-FUNC-THERMOELECTRIC-001-c861a777de27b860.json; empirical validation sha256:6c525fdf2f2a8591cbb0fe07899e059cc14b5446ff5d88a1b60538aa0f4d771b; measurement receipt sha256:827f8e8fa21a88f168df8e3788a9549165f2e074652d33693846f2b23bb5e6df; isolation sha256:81aad8c3ba087051b0eca6244280ea87263116cda7f57708077f8efdcc15aba9; custody sha256:edaa008f449455c72fd96ef412f4bab31a0314fc502f8bea8e75a80e3280d417.

80. Photonic material organization

Claim identity: SFT-MAT-FUNC-PHOTONIC-001

Question and exact theorem. A photonic material organizes electromagnetic recurrence support so that allowed propagation classes and structural gaps follow from complete generated geometry and boundary relations.

```
electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequenc
y-and-boundary-retained__spectral-scale-bounded__complete-state-transition-bound
```

```
ary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-ext
ra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves electromagnetic-cell-network, wave-recurrence-and-gap-support, geometry-frequency-and-boundary-retained and spectral-scale-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`electromagnetic-cell-network`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`wave-recurrence-and-gap-support`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`geometry-frequency-and-boundary-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`spectral-scale-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	authority-or-target-selected-law	root-bound-forward-forcing	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
generality	single-specimen-or-instance-lookup	positive-finite-successor-closure	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
extension	free-fit-exception-or-extra-rule	no-extra-rule	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequency-and-boundary-retained__spectral-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c32943d095b89193d400a5b2631eca6463a05ce83ff023c092e18e7ba3433f6f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:50acfe0be40722d399943981a25721636a151897789b475bafcde1015ad34e87`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:61b3cdacb55eb4a710cc30fadeb2ad4d1adc5a38bc277fcf964e0cb41b326b6f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ee7373f233a08e9471fb6686dcb5b89984f07cb5f74d32b13cd25379e37c4c9b`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:22bfd80dd14b7366e546e51abc799a76a5ab463562e49218bb9c373fce71cd15`; certificate `sha256:f38b1b5ef15f21b39d7a46ca08c82d87709a63df3623fc8d55d110c32d94ea34`; external-validation hash `sha256:65befe7fa71ec4b61efbb9c9025b153fa0b8b56543006349263735440208c414`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is

sha256:8f8bf66262959ed2118bcb56bb0a0fa706206788b7efc3da91b17c7d057f1c92. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-OPTICAL-MATERIALS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/theory-optical-properties-materials; snapshot experiments/external_sources/materials/snapshots/nist-optical-materials.html; sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b; scope: index of refraction, absorption, reflectance, dielectric response and photonic crystals.

Comparison and custody records:

- sft-mat-func-photonic-001-authority-correspondence: predicted electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequency-and-boundary-retained__spectral-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived electromagnetic-cell-network__wave-recurrence-and-gap-support__geometry-frequency-and-boundary-retained__spectral-scale-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered photonic material organization observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves electromagnetic-cell-network, wave-recurrence-and-gap-support, geometry-frequency-and-boundary-retained and spectral-scale-bounded.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:b8359125c415f70252bad5e8662f32a4a0411fd4ecab90b210f238e5d4eff3f7; derivation seal sha256:d5719938752bff0b363ba54eb5c0394c4b5fa74ca27361fee3d5493208f569e9; engine receipt sha256:b63114cadba712eeb0a65b991315d9035f61024e44e181fc9dbb26f93b2947c7 at receipts/engine/model_admitted/SFT-MAT-FUNC-PHOTONIC-001-b63114cadba712ee.json; empirical validation sha256:d70953f2830575c85ac5367761e35a0a7daa230e39683dbb25e951db3f69a00b; measurement receipt sha256:076b47a085e881f1daba570530905b12afa40b4cfd62cb8dd1c26da66228211a; isolation sha256:95a9d5ee8ac6b7e943fc57db2431acee31ac9e4115a590b4eae9b3b301dc8a94; custody sha256:6a0be96588221e8b3bbc5e6b3e84565643db68aecf32f4b3ee6a428c2135b7bb.

81. Functional magnetic material response

Claim identity: SFT-MAT-FUNC-MAGNETIC-001

Question and exact theorem. A functional magnetic response is the complete relation among applied field, domain transition path, retained orientation, loss and remanence at declared frequency and temperature.

```
moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-an
d-remanence__frequency-temperature-bounded__complete-state-transition-boundary-t
```

```
race__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-ru
le
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves moment-domain-and-field-carrier, field-history-response-map, domain-wall-loss-and-remanence and frequency-temperature-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`moment-domain-and-field-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`field-history-response-map`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`domain-wall-loss-and-remanence`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`frequency-temperature-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:db19f1a2cc9d5f45d18837e5d0172d17a4f6e0b27b08b0096dc0622a8fdfec16`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5ee5c3258afeb367bb0cd4365b9db18fdb8207765c46dad0c76f440d2f2a0614`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ae310bd447bd0da6fef8f21bcd1ce0c0df36b3523c38cc3dff2dd5e5be48d62f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:334673e317625efa1bbb433ce8d7cdec742cf726cb5311cb9c5b0b1bbd9bd85f`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:cd8911fe8913dbfca7c344e88e4917190bbc07e7d04e1403ca7f513d06fce2f7`; certificate `sha256:a0b1af464cab9fc6344d8193cfe38a2651617f9c887720c78ad0eae55b49a065`; external-validation hash `sha256:d28ec86f9c2ea983d46ed613c328135ead66046bc3f0baf7bb15769aeeafd3b8`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is

`sha256:8ab5d335d373302646984a94eb1f45dccc0a137eea7206abf17029cac4627ff1`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot `experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html`; `sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e`; scope: dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.

Comparison and custody records:

- sft-mat-func-magnetic-001-authority-correspondence: predicted moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived moment-domain-and-field-carrier__field-history-response-map__domain-wall-loss-and-remanence__frequency-temperature-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered functional magnetic material response observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves moment-domain-and-field-carrier, field-history-response-map, domain-wall-loss-and-remanence and frequency-temperature-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:6d22b16de46e8b5d63aa64b2a61e80240e18216a5feb89964e7099a2854d017c`; derivation seal `sha256:395244dc506ff152b9c09887ab33883310f04880d049893f0e8d4cd625914bed`; engine receipt `sha256:1ed29896c9091b1948b2d855f0c0d51738df688e864db22fcde9f344da2dfca4` at `receipts/engine/model_admitted/SFT-MAT-FUNC-MAGNETIC-001-1ed29896c9091b19.json`; empirical validation `sha256:b117f217af70207aaacd9b09bb785b027e22a059780fa54a291a85793545b6cc`; measurement receipt `sha256:5d4519d974b7c1ab4790a14e7d29d2343daa1b3b423846555bf212c4f5e7716a`; isolation `sha256:58661800655851db8c9f5dc1c1717efb27bcdfc0624e4d19967f68b2400e53f2`; custody `sha256:2b92d674f030c9acc3cb3072a05f64cf6665bb62637a7c91f52e7aad01229652`.

82. Nanoscale material boundary

Claim identity: `SFT-MAT-FUNC-NANOMATERIAL-001`

Question and exact theorem. A nanomaterial regime begins when boundary and finite-support distinctions remain in the material state/property census and cannot be closed by the bulk observation quotient.

```
finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__di
screte-state-and-interface-retained__size-shape-resolution-bounded__complete-sta
```

```
te-transition-boundary-trace__root-bound-forward-forcing__positive-finite-succes
sor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves finite-constituent-boundary-carrier, boundary-support-comparable-to-interior, discrete-state-and-interface-retained and size-shape-resolution-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`finite-constituent-boundary-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`boundary-support-comparable-to-interior`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`discrete-state-and-interface-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`size-shape-resolution-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:31c76c3f386bed3a498183b0f457baf72566e4999e3bbfde6be9cf01415a227d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:dc40a0ec44efafc6e16bc5818d4e5876e2169c4a4d15e75a7c44b057a6232ce2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:42da32bcf34d016e4fb04da37d8e1a77bed1c2942845d0bbdb58abb5d23c657f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e95636207c13c2d934d5cc7503c8af418875ef0267cb94376f56979200802167`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:69caf8202ee5680eb52cb0ab81104a75f12ccbaf87e08f079578e9953f1ba93e`; certificate `sha256:77995cabcde5179d50722c7318e402d37e31b3cbe2ac7ad4afa6a53c415d9547`; external-validation hash `sha256:5a9764c3798b5bc1028e1d681d140e937971b971a0207684cd14a01d4c057987`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, `2` claim-specific source discriminators were required; their ordered identity is

`sha256:453e167dbba6de5524a9410a9889cf4b02c39e7b2c962716c0a287338c24ee3e`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-NANO-PROTOCOLS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/nano-measurement-protocols; snapshot `experiments/external_sources/materials/snapshots/nist-nano-protocols.html`; `sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51`; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- sft-mat-func-nanomaterial-001-authority-correspondence: predicted finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived finite-constituent-boundary-carrier__boundary-support-comparable-to-interior__discrete-state-and-interface-retained__size-shape-resolution-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered nanoscale material boundary observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves finite-constituent-boundary-carrier, boundary-support-comparable-to-interior, discrete-state-and-interface-retained and size-shape-resolution-bounded.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest `sha256:a772561a12e9f1e33384ed3b074205b589ac789c8348c602e27c320ce8c2b925`; derivation seal `sha256:2cc9bc4362cbc4dee60c592336a2df482696e5330953579100fc5e1f15ceb9b1`; engine receipt `sha256:a815d0fcf264919778c652d76e2b46dd60f5f75df6e79c325e98bc7192cc85b3` at `receipts/engine/model_admitted/SFT-MAT-FUNC-NANOMATERIAL-001-a815d0fcf2649197.json`; empirical validation `sha256:092002f7236e19aaf27f182f5f1ae15d177a80a287affd058a15195835790be3`; measurement receipt `sha256:3d6f316b710b89615ba3a69f4c6b1bff427dcfdbf675756a2f8359d144c7f311`; isolation `sha256:35d270615bc1e69314b08304d5ffda3838eaf6b7e856a0a86c3d8d89b2c1b02c`; custody `sha256:7d55c2da7e6abd7970c8a0ba99fd51def81599e28c42cc318be5f4e2b0bcaa3a`.

83. Biomaterial interface boundary

Claim identity: `SFT-MAT-FUNC-BIOMATERIAL-001`

Question and exact theorem. A biomaterial claim is a joint material/biological interface relation with explicit exposure, transfer, response and function boundaries; biological outcomes cannot select the material law.

```
material-biological-interface-carrier__bidirectional-response-and-transfer__mate
rial-and-biological-identities-retained__exposure-function-bounded__complete-sta
te-transition-boundary-trace__root-bound-forward-forcing__positive-finite-succes
sor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-DYNAMICAL-SYSTEMS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-WAVE-PROPAGATION-001`
- `SFT-PHYS-CONDENSED-PHASE-ORDER-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`

The repository graph audit independently confirms that this claim reaches `SFT-ROOT-THERE-IS-NO-NOTHING`.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves material-biological-interface-carrier, bidirectional-response-and-transfer, material-and-biological-identities-retained and exposure-function-bounded.

The Cartesian product contains `256` candidates. The evidence retains `256` candidate records and `256` decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`carrier`	`carrier-erased-or-answer-only`	`material-biological-interface-carrier`	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
`relation`	`relation-imported-fitted-or-erased`	`bidirectional-response-and-transfer`	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.
`organization`	`organization-collapsed`	`material-and-biological-identities-retained`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`exposure-function-bounded`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `material-biological-interface-carrier__bidirectional-response-and-transfer__material-and-biological-identities-retained__exposure-function-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c77ecdcc4cee7390b374f11dac9a22f81f990b7789c59a924c07fe55f79a889e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:cb1ea4682cb262aad2796a49a64cedd62acb16f9534428567b7cd8dc5a118235`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1d101e88432f57335048d4d9f0a2b09fe305b1cf62dd6fb95fe211598e2553a4`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:da528078953f494ae3b594ae054f42104f1e21c78ec26b6bae2a906ca4dc1689`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:01f66c16a5ba692961fbabd31a4335924f06b60bb42767fddf09c7682a57a104`; certificate `sha256:eaa0c22c327dabffe543439a4565a94849959aebfa0e4fe2e9c48504b4de0085`; external-validation hash sha256:4f1fda699d5d3bb69d2a10eac5fcdeb43aa1257bb54bf96f88d3f002556b7209.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is sha256:bf5574013b2ace3aec4970b1b058bd142529651a68815439d9d26d155166b6f2. Target content opened after the derivation seal: true. All rows preserved: true. Exact comparison passed: true. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- NIST-NANO-PROTOCOLS-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/mml/nano-measurement-protocols; snapshot experiments/external_sources/materials/snapshots/nist-nano-protocols.html; sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.
- NIST-POLYMER-COMPOSITES-2026-07-24 - National Institute of Standards and Technology; https://www.nist.gov/programs-projects/polymer-composites; snapshot experiments/external_sources/materials/snapshots/nist-polymer-composites.html; sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; scope: polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.

Comparison and custody records:

- sft-mat-func-biomaterial-001-authority-correspondence: predicted material-biological-interface-carrier__bidirectional-response-and-transfer__material-and-biological-identities-retained__exposure-function-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived material-biological-interface-carrier__bidirectional-response-and-transfer__material-and-biological-identities-retained__exposure-function-bounded__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered biomaterial interface boundary observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves material-biological-interface-carrier, bidirectional-response-and-transfer, material-and-biological-identities-retained and exposure-function-bounded.

Immutable evidence identities. Pre-source branch seal experiments/sealed_predictions/materials_complete_branch_pre_source.json; source manifest sha256:1f6cf609fdd14e3ef9884853b588680c81246b326cb6628497bb7524f680b695; derivation seal sha256:e04744de5df085dc908548847c8701d22343f9735c7e28a67250206beeddc22f; engine receipt sha256:5ed0cf28440ad78632017718d57debfecdda937fdc7acb83e53a782aefad24d3 at receipts/engine/model_admitted/SFT-MAT-FUNC-BIOMATERIAL-001-5ed0cf28440ad786.json; empirical validation sha256:9810925c7e30dd79c38a3577eca33c9fbd681aded4f9d2526db42d8ebbf8359c; measurement receipt sha256:b61bf887f8ab6b53a09748fbffeff3dad5477ea07a50a081cc25e46e1899ac14; isolation sha256:d4685e1d38fb5855d3c4519cb602ff7498c3fc36794645bd703176a74392c664; custody sha256:c8acfc9ecf4861523b5dcfff4402a370a9b10fde0758517ffbf0bd42fbd9ed50.

84. Material lifecycle and recyclability

Claim identity: SFT-MAT-SUST-LIFECYCLE-001

Question and exact theorem. A material lifecycle is the complete source-to-processing-to-use-to-recovery flow ledger; recyclability requires an explicit identity-preserving recovery path and records every transformation, loss and quality boundary.

```
source-process-use-recovery-carrier__complete-material-flow-path__loss-transformation-and-quality-ledger__system-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule
```

Rooted dependency chain. The engine registration names the one premise-free root theorem and requires these already admitted receipts:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-PHYS-CONDENSED-PHASE-ORDER-001
- SFT-CHEM-STOICH-CONSERVATION-001

The repository graph audit independently confirms that this claim reaches SFT-ROOT-THERE-IS-NO-NOTHING.

Generated grammar and boundary. Generate the literal Cartesian product of the registered content-specific carrier, relation, organization and observation choices with record, provenance, generality and extension choices.

Boundary: Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves source-process-use-recovery-carrier, complete-material-flow-path, loss-transformation-and-quality-ledger and system-boundary-declared.

The Cartesian product contains 256 candidates. The evidence retains 256 candidate records and 256 decisions. Exactly one decision survives. The other 255 are rejected at their first non-preserving coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-erased-or-answer-only	source-process-use-recovery-carrier	Erasing the material carrier prevents reconstruction of the observed distinction. The complete generated material carrier is retained.
relation	relation-imported-fitted-or-erased	complete-material-flow-path	An imported, fitted or absent relation can select a familiar answer without forcing it. Only the generated constituent, adjacency, transfer or response relation is admitted.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
`organization`	`organization-collapsed`	`loss-transformation-and-quality-ledger`	Collapsing organization merges materials states that the question requires to remain distinct. Every organization, interface, history or boundary distinction required by the claim remains held.
`observation`	`observation-boundary-unrecorded`	`system-boundary-declared`	An unrecorded method, scale or condition cannot identify the Materials observation class. The exact declared observation boundary is retained and cannot select the law.
`record`	`result-without-state-transition-record`	`complete-state-transition-boundary-trace`	A result label without its state, transition and boundary trace is not auditable. The complete initial state, transition, result and retained/lost distinctions are recorded.
`provenance`	`authority-or-target-selected-law`	`root-bound-forward-forcing`	Authority, consensus, a handbook or target data may test but cannot select a Fold law. Every decision is traced through admitted dependencies to the premise-free root theorem.
`generality`	`single-specimen-or-instance-lookup`	`positive-finite-successor-closure`	One favorable specimen or instance does not close the generated class. The base carrier and every positive finite successor preserve the relation at the declared boundary.
`extension`	`free-fit-exception-or-extra-rule`	`no-extra-rule`	A free coefficient, fit, exception or added constitutive law can force a desired answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is present.

Unique survivor, base and successor. Sole survivor: `source-process-use-recovery-carrier__complete-material-flow-path__loss-transformation-and-quality-ledger__system-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule`.

Base: One complete material carrier retains its generated relation, organization, observation boundary and proof record.

Successor: Adding one generated constituent, cell, interface, transition or observation row preserves every earlier distinction and appends its exact source-bound relation without an extra rule.

Closure scope: `depth_independent`. Minimality and named-shape uniqueness are preserved in the closure certificate.

Operational witnesses.

- `carrier`: the generated carrier label is held and nonempty; passed `true`.
- `relation`: the generated relation and organization are separately retained; passed `true`.
- `observation`: the observation boundary is distinct from the material organization; passed `true`.

Detailed mathematical and physical meaning. The law's exact result is relational. Any material-specific magnitude remains conditional on the specimen, method, direction, scale, history and declared conditions; no such magnitude is promoted into a universal constant.

Functional and lifecycle laws compose coupled response and complete material-flow paths while keeping application and biological outcomes outside law selection.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:42532e43e708387eab368ef316263c2ed870195ef765bab6a76e7b16dd4a334b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1360d7af1f0267dfc482885e6b6289c9a952d85d977db53826e457f0fe3d337a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:599f2eca8225f1d225f32ee678bb4237582333e452455db62034e12fdbfc6981`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:29bef3fbd7454e3857b3b0be39cd76cdc6725365e836fe943c4c620e0cc2da2d`.

Independent implementation. The implementation-distinct validator regenerated the literal eight-axis product, candidate order, all decisions, one survivor, closure fields and four control classes. Implementation hash `sha256:8c2a8e8674efd78cc9aeb6e6ff178591c51fdae5c0f7573769bd763ad4c69bca`; certificate `sha256:7271f451ff5b28ae59d344e9b829463616b6e8225c85562520df817c592ef606`; external-validation hash `sha256:a34eca90ac5590c6a6444768b73d44224b164109b923c033cce4da264b285775`.

Post-seal empirical correspondence. The complete 84-law prediction set was sealed before any Materials source identity was selected. For this claim, 4 claim-specific source discriminators were required; their ordered identity is `sha256:d1aa7f6fa9c4241ced6e2ef6f2f5900b28d79e8a963460e50d55a0fabda4f634`. Target content opened after the derivation seal: `true`. All rows preserved: `true`. Exact comparison passed: `true`. The deliberately changed unfavorable record was rejected.

Measurement-body sources:

- `NIST-MATERIALS-DIVISION-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/materials-science-and-engineering-division; snapshot `experiments/external_sources/materials/snapshots/nist-materials-division.html`; `sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3`; scope: materials measurement, composition, interfaces, processing, damage, classes and lifecycle.
- `NIST-NANO-PROTOCOLS-2026-07-24` - National Institute of Standards and Technology; https://www.nist.gov/mml/nano-measurement-protocols; snapshot `experiments/external_sources/materials/snapshots/nist-nano-protocols.html`; `sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51`; scope: nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

Comparison and custody records:

- sft-mat-sust-lifecycle-001-authority-correspondence: predicted source-process-use-recovery-carrier__complete-material-flow-path__loss-transformation-and-quality-ledger__system-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; source-derived source-process-use-recovery-carrier__complete-material-flow-path__loss-transformation-and-quality-ledger__system-boundary-declared__complete-state-transition-boundary-trace__root-bound-forward-forcing__positive-finite-successor-closure__no-extra-rule; exact match True
- every required authoritative fragment and snapshot hash reproduced
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if a complete registered material lifecycle and recyclability observation lacks any sealed carrier, relation, organization or boundary feature, any required row is omitted, or a tampered row is accepted.

Explicit exclusions.

- no V1/V2 result, handbook, named material, database row, measured property or application outcome may select a survivor
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- structural absence is a held empty form and opposed direction is a held orientation, never a prohibited scalar
- no axiom, free parameter, fit, learned coefficient, constitutive exception or target-derived rule
- no opaque prediction, selected favorable specimen, omitted failure, or unrecorded method/condition boundary
- external target content remains inaccessible until the complete prediction set is sealed
- Every positive finite generated Materials carrier in the registered advanced_functional_sustainable grammar that preserves source-process-use-recovery-carrier, complete-material-flow-path, loss-transformation-and-quality-ledger and system-boundary-declared.

Immutable evidence identities. Pre-source branch seal `experiments/sealed_predictions/materials_complete_branch_pre_source.json`; source manifest

sha256:ae6eca4c43b073a3ed7eda20aa9836c009c9bb3b05a6f443ab2d24738fd48e57; derivation seal sha256:3e9a7308036c4ef8d2b7fc56f68e6b33cff28eab1ed004ae9f243609dc70434e; engine receipt sha256:0046611012fa957e6166fede124a4a43e24de052957566bb7411afb54ed2bc43 at receipts/engine/model_admitted/SFT-MAT-SUST-LIFECYCLE-001-0046611012fa957e.json; empirical validation sha256:06b1e6c1f4a57c40bb00be257aff0e6c57a0e88208a7fb723ec9fb26800cfbc4; measurement receipt sha256:4feb710f4285d4f1a27cabaeca9e8b34fe808c3da8ccd7e9033c74c077225620; isolation sha256:b568d984120dc112f697754f4fb3c44d6f111e8acca6c1dcdc953243ff61c6f1; custody sha256:33b0dc3bcf6a48ff0c1dd145f2729a0d5e168d02649d0ced23e6809078094545.

21. Integrated reconciliation table

- Step 47: cubic lattice and six nearest neighbours -> SFT-MAT-CRYST-CUBIC-COORDINATION-001
- Step 49: crystalline order and crystallographic restriction -> SFT-MAT-CRYST-TRANSLATION-001, SFT-MAT-CRYST-ROTATION-RESTRICTION-001, SFT-MAT-CRYST-SYSTEMS-001, SFT-MAT-CRYST-BRAVAIS-001
- Step 52: paired-carrier superconducting response -> SFT-MAT-SC-PAIR-001, SFT-MAT-SC-ZERO-RESISTANCE-001, SFT-MAT-SC-MEISSNER-001, SFT-MAT-SC-FLUX-QUANTIZATION-001, SFT-MAT-SC-JOSEPHSON-001
- Step 54: electronic bands, gaps and transport classes -> SFT-MAT-ELEC-BAND-GAP-001, SFT-MAT-ELEC-CONDUCTOR-CLASS-001
- Step 72: three acoustic lattice-excitation branches -> SFT-MAT-CRYST-PHONON-001
- Step 74: aligned and opposed magnetic order -> SFT-MAT-MAG-FERROMAGNETISM-001, SFT-MAT-MAG-ANTIFERROMAGNETISM-001
- Step 75: semiconductor carrier types, doping and junction balance -> SFT-MAT-ELEC-CARRIER-DUALITY-001, SFT-MAT-SEMI-DOPING-001, SFT-MAT-SEMI-PN-TYPE-001, SFT-MAT-SEMI-JUNCTION-001, SFT-MAT-SEMI-TRANSPORT-001
- Step 133: quasicrystalline order without periodic translation -> SFT-MAT-CRYST-ROTATION-RESTRICTION-001, SFT-MAT-CRYST-RECIPROCAL-001, SFT-MAT-CRYST-QUASICRYSTAL-001
- Step 137: superfluid flow and quantized circulation -> SFT-MAT-SF-SUPERFLUID-001, SFT-MAT-SF-CIRCULATION-001
- Step 143: topological class and bulk-boundary protection -> SFT-MAT-TOPO-INVARIANT-001, SFT-MAT-TOPO-BULK-BOUNDARY-001
- Step 193: elastic, plastic, slip, strength, fracture, fatigue and creep response -> SFT-MAT-MECH-STRESS-STRAIN-001, SFT-MAT-MECH-ELASTICITY-001, SFT-MAT-MECH-PLASTICITY-001, SFT-MAT-MECH-SLIP-001, SFT-MAT-MECH-MODULUS-001, SFT-MAT-MECH-STRENGTH-HARDNESS-001, SFT-MAT-MECH-FRACTURE-001, SFT-MAT-MECH-FATIGUE-CREEP-001
- Step 291: quasicrystal component of the mixed materials-and-astrophysics step -> SFT-MAT-CRYST-QUASICRYSTAL-001; routed remainder: Planetary and Tully-Fisher claims are not Materials results and remain assigned to Astronomy and Cosmology.

22. External-source ledger

- BIPM-JCGM-VIM-TRACEABILITY-2026-07-24 - https://jcgm.bipm.org/vim/en/2.41.html; snapshot experiments/external_sources/materials/snapshots/bipm-vim-traceability.html; sha256:3c06dca2ea8ad71516b027b9979e7a8381edd1576606ad10a379591c4cb6dba4; measurement result, reference chain, calibration and uncertainty.
- NIST-REFERENCE-MATERIALS-2026-07-24 - https://www.nist.gov/reference-materials; snapshot experiments/external_sources/materials/snapshots/nist-reference-materials.html; sha256:2ff393988c4880068ca103e365b0f1e6aed5a02a493b33507f6684a2d9271db9; material identity, property characterization, uncertainty and traceability.

- NIST-MATERIALS-DIVISION-2026-07-24 - https://www.nist.gov/mml/materials-science-and-engineering-division; snapshot experiments/external_sources/materials/snapshots/nist-materials-division.html; sha256:fac19ae0959946a8e73f970bf76c73325d93f4671ded97f8699dfcc5e1bf53e3; materials measurement, composition, interfaces, processing, damage, classes and lifecycle.
- NIST-WULFFMAN-CRYSTALLOGRAPHY-2026-07-24 - https://www.ctcms.nist.gov/wulffman/docs_1.2/; snapshot experiments/external_sources/materials/snapshots/nist-wulffman-crystallography.html; sha256:52558a4ed54b8ecc464b0ba83d879a0cbbffcb7ed25385f7742d021e87b43cb6; lattice generation, rotation, seven crystal systems and fourteen Bravais classes.
- NIST-BRAVAIS-CLASSIFICATION-2026-07-24 - https://www.nist.gov/publications/semi-supervised-approach-automatic-crystal-structure-classification; snapshot experiments/external_sources/materials/snapshots/nist-bravais-classification.html; sha256:82fc7e011a992352d5200131028339a83eca0938c7462131ada2dcc059f38c91; diffraction classification across all fourteen Bravais lattices.
- NIST-SHECHTMAN-QUASICRYSTALS-2026-07-24 - https://www.nist.gov/nist-and-nobel/dan-shechtman; snapshot experiments/external_sources/materials/snapshots/nist-shechtman-quasicrystals.html; sha256:911c06e466a95d58749422c56cd4e33fa1fbece04ec06977af428d7dac23ae0e; periodic rotation orders and fivefold quasicrystal observation.
- NIST-QUASICRYSTAL-MEASUREMENT-2026-07-24 - https://www.nist.gov/news-events/news/2025/04/rare-crystal-shape-found-increase-strength-3d-printed-metal; snapshot experiments/external_sources/materials/snapshots/nist-quasicrystal-measurement.html; sha256:a1b7cc0a5882502bb31b9706225b926c9ffb1e8d93ea5c0043249376c4d6ff18; microscope confirmation, nonrepeating order and fivefold symmetry.
- NIST-POINT-DEFECTS-2026-07-24 - https://www.nist.gov/programs-projects/measurements-point-defect-chemistry-complex-oxides; snapshot experiments/external_sources/materials/snapshots/nist-point-defects.html; sha256:b885250bd645ed8eaefa55a29d99ea4436b691f77c17c74a457acb6acf88335a; vacancies, interstitials, substitutions, diffusion, composition and defect metrology.
- NIST-TEXTURE-PHASE-FRACTION-2026-07-24 - https://www.nist.gov/programs-projects/ncal-quantifying-crystallographic-texture-and-phase-fraction; snapshot experiments/external_sources/materials/snapshots/nist-texture-phase-fraction.html; sha256:b5db09c98c668fbc3c4b7b8f1cf8b83c1b093fea6c42b7846316fb10004edabc; grains, phase fractions, texture, deformation and uncertainty.
- NIST-MULTISCALE-MATERIALS-2026-07-24 - https://www.nist.gov/programs-projects/multiscale-structure-and-dynamics-advanced-technological-materials; snapshot experiments/external_sources/materials/snapshots/nist-multiscale-materials.html; sha256:2de46a8509f29f4ad9367652b9b4a4740b6cb0e1c7be1bf4cd3087b25e2eed2a; multiscale microstructure, phase evolution, heat treatment, porous materials and in-situ processing.
- NIST-SEMICONDUCTORS-2026-07-24 - https://www.nist.gov/mml/mmsd/primary-focus-areas/semiconductors; snapshot experiments/external_sources/materials/snapshots/nist-semiconductors.html; sha256:ef9fac0f8cf33879c90fd5914dfd0ed1ac70c076972dbdb94b1ba3843be91325; semiconductor structural, nanomechanical, thermal and spectroscopic measurement.
- NIST-FUNCTIONAL-ELECTRONIC-MATERIALS-2026-07-24 - https://www.nist.gov/programs-projects/genomics-electronic-materials; snapshot experiments/external_sources/materials/snapshots/nist-functional-electronic-materials.html; sha256:355b4893ea38080dd5bc36430beedf4631667ed50d465b08bb1c2aefd91e544e; dopants, defects, piezoelectric, ferroelectric, magnetic, topological and superconducting materials.
- NIST-SUPERCONDUCTING-GRAIN-BOUNDARIES-2026-07-24 - https://www.nist.gov/news-events/news/2009/06/nist-discovers-how-strain-grain-boundaries-suppresses-high-temperature; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-grain-boundaries.html; sha256:1309d6599f6894e517dac8f3adc0b6b8506842cdcccc680329007351cde6abf2; grain boundaries, strain, dislocations, current flow and superconducting response.

- NIST-SUPERCONDUCTING-FLUX-2026-07-24 - https://www.nist.gov/ncnr/flux-lattice-superconductors-and-melting; snapshot experiments/external_sources/materials/snapshots/nist-superconducting-flux.html; sha256:47bbe8e690eb0c796876e0a0c808d131fd9f8fee98aee7ed4ab645b5b5de4cd6; zero resistance, Meissner expulsion, quantized flux and vortices.
- NIST-JOSEPHSON-STANDARD-2026-07-24 - https://www.nist.gov/news-events/events/josephson-volt-nists-first-quantum-electrical-standard; snapshot experiments/external_sources/materials/snapshots/nist-josephson.html; sha256:371bb54f09d8e66d74cda08e66518b44ebf752445ee2a049b26a76fd669487e0; superconductive tunnel junctions and quantized Josephson voltage.
- NIST-SUPERFLUID-CIRCULATION-2026-07-24 - https://www.nist.gov/news-events/news/2012/11/first-controllable-atom-squid; snapshot experiments/external_sources/materials/snapshots/nist-superfluid-circulation.html; sha256:49a5edb2dc307d0849d5fb5a245be46166dbffe2c50f2e9422a78f139aa5f8a0; superfluid flow, vortices and discrete quantized circulation.
- NIST-TOPOLOGICAL-INSULATORS-2026-07-24 - https://www.nist.gov/programs-projects/topological-insulators; snapshot experiments/external_sources/materials/snapshots/nist-topological-insulators.html; sha256:4cbe823dffeabd728844a897aa1c5a222dfc3aef465263a06501974e194df449; topological invariants, insulating bulk, band gap and metallic boundaries.
- NIST-FATIGUE-FRACTURE-2026-07-24 - https://www.nist.gov/programs-projects/additive-manufacturing-fatigue-and-fracture; snapshot experiments/external_sources/materials/snapshots/nist-fatigue-fracture.html; sha256:449a86d3ae74591957dd1a22cd314274c0a4b7acd05d9d6a9f9671af4ea06a11; processing-structure-property-performance, defects, fatigue and fracture.
- NIST-POLYMER-COMPOSITES-2026-07-24 - https://www.nist.gov/programs-projects/polymer-composites; snapshot experiments/external_sources/materials/snapshots/nist-polymer-composites.html; sha256:0ca34811fdf8880ce744219b7ad7f9574bacfc01e5ed0b073f8c64a225f692c5; polymer matrix composites, interfaces, strength, toughness, fatigue and recycling.
- NIST-TRANSPORT-THERMOELECTRIC-2026-07-24 - https://www.nist.gov/programs-projects/transport-property-measurements-semiconductors-and-energy-materials; snapshot experiments/external_sources/materials/snapshots/nist-transport-thermoelectric.html; sha256:4db74747d716cccfc34518d4885187f19dfdbb8b254eef29c92e6ed32f1f20a0; thermal conductivity, heat capacity, electrical transport and thermoelectric conversion.
- NIST-CERAMIC-ADDITIVE-2026-07-24 - https://www.nist.gov/programs-projects/additive-manufacturing-ceramics; snapshot experiments/external_sources/materials/snapshots/nist-ceramic-additive.html; sha256:540ef912a827c1834a113352ac0305c7e8d25a76283dcc1bcbab48f0b826a3e9; ceramic feedstock, defects, sintering, densification, phases and microstructure.
- NIST-POLYMER-PROCESSING-2026-07-24 - https://www.nist.gov/programs-projects/polymer-advanced-manufacturing-and-rheology; snapshot experiments/external_sources/materials/snapshots/nist-polymer-processing.html; sha256:11279429040fb5daff9108435f0c510fc12d89e09710e799b1a6182f63a63a18; polymer processing, crystallization, rheology, kinetics and recycling.
- NIST-SURFACE-DAMAGE-2026-07-24 - https://www.nist.gov/programs-projects/surface-damage-polymer-nanocomposites-project; snapshot experiments/external_sources/materials/snapshots/nist-surface-damage.html; sha256:7f0a61e225462f1514a9b2b9e459f5a0727ef2dc291bbcf5e83626f52359169d; surface/bulk distinction, degradation, radiation, moisture, cracking and nanofiller release.
- NIST-NANO-PROTOCOLS-2026-07-24 - https://www.nist.gov/mml/nano-measurement-protocols; snapshot experiments/external_sources/materials/snapshots/nist-nano-protocols.html; sha256:366ea750d8ed231893eae786234339a56067e86d14d8e282496d130680df5c51; nanomaterial property measurement, reproducibility, preparation, lifecycle and biological/environmental matrices.

- `NIST-SOLIDIFICATION-2026-07-24` - https://www.nist.gov/publications/solidification-0; snapshot `experiments/external_sources/materials/snapshots/nist-solidification.html`; `sha256:d5e7cf15296ae0a6f1df3e022a8b4bb4c618a46d6b4f9b36ab2f49f7532a9d6c`; solidification transport, nucleation, growth, interfaces, segregation and processing.
- `NIST-METAL-ADDITIVE-CORROSION-2026-07-24` - https://www.nist.gov/programs-projects/additive-manufacturing-metals; snapshot `experiments/external_sources/materials/snapshots/nist-metal-additive-corrosion.html`; `sha256:a1bb9a6bc22eb85fb7a1b3c7d0ac4bc0837d68832b7b7c6f357a68ea6cde0322`; phase transformation, heat treatment, corrosion and environmental cracking.
- `NIST-NANOTRIBOLOGY-2026-07-24` - https://www.nist.gov/programs-projects/nanotribology-nanomanufacturing-archived; snapshot `experiments/external_sources/materials/snapshots/nist-nanotribology.html`; `sha256:3823f313343779c2a4a74a3e251a620319c89fef2142f2b1e58fa03250fec523`; interacting surfaces, relative motion, friction, adhesion, lubrication and wear.
- `NIST-OPTICAL-MATERIALS-2026-07-24` - https://www.nist.gov/programs-projects/theory-optical-properties-materials; snapshot `experiments/external_sources/materials/snapshots/nist-optical-materials.html`; `sha256:3cd2616a7d74445fda880027ee0d7659d112e27a48e9a86616e766b41f0c5f1b`; index of refraction, absorption, reflectance, dielectric response and photonic crystals.
- `NIST-SIX-NEIGHBOUR-MESH-2026-07-24` - https://math.nist.gov/oommf/doc/userguide20a0/userguide/Standard_Oxs_Ext_Child_Clas.html; snapshot `experiments/external_sources/materials/snapshots/nist-six-neighbour-mesh.html`; `sha256:505ca000d91c5ef134836cbc75307df1584750dfbfb2faf5c4e55756d054f5c2`; six nearest neighbours as forward/backward positions on three coordinate axes.
- `NIST-POLYMER-INTERFACE-CONSORTIUM-2026-07-24` - https://www.nist.gov/el/mssd/polymer-surfaceinterface-consortium; snapshot `experiments/external_sources/materials/snapshots/nist-polymer-interface-consortium.html`; `sha256:2d48772b1b901e657fa86e4ee28fad15e176dd675911ce8322b54cc718c32635`; surface damage, interfacial adhesion and interface-property measurement.
- `NIST-SEMICONDUCTOR-CARRIERS-2026-07-24` - https://www.nist.gov/news-events/news/2017/03/testing-performance-semiconductors-light; snapshot `experiments/external_sources/materials/snapshots/nist-semiconductor-carriers.html`; `sha256:05c774e4a30e6009f031b3064b16319d181ddac92e6d088a01bb6fe362172cc5`; electron and hole carrier signs, doping, transport and Hall measurement.
- `NIST-ANTIFERROMAGNETIC-COUPLING-2026-07-24` - https://www.nist.gov/ncnr/acns-2020-tutorial-ii-practical-approach-fitting-neutron-reflectometry-data/understanding; snapshot `experiments/external_sources/materials/snapshots/nist-antiferromagnetic-coupling.html`; `sha256:b8ce5105622c02b84bffe643693c05c199df57beae79bd9c5e29f26c4360e85e`; polarized-neutron measurement of magnetic structure and antiferromagnetic coupling.
- `NIST-GLASS-TRANSITION-2026-07-24` - https://www.nist.gov/publications/glass-transition-its-measurement-and-underlying-physics; snapshot `experiments/external_sources/materials/snapshots/nist-glass-transition.html`; `sha256:2d6b99e46aae9d3c6a503494eedd17db45d4b48a633bb254cd3a6a5195440777`; thermodynamic and kinetic glass-transition measurement in glass-forming materials.

23. Reproducibility and falsification

The repository's one-command validation route parses every schema, verifies snapshot and source hashes, checks the frozen inventory, exercises the engine and replays every admitted execution manifest entry. The Materials publication gate separately requires all 84 evidence packages, complete candidate/decision parity, one survivor each, four passing controls each, 84 empirical validations, paper inclusion of each receipt identity, the pre-source seal, V2 reconciliation, evidence maps and a rendered PDF.

The branch fails if any registered source hash changes without an explicit new version, any required discriminator is absent, any prediction opens a target before sealing, any row is omitted, any tampered record passes, any dependency no longer resolves to the root, any census is incomplete, any claim has other than one survivor, or an independent validator fails to reconstruct the result.

24. Limitations

- Closure is relative to the declared positive-finite grammars and question surface, not every conceivable future material.
- The external correspondence records are independent tests, not derivational premises.
- Property laws are conditional on specimen, preparation, direction, method, scale, history and environment; this paper does not claim universal numerical moduli, conductivities or thresholds.
- Biomaterial claims stop at the material/biological interface. Biological functions and clinical outcomes require their own branches.
- Industrial performance can test a law but cannot select it.

25. Conclusion

The Materials branch is complete at its frozen V3 boundary: 84 required laws, 21,504 exact candidates and decisions, 84 unique survivors, 336 adverse controls, 84 independent reconstructions, 84 post-seal authority checks and 84 root traces. The result is an openly inspectable computational account of material identity, organization, defects, collective response, classes, processing, degradation and function. Its evidential authority lies in the complete public chain from the premise-free root theorem to immutable receipts and falsifiable external checks.

26. Publication and repository links

- Canonical repository: https://github.com/MettaMazza/ernos-labs-sft-platform
- Materials release: https://github.com/MettaMazza/ernos-labs-sft-platform/releases/tag/materials-v1.0.0
- Zenodo DOI: https://doi.org/10.5281/zenodo.21532482

- Author: Maria Smith, Ernos Labs
- Contact: Maria.Smith.Sftoe@gmail.com

- Submissions: https://discord.gg/ucwGryVxGr

27. References

- Bureau International des Poids et Mesures / Joint Committee for Guides in Metrology, International Vocabulary of Metrology.
- National Institute of Standards and Technology, Materials Science and Engineering Division records listed in the external-source ledger.
- Smith, Maria. *From Nothing to Fold*. Ernos Labs Foundation Branch Paper 001. doi:10.5281/zenodo.21515629.
- Smith, Maria. *From Fold to Mathematics*. doi:10.5281/zenodo.21516146.
- Smith, Maria. *From Distinction to Information*. doi:10.5281/zenodo.21516916.
- Smith, Maria. *After Turing: The Fold Machine*. doi:10.5281/zenodo.21518311.
- Smith, Maria. *The Quantum Fold Machine*. doi:10.5281/zenodo.21518313.
- Smith, Maria. *From Fold to Physics*. doi:10.5281/zenodo.21520881.
- Smith, Maria. *From Fold to Chemistry*. doi:10.5281/zenodo.21531455.